

UNIVERSITI TEKNOLOGI MARA

**DATA VISUALIZATION OF THE IMPACT OF
RADICAL ARTIFICIAL INTELLIGENCE
TECHNOLOGY TOWARD JOBS**

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SUPERVISOR APPROVAL

DATA VISUALISATION OF THE IMPACT OF RADICAL ARTIFICIAL INTELLIGENCE TECHNOLOGY TOWARDS JOBS

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The thesis was prepared under the supervision of the project supervisor, Dr Aznoora binti Osman. It was submitted to the College of Computing, Informatics, and Mathematics and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Science (Hons) in Information Technology.

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STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise, are fully acknowledged in accordance with the standard referring practices of the discipline.



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ABSTRACT

Artificial intelligence technologies have the capacity to perform a wide range of tasks across multiple industries, raising concerns about job displacement. This research project aimed to develop a dashboard that visualizes the impact of AI technology on various job domains. Key objective is to spread awareness about the AI impact on jobs through visualization dashboard. The research utilizes Agile principles for iterative data exploration. Data extract and transform was done using Microsoft Excel and Jupyter Notebook. The analysis and visualization are conducted with Microsoft Power BI, following data warehousing with Apache Hive. The findings revealed insights into the progression of AI models, the average AI impact across job categories, and the distribution of job tasks and AI models by job category, which can help stakeholders better prepare for the future of work. The research contributes to understanding the impact of radical AI technology on jobs, enabling informed decisions and strategies. Future work could explore the integration of real-time data sources and the development of predictive models to forecast the long-term effects of AI on employment.

Keywords: Artificial Intelligence, Big Data, Data Visualisation, Employee, Job Displacement, Robot, Technology, Unemployment.

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CHAPTER 1

INTRODUCTION

This chapter provides the background of the studies, which includes the details of the research significance, problem statements, research objectives, and the scope of the research. Subchapter 1.1 introduces the background of the research. Subchapter 1.2 describes the problem statement of this research. Subchapter 1.3 mentions the project objectives, Subchapter 1.4 discusses the project scope, Subchapter 1.5 describes the significance of the research, Subchapter 1.6 mentions the thesis outline, and finally, Subchapter 1.7 is the summary for chapter one.

1.1 Background of Study

Machine thinking, or artificial intelligence as we know it now, started in the 1950s. One of the early documentations of attempts to go for AI is by Alan Turing, who asked the question, “Can machines think? ” in his book “Computing Machinery and Intelligence.”. Normal computers are supplied with sets of instructions for them to follow exactly as ordered, preventing unnecessary work and errors and resulting in consistent output (Turing, 1950). Artificial intelligence, however, refers to the ability of computers to mimic mankind's “thinking” method, delivering uniqueness in every possible option and outcome. AI is a rapidly developing field, with new algorithms and applications being developed all the time to revolutionise many industries, and change many aspects of our lives.

While certain companies like Amazon, Dropbox, and Tencent are starting to replace physical labour with AI, some companies like Google and Microsoft are excited to embrace the new technology capabilities. Therefore, many people are still worried that AI technologies might cause widespread job losses in the next 10 to 20 years.

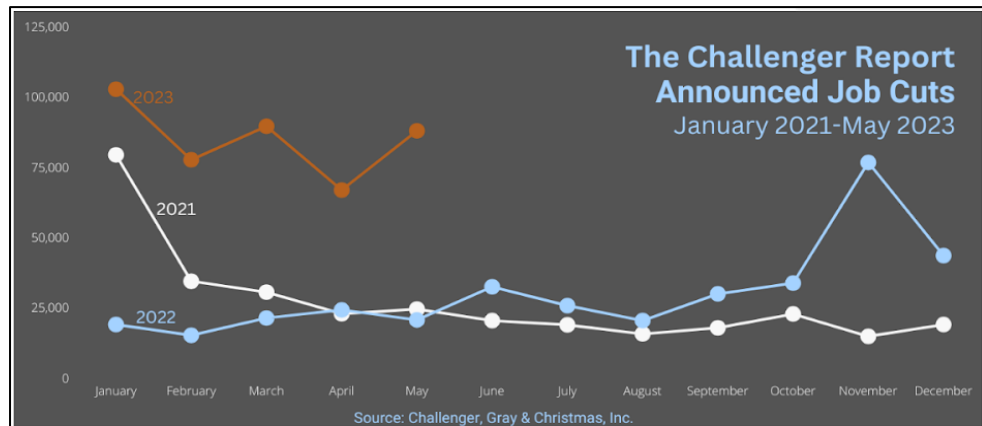


Figure 1.1 Announced Job Cuts from January 2021 to May 2023
(source: The Challenger, Gray & Christmas, 2023)

Figure 1.1 shows a graph of announced job cuts by companies from January 2021 to May 2023. According to Challenger, Gray and Christmas (2023), AI was responsible for nearly 4,000 job losses in May 2023, making it the seventh-highest contributor to employment losses in May of 2023. This is certainly a threat to the future jobs of younger generations if they wrongly choose their study and work courses.

To know which job categories are most affected by AI, we need to scan through all job categories and identify how they are connected with AI automation. The best way to do this is through visualisation and big data research. Big Data refers to the processing and analysis of large and complex datasets to gain business insights. Big Data Visualisation means presenting large sets of data through visual aids, allowing better perception in decision-making (Tiburca, 2021). The visualisation of big data allows the interpretation of the dataset in a more meaningful way, identifying possible patterns and boosting its usability.

1.2 Problem Statement

The purpose of this study is to create a web-based dashboard system that can analyse and visualise the impacts of AI on different job domains. Using dynamic data visualisation and dashboards, these data should be able to be displayed for specific jobs requested by the user and the overall impacts of AI on each domain of the category of jobs. It simplifies the understanding of

complex data, making it easier to review. Users should be able to achieve their goals, like making predictions on the next work domain to be automated by AI, as well as helping them make decisions and be adept through the changes.

Ideally, assessing the impacts of AIs on jobs is essential for ensuring a successful transition to the future of work, as new jobs may emerge as technologies and AI continue to grow. Good insights allow people and companies to understand how AI might affect their job scope and employees, improving decision-making and reducing costs in terms of time spent, expense payments, and job cuts. Governments can develop policies to support workers who might be displaced, ensuring a smooth transition to new job roles, and educational institutions can update their curricula to prepare students for jobs that are less susceptible to automation (ITU, 2018). Through proper understanding, we know that there are still ethical considerations in terms of implementation, fairness, and biases because AI automation cannot deliver the uniqueness of human thinking in its outputs. A judge can be assisted by AIs, performing as lawyers for the defendant, but to a certain extent only as there are still ethical obligations (Greenstein, 2020).

Nevertheless, the development of AI has affected almost all categories of employment. Houston (2023) stated that implementation of AI is much more efficient, thus, Dropbox is reducing its workforce by 16%, or about 500 people, in order to implement AI technologies. This shows that AIs nowadays are powerful enough to be able to perform certain jobs much better, more cost-efficiently, and quicker than manpower could. A survey done on US graduates shows that they are divided on whether AI will replace their jobs, with 46% believing that AI will threaten their future jobs and the remaining 54% stating that AI will not replace them, but most of both sides agreed they need to develop digital soft skills and proper training on working with AI (Cengage Group, 2023). Employers are aware that new technologies are rapidly changing the hiring landscape, and as a result, they are looking for new hires who can combine soft skills such as emotional intelligence, negotiation, and communication with the ability to work with AI (Nietzel, 2023). Those who

lack digital soft skills and have no experience with AI are less likely to be chosen.

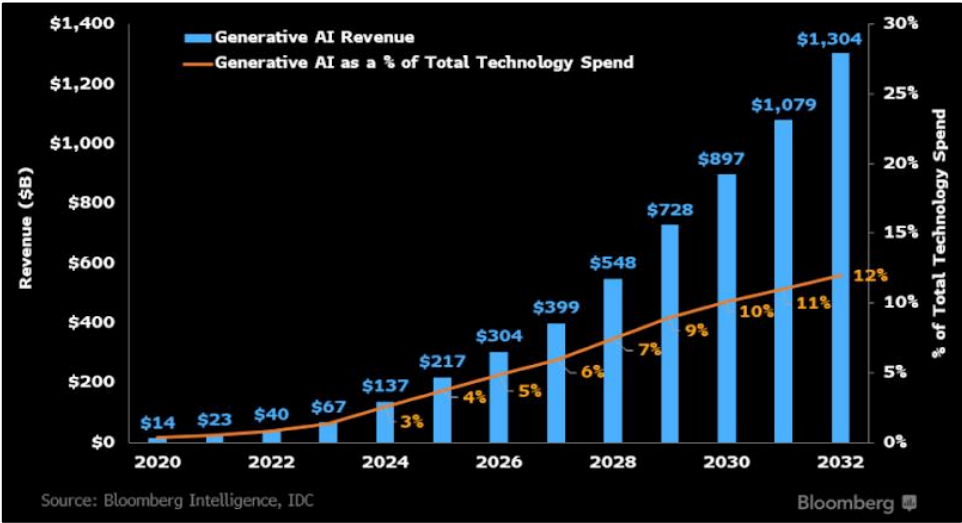


Figure 1.2 Prediction of Revenues about Technology Spent from 2020 to 2032 (source: Rudnitsky, 2023)

Figure 1.2 shows the prediction made by Bloomberg about the revenues gained based on technology spent by the companies from 2020 to 2032. Rudnitsky (2023) stated that the introduction of consumer-focused AI systems such as ChatGPT and Google's Bard is expected to initiate an exponential increase in the generative AI market, which is expected to grow from \$40 billion in revenue last year to an estimated \$1.3 trillion by 2032. The companies that stand to benefit the most from this growth are Amazon, Microsoft, Google, and Nvidia.

Ignoring the growth of AI could lead to several negative consequences, including job displacement, skills mismatch in employment, economic inequality, social unrest, a lack of preparedness, missed opportunities, and a lack of policy response (Jane, 2023). AI is expected to replace many jobs, leading to job displacement and unemployment. Without proper research and training, workers may not have the necessary skills to transition to new job roles, leading to a skills mismatch. The impact of AI on jobs may not be felt evenly across all workforces, with office workers likely to be more affected than manual labourers, leading to economic inequality. Job displacement and

economic inequality could also lead to social unrest and political instability. Individuals and organisations may not be prepared for the future of work, leading to a lack of preparedness. They may also miss out on new opportunities for employment and skill development. Policymakers may not develop policies to mitigate the negative effects of automation and ensure that the benefits of AI are shared more equitably.

1.3 Objective

The proposed study of this big data is to create data visualisations to spread awareness about the impacts of AI technologies on current job domains. This study has several objectives, which include:

RO1: To determine the data required to analyse the impacts of the radical artificial intelligence technology towards jobs.

RO2: To design and develop dashboards that include dynamic visualisations of the impacts of radical artificial intelligence technology towards jobs using Figma and Microsoft Power BI.

RO3: To evaluate the user acceptance of the developed web-based dashboard using the Technology Acceptance Model (TAM).

1.4 Project Scope

The scope of this project is to analyse and visualise the impacts of AI towards job domains by calculating and comparing the cumulative value of the impact of AI on each job domain using the dataset acquired. The data will be displayed in a more interactive, meaningful, and understandable way using dynamic visualisation technology such as Microsoft Power BI and Python libraries, allowing the user to choose specific output and display requested results. The dataset used in this project was obtained from the Kaggle website, but the source was from the website called "There's an AI for that." The website keeps track of the number of AI models developed and registered online and

simultaneously records real-time changes in the number of job tasks, and the percentage of AI impact on job titles.

The primary target users for this project are graduates, policymakers, and business owners, who are individuals who are directly connected to employment. Resources or information gathered from this project will help graduates prepare and find suitable jobs in the future, policymakers to develop effective employment policies to protect workers from being replaced, and business owners to train, hire, and retain qualified employees suitable to the company’s changes towards AI automation. Figure 1.3 shows a mind map of the scope of this research project.

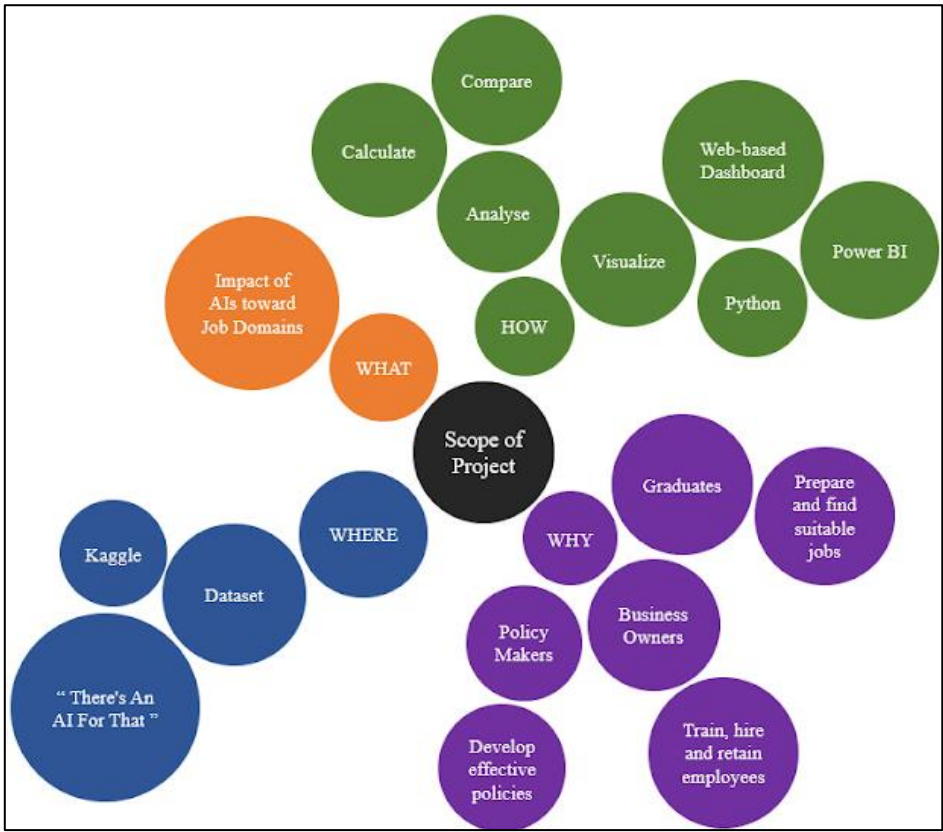


Figure 1.3 Mind Map of the Research Project Scope

1.5 Project Significant

This project will prove to be of help to various individuals and industries. Policymakers can develop policies that mitigate the negative effects of

automation and ensure that the benefits of AI are shared more equitably. Governments can develop policies to support workers who might be displaced, ensuring a smooth transition to new job roles. Academicians and educational institutions can update their curricula to prepare students for jobs that are less susceptible to automation and learn how to work together with AI in the future. Employers could use the research to evolve their hiring habits and prioritise softer skills and previous job experience. Workers and employees could use the research to understand the potential impact of AI on their jobs and prepare for the future of work by developing new skills, training, and exploring new job opportunities. Business owners can prepare for the future by understanding which job roles might be automated and strategizing by upskilling or reskilling their workforce accordingly. For graduates, both undergraduates and postgraduates, they can plan on considering digital soft skill classes and choose alternate courses to make sure that they can get better and more suitable qualifications for their resume in their future job hunting. And practitioners can use the visual data to improve their research and decision-making. They can also expect future job opportunities that might rise as AI continues to grow. Table 1.1 shows the list of individuals who benefit from this research project.

Scope	Significant
Businesses	Able to strategize for upskilling or reskilling employees.
Policymaker	Allow policies to be developed to protect workers' rights.
Employers	Can prioritise hiring workers with suitable required skills.
Workers	Train themselves to match future job qualifications.
Educators	Can update their curricula and learn to work with AIs.
Graduates	Better in planning for their future to match job scaling.
Practitioner	Allow research and improve decision-making.

Table 1.1 List of Individuals who Benefit from this Research Project

1.6 Project Outline

This thesis consists of six main chapters, and all the chapters are briefly introduced as follows:

Chapter 1 explains the introduction of the project, which includes the background of the study, problem statement, objective, project scope, project significance, project outline, and the summary for Chapter 1. Chapter 2 discusses the literature review. These literature reviews have been used to gather information and requirements for the development of this project. Chapter 3 clarifies the methodology used for the project plan and development. This chapter explains the methodology and provides a detailed explanation for each phase. It also consists of a project development schedule and a summary of this chapter. Chapter 4 describes the development process of this project in detail. It also includes details of all the tools and software that were used in this project's development. Chapter 5 explains the results and analysis of this project by providing questionnaires using usability testing. Chapter 6 discusses the conclusion and recommendations suggested for the project, as well as the conclusion from the research and analysis.

1.7 Summary

The focus of this research project is to spread awareness of how much AI impacts every job domain. Although AI is a growing title and almost impossible to miss, some of us might still be unaware of the pros and cons of these AI implementations. Our gathered data will be analysed and visualised on a web-based dashboard to further improve its usability and understanding to the general public further improving their decision-making, better predictions, and a better understanding of future expectations.

CHAPTER 2

LITERATURE REVIEW

This section reviews related topics and issues regarding the project research which is the impact of radical artificial intelligence technology towards jobs. It primarily explains the concepts, methods, and previously researched work that are related to this research project.

2.1 Big Data

Big data is the large, complex, and varied sets of data that are difficult to store, process, and visualise using traditional data-processing applications, and are used to get specific information. The data can be structured, unstructured, or even semi-structured. These data were collected from many sources. There are six, but these data are often characterized by three V's, which are "volume" of data, "variety" of data types, and "velocity" of the data generated, collected, and processed (Botelho & Bigelow, 2022). Volume refers to a large amount of data, variety refers to the different types of data, and velocity is the speed of the data received or processed. Big data is used in machine learning, predictive modelling, and other advanced analytics to solve business problems and make informed decisions. Figure 1.2 below shows the six Vs of characteristics of big data.

VOLUME	VARIETY	VELOCITY	VERACITY	VALUE	VARIABILITY
The amount of data from myriad sources.	The types of data: structured, semi-structured, unstructured.	The speed at which big data is generated.	The degree to which big data can be trusted.	The business value of the data collected.	The ways in which the big data can be used and formatted.
					

Figure 2.1 The Six Characteristics of Big Data

(source: Botelho & Bigelow, 2022)

2.1.1 Importance of Big Data

Big data is important to various industries, including management, finance, and healthcare. It collects, stores, and analyses large volumes of data from various sources, using them to identify patterns, trends, and insights that can help organisations to make better decisions, improve operations, and increase revenue and profits.

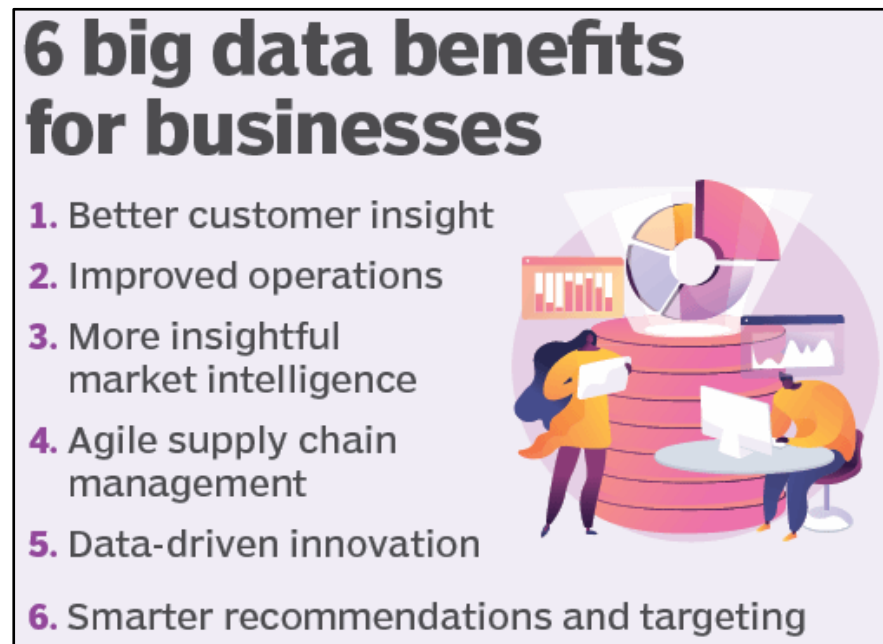


Figure 2.2 The Six Big Data Benefits for Businesses

(source: Botelho & Bigelow, 2022)

Figure 2.2 shows the six big data benefits for businesses, which are better customer insight, improved operations, more insightful market intelligence, agile supply chain management, data-driven innovation, and smarter recommendations and targeting. Businesses that effectively apply big data will have a significant competitive advantage because they can make faster and more informed business decisions (Botelho & Bigelow, 2022).

2.1.2 Challenges of Big Data

Big data offers a variety of advantages in almost all industries, providing greater opportunities for companies to better excel in their productivity, expenses, and businesses, however, there are challenges in implementing big data. One of the biggest challenges in big data is the large amount of data that requires quite a significant amount of storage and space to save as the data keeps growing (Simplilearn, 2021). All the data collected comes in different types and formats, making it hard to keep. For example, traditional databases cannot store unstructured data. These issues can be solved by investing in big data storage services such as Microsoft Azure, and Google Cloud. Other than providing backup and disaster recovery options, these technologies are highly scalable and capable of handling petabytes of data. Next, big data can only be processed with advanced processing techniques and applications as the raw information is huge and comes in different formats (Alcyone Technologies, 2023). Organisations use Apache Spark because it can support multiple data formats, and Hadoop can process both structured and unstructured raw data. Collecting diverse sets of information from a variety of inputs usually results in the issue of security. Security and privacy may be jeopardised by combining data sources that were not meant to be integrated (MongoDB, N.D.). Building an effective and secure database architecture is important for protecting sensitive information from hackers and cybercriminals. There are also challenges in data governance and integration. Data needs to be in compliance with regulations and policies. The best solutions are to establish a data governance framework and utilise data integration tools to guarantee data consistency between various systems, automate the data integration process, and save manual labour (Alcyone Technologies, 2023).

2.1.3 Application of Big Data in Artificial Intelligence

Big data and artificial intelligence (AI) are inseparable and beneficial to one another, since AI requires vast amounts of data to learn and improve its decision-making processes, whereas big data provides the volume of data

required for AI to analyse and extract insights. Big data and AI have extensive uses in a variety of industries, including healthcare, finance, retail, and transportation.

2.1.3.1 Healthcare and Health Technology

Natural language processing and machine learning methods are used by AI systems, such as IBM's Watson, to help doctors diagnose complex illnesses and recommend personalized treatment options. This improves patient treatment outcomes and lowers medical errors. With the aid of this cutting-edge technology, medical practitioners can now access a vast medical knowledge base and make quicker, more accurate judgements, which in turn improves patient outcomes and lowers medical errors (Bahrynovska, 2023). Figure 2.3 below shows a photo of IBM's Watson that uses AI technology.



Figure 2.3 Photo of the IBM's Watson

(source: Mlot, 2014)

2.1.3.2 Marketing and Business Intelligence

Businesses may leverage artificial intelligence (AI) and big data to analyse consumer data, gain insights into client preferences and behaviour, and adjust their marketing campaigns appropriately. Subsequently, they can utilise such patterns for product improvements, generate better conversions, enhance brand loyalty, identify trends ahead of time, or discover other methods to augment overall customer satisfaction (Walch, 2023).

2.1.3.3 Unmanned Vehicles and Automated Logistics

Drones and autonomous vehicles can explore complex environments and make real-time decisions when needed with the help of artificial intelligence and big data. Route optimisation, traffic prediction, and driver behaviour monitoring are among the applications of AI-powered systems that improve user experiences and operational efficiency (Bahrynovska, 2023). AI algorithms are used by ride-hailing services like Uber and Lyft to improve their matching algorithms, ensuring efficient and timely pickups for passengers.

2.1.3.4 Natural Language Processing

Machine learning in AI refers to the use of algorithms to carry out tasks autonomously and without human intervention. Deep learning uses neural networks to detect complex patterns in vast amounts of data, cognitive computing to mimic the workings of the human brain in addressing complex problems, and natural language processing, which aids in the comprehension and interpretation of human language (Qlik Team, n.d.). This leads to the innovation of AI chatbots, sentiment analysis tools, and other applications that rely on understanding and interpreting human language.

2.2 Data Visualisation

Data visualisation is a graphical or imagery-based representation of data information using visual elements such as charts, graphs, and maps. It allows for a better understanding of the complex and robust set of information by simplifying it. It allows users to identify trends and patterns, leading to better insights and data-driven decisions (Tableau, 2023). There are three categories of data visualisation, which are multidimensional, temporal, and geospatial. Multidimensional are bar charts, pie charts, and tables. Line charts are an example of temporal, and map visualisation is an example of geospatial. Data visualisation is crucial in this project's research because the raw data set obtained can be quite confusing, as it includes percentages and numeric values that are hard to use to make calculations directly from the CSV file.

2.2.1 Data Visualisation Techniques

Using dashboards and other common visualisation techniques helps deliver better tracking and understanding of the data, aid in decision-making, draw conclusions, and capture users' attention. Pie charts, bar charts, and line charts are some of the most common ways to illustrate big data (Tiburca, 2021). Each data visualisation technique has specific usage and requires specific sets of information or data for it to deliver the visual effectively to the users.

2.2.1.1 Dashboards

Dashboards are effective data visualisation tools as they help in displaying, visualizing, and tracking multiple datasets. There are many types of dashboards, some are strategic dashboards, operational dashboards, and analytical dashboards. They can be made dynamic, interactive, and even have real-time data tracking. Seeing every piece of data presented as a whole, users will be able to make more data-driven decisions and understand complex data (Tableau, 2023). Figure 2.4 below shows an example of big data visualisation using a dashboard.

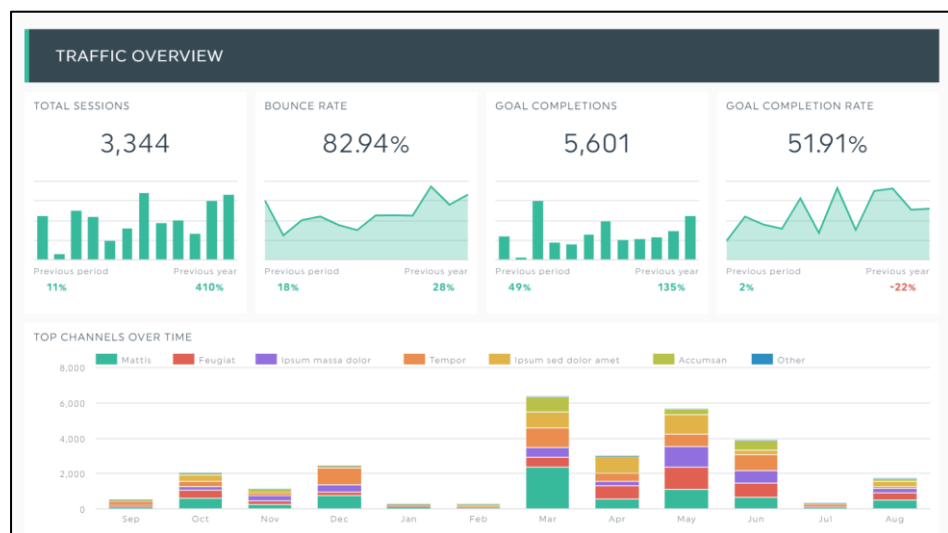


Figure 2.4 Example of Data Visualisation Dashboard

(source: DashThis Team, n.d.)

2.2.1.2 Graphs, Charts and Maps

Graphs, charts, and maps alike are used in visualizing data on the dashboards, to present data more effectively (IBM, n.d.). This will help users understand and interpret complex data, especially non-technical people. Graphs are used to show relationships among the data in a dataset, while maps help to show data hierarchy, geographical area, and related aesthetics (Shoderu, 2022). Charts uniquely present the data, allowing for much easier comparison, categorization, and quicker data comprehension (Slater, 2023). Figure 1.6 shows examples of data visualisation using graphs, charts, and maps.



Figure 2.5 Examples of Data Visualisation Charts

(source: Shoderu, 2022)

Pie charts, a multidimensional data visualisation that uses more than two data dimensions, display percentages of numbers by using a series of slices in circular statistical graphics. The size, central angle, and arc length of each slice correspond to the quantity it represents (Bock, 2018). The pie chart is effective in representing simple data, especially when showing the contribution of each part of the whole data. For this project, pie charts can be used to show the percentages of each job domain that is affected by AI. This will be efficient as it classifies each job into their respective job domains or categories, making users see the whole view from each job domain.

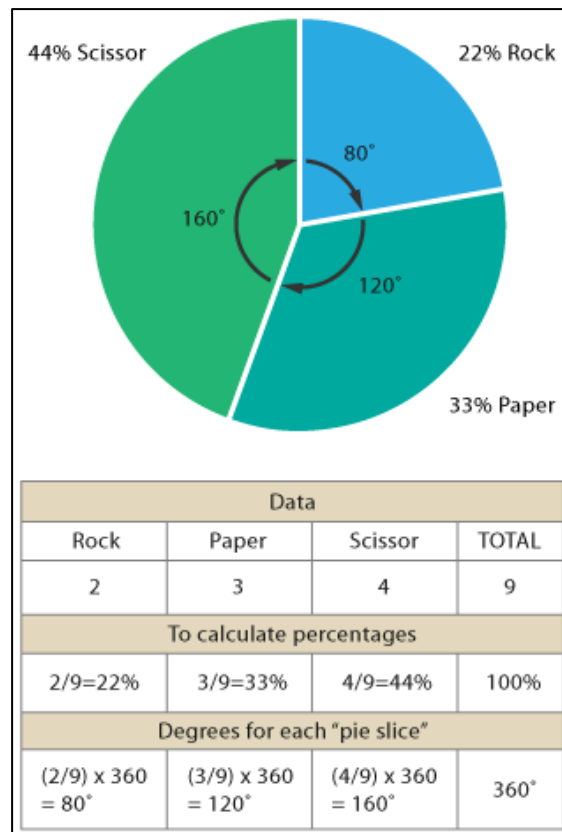


Figure 2.6 Example of Data Visualisations using Pie Chart.

(source: Ribecca, n.d.)

The bar chart is a temporal data visualisation as it is one-dimensional and linear. The bar chart is a graph that uses rectangular bars whose lengths or heights correspond to the values they represent to show categorical data. It can be plotted vertically and horizontally. It is used to display the distribution of data points and makes it easier to compare metric values across different subgroups of the data (Yi, 2019). Several types of bar charts include vertical bar charts, horizontal bar charts, grouped bar charts, and stacked bar charts. Bar charts can be used in this research project to make comparisons between job domains that are almost similar to each other, like administrations, businesses, and management, to identify which job domain is prone to being automated by AI.

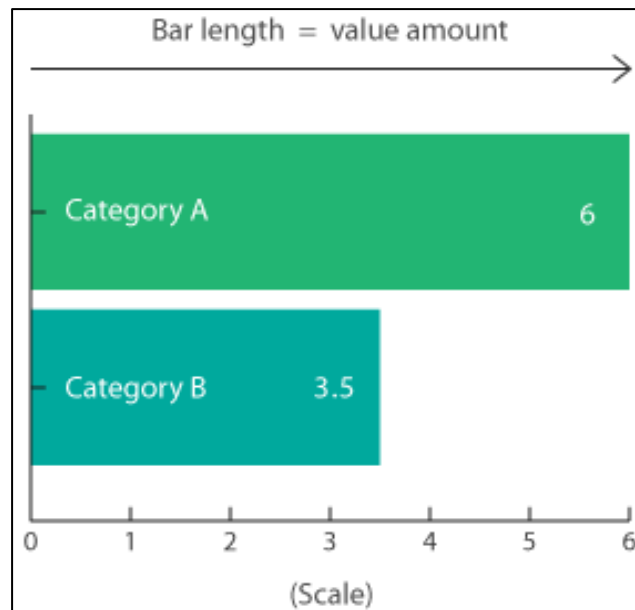


Figure 2.7 Example of Data Visualisations using Vertical Bar Chart
(source: Rebecca, n.d.)

2.2.2 Big Data Visualisation Tools

Visualizing thousands of complex datasets can be time-consuming and very hard. However, there are a variety of data visualisation tools available, allowing users to create visual data representations of large datasets easier and faster.

2.2.2.1 Microsoft Power BI

Microsoft Power BI is part of the Microsoft Power Platform, which contains software services, applications, and connectors, to create both static and interactive data visualisations from a variety of data sources. Developed by Microsoft, these visualisation tools primarily focus on business intelligence. With Power BI, users can connect to any type of data, visualise it, and then easily integrate the visualisations into the daily apps they use. Via an easy-to-use dashboard interface, it provides a broad range of data visualisation tools and software to create different kinds of charts, graphs, and maps (Microsoft, n.d.). Microsoft Power BI can be used in this project research to visualise data as it is simple and easy to use. With its features, the dataset used can be visualised in many different ways, like a pie chart of the contribution of AI among all job domains and a bar chart of the comparison between job domains

that are related to business. Figure 2.8 shows an example of a data visualisation dashboard using Microsoft Power BI.

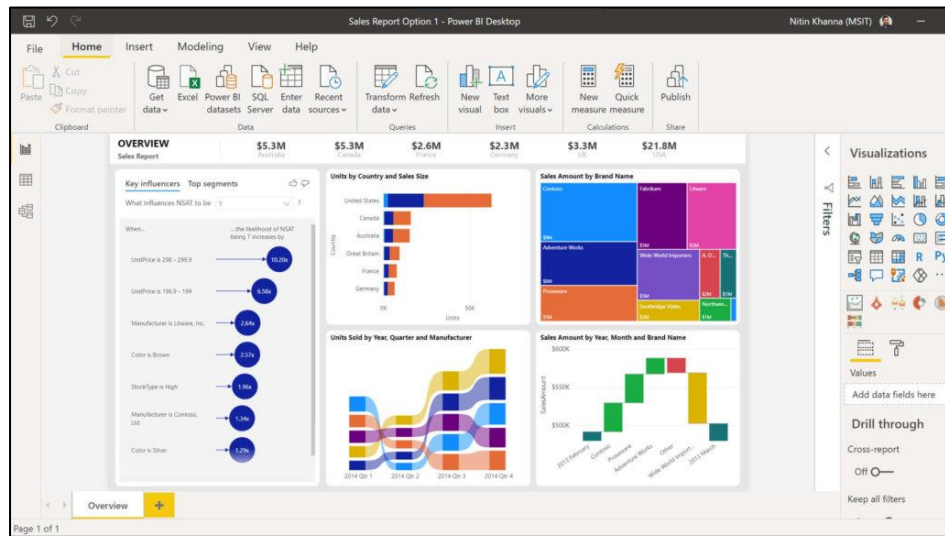


Figure 2.8 Microsoft Power BI Sample Dashboard
(source: Daxonet Team, 2021)

2.2.2.2 Tableau

Founded in 2003, Tableau is now one of the leading software programmes for data visualisation and business intelligence tools used for large-scale data analysis and reporting. Tasks like data sorting, comparing, and analysing are made easier and faster with the Tableau drag-and-drop interface feature. A variety of dashboards, charts, graphs, and maps can be made to help in visualizing and analysing data for business decisions. Tableau supports sources from Excel, SQL servers, and cloud-based data repositories. With this, the dataset CSV file can be directly imported into the tool, along with faster data visualisation. However, the dataset needed to be clean or missing values to make sure that this data visualisation tool does not create incorrect graphs or charts. Figure 2.9 shows an example of data visualisation using Tableau.



Figure 2.9 Example of Data Visualisation using Tableau
(source: Yu, 2016)

2.2.2.3 Google Charts

Google Charts allows users to visualise data directly on their website via simple JavaScript programming. Other than that, Google Charts provides many ready-to-use chart types for users. After loading some Google Chart libraries, the user needs to list the data to be plotted, make customization choices, and then construct a chart object with a desired ID. The Google Chart will be displayed on the user's website via a <div> with the id created. As I decided to create a web-based dashboard system, Google Charts can be very helpful in displaying the visualised data. Rather than importing images into the website, which is a lot of work as the image needs to be in the correct size and resolution, Google Charts only provides a set of codes to be applied to the website, importing and visualizing it directly online. Figure 2.10 shows multiple examples of data visualisation using Google Charts.

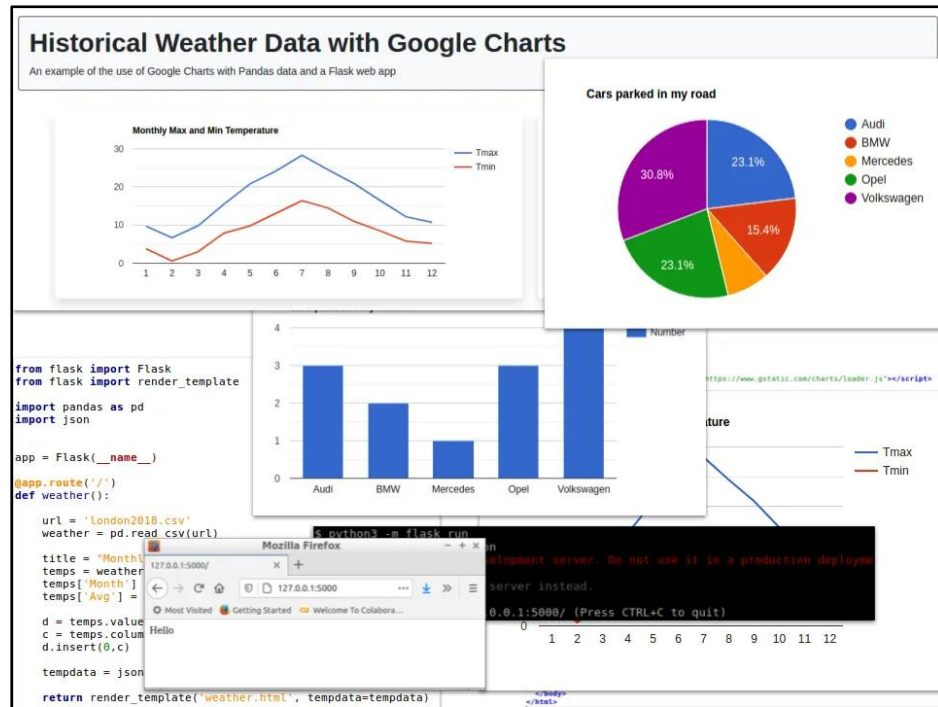


Figure 2.10 Examples of Data Visualisation using Google Chart
(source: Jones, 2021)

2.2.3 Data Processing Framework

Data processing frameworks are tools used to make the processing of large sums of data easier, especially for big data analytics. Data processing speed, efficiency, and security are the main features of these frameworks (FunctionUp, 2023). In addition, they are also capable of handling batch and stream processing as well as real-time analytics. This project requires data processing in order to get effective visualisation as the data requires cleaning and integration. Table 2.1 shows the data processing framework of this research project.

Framework	Description	Tools Used
Data Extraction	Process of gathering data from multiple sources like websites, data lakes and warehouses.	• Microsoft Excel • Python Libraries
Data Cleaning	Process of cleaning and preprocessing of the data for analysis.	• Python Libraries • Cloudera
Data Storage	Processing of storing the data in a storage system.	• Google Cloud • Apache Hive • Hadoop Distribution File System
Data Analysis	Process of extracting useful information from the analysed data.	• Apache Hadoop • Apache Spark • Apache Hive
Data Visualisation	Process of transforming the extracted data into visual charts or graphs.	• Microsoft Power BI • Tableau • Google Charts

Table 2.1 Data Processing Framework of this Research Project

2.2.3.1 Data Extraction

Data extraction, or data collection, was the first step in the data processing framework. These data are gathered from various sources, like websites, data lakes, and warehouses. It can be unstructured or structured data, either physically or digitally documented. The data gathered needed to be in scope with the objectives, trustworthy, and well-built to ensure accurate output when used in data analysis (Talend Team, n. d.).

The dataset used for this research project contains information detailing the impact of AI toward jobs titles and job domains. It was scraped from the website "theresanaiforthat.com," compiled into a dataset, and then uploaded to Kaggle under the title "The AI Job Threat Index" by the user named Tensor Boy. It was in the Microsoft Excel CSV format. However, the dataset was not suitable enough to be used in this research project.

Since the source of the dataset has already been identified, it is much more reliable and trustworthy to retrieve the data directly from the source. The owner or administrator of the website was contacted, and I received three files worth of datasets in the form of a CSV file. These datasets were named after the year they were recorded, which are AI_Impact_2022, AI_Impact_2023, and AI_Impact_2024. There are 6 columns for each dataset, which are ID, job titles, percentage of AI impact, job tasks and AI models dedicated to that job, and the job domain. It also contains over 4800 rows of data. Table 2.2 shows the data dictionary of the datasets used.

Column	Description	Example Data
ID	Identification number that was assigned to be unique for each row.	839
Job Titles	Names of different job titles across various industries.	Communications Manager
AI Impact	Percentage representing AI's influence on each respective job title, where value indicates the extent to which AI technology has impacted or transformed the job role.	98%
Domain	The category or industry to which each job title belongs, providing context about the specific field or sector in which each job role operates.	Communication & PR
Tasks	Numerical count of human-performed tasks associated with each job title, providing an insight into the range and complexity of tasks performed by professionals in each role.	365
AI Model	Count of AI models developed and associated with each job title, where the columns indicate the extent to which AI technologies have been integrated into the workflow of each job role.	2546

Table 2.2 Data Dictionary

2.2.3.2 Data Cleaning

Data cleaning, which also includes data transforming, was where the raw data was cleaned and pre-processed to be uploaded into the data warehouse and used for analysis. This process eliminates bad data values like redundancies, incomplete or null values, incompatibility data, incorrect data, and other forms of data error. Data transformation also includes creating new data columns in which the data values are driven from other existing columns. The data cleaning and transformation for this research project was done using the Python library environment and the Cloudera environment.

2.2.3.3 Data Storage

Data storage refers to the process of saving or storing data in a secure location where the data can be protected and accessed when needed. Processed data also needed to be stored for future reference. The data gathered from this project was recorded using Microsoft Excel and stored using Google Cloud as backup storage. For analysis purposes in this research project, the processed datasets will be duplicated and stored inside the Hadoop Distributed File System (HDFS), Apache Hive, in the Cloudera environment.

2.2.3.4 Data Analysis

In data analysis, the data that was cleaned, transformed, and pre-processed from the previous step, was then analysed to extract meaningful and useful information. There are many techniques used in data analysis, including both quantitative and qualitative methods such as regression analysis, cluster analysis, and factor analysis, with each having its own purpose and requiring specific data (Stevens, 2021). This is because there are many types of data, and each of them can be interconnected with each other, resulting in different output information. For example, the dataset used in this project research can be analysed using cluster analysis because the list of jobs can be categorized by their job domains and evaluated as a whole management domain, including human resource managers, construction managers, and financial managers.

2.2.3.5 Data Visualisation

Lastly, after the datasets were analysed and suitable information had been recorded for each chart, the datasets were visualised to identify trends and patterns that would otherwise be unclear or difficult to understand. These visualisations include charts, graphs, and maps. The data visualisation tools that will be used in this project are Microsoft Power BI and Google Charts. However, I am also considering using Tableau if I am able to understand and master that tool.

2.2.4 Data Processing Tools

Data processing tools are software applications made to assist users with data collection, archiving, cleaning, analysis, and visualizing. To help organisations get insightful knowledge and make wise decisions, these software applications are critical for managing big, complicated datasets and information.

2.2.4.1 Python Libraries

One of the easier ways to visualise data visualisation by using Python programming libraries, which are in the Python environment. Python is a popular programming language known for its ease of use. It can access many powerful programming libraries, such as Matplotlib, Seaborn, and Plotly. Data processing, data analysis, and machine learning can also be done using Python programming.

2.2.4.2 Apache Hive

Utilizing the Hadoop platform, Apache Hive is an application that offers a querying and processing interface similar to SQL. This Hadoop-based data warehouse uses the underlying Hadoop infrastructure to translate HiveQL queries into MapReduce or Tez operations (Gaurav, 2023).

2.2.4.3 Apache Spark

Apache Spark is an open-source unified analytics engine used for large-scale data processing that can handle both batches as well as real-time analytics (Databricks, 2019). Programming clusters with implicit data parallelism and

fault tolerance is made possible by Spark's interface. This multilingual engine can be used on single-node computers or clusters to do data science, machine learning, and data engineering tasks.

2.2.4.4 Microsoft Excel

Microsoft Excel is known to allow users to import, explore, clean, analyse and visualise data easily and directly. Microsoft Excel has built-in pivot tables, which are a powerful tool for data analysis, and enable users to do data processing including Power Query, Power Pivot, conditional formatting, and the ability to remove duplicates (Richardson, 2022).

2.3 Artificial Intelligence (AI)

The world we live in nowadays is familiar with the concepts of artificial intelligence and robots because of the centuries-old science fiction shows on television such as Star Trek and Star Wars. This made mankind think of the possibility of such technology as computations start to stretch across all industries. Alan Turing, a British polymath, explores the mathematical possibility of artificial intelligence. Mankind uses information aided by reasoning to draw a conclusion or decision, so why can't machines do the same thing? (Turing, 1950).

In general, artificial intelligence, or AI, is the capability of computers to mimic human behaviours and thinking methods. It is also the simulation of human intelligence by software-coded heuristics (Frankenfield, 2016). Nowadays, this code is present in a wide range of applications, including consumer, cloud-based, business, and embedded firmware applications. The capacity for reasoning and making decisions that maximize the likelihood of accomplishing a given objective is the ideal quality of artificial intelligence (Frankenfield, 2016).

To achieve these, artificial intelligence needs access to large sets of data. The AI then processes them through its complex yet manipulative algorithms to learn the patterns. The algorithm is manipulative, as both humans and AI itself

can make changes to fit its usage and requirements (Wilson, 2023). After the data is processed, AI then predicts the possible outcomes and aligns them with the data processed to determine whether the outcomes are a failure or a success. If the data set produces a failure result, the AI learns from that mistake and repeats the process differently. The algorithms' rules may need to be adjusted based on the learning from the failed data set to achieve a successful outcome (Wilson, 2023).

2.3.1 Definitions in AI

Here are some other definitions related to AI. These definitions help us understand what Artificial Intelligence truly means and how it works.

2.3.1.1 Artificial

Artificial means something that is made, produced, or invented by human beings rather than occurring naturally, especially from a copy of a naturally made model (Oxford English Dictionary, n.d.). Artificial sweeteners, which are a sugar substitute, and artificial grass, also known as synthetic turf, are examples of artificial objects.

2.3.1.2 Intelligence

Intelligence is the ability to acquire and apply knowledge and skills (Oxford English Dictionary, n.d.). This includes judgements, opinions, reasoning, and emotions. Examples of intelligence are rational thinking and intentionality.

2.3.1.3 Algorithm

The algorithm is a process or set of instructions and rules to be followed in calculations or other problem-solving operations, especially by a computer (Oxford English Dictionary, n.d.). The algorithm in a computer can be written using programming languages like Python, HTML, and JavaScript, however, mastering these programming languages is the key to a successful algorithm, as its effectiveness may vary in its effectiveness.

2.3.1.4 Machine Learning

Machine learning refers to the use and development of computer systems that can learn and adapt without following explicit instructions by using algorithms and statistical models to analyse and draw inferences from patterns in data (Oxford English Dictionary, n.d.). Machine learning often leads to automation. An example of machine learning is deep learning, which is used by machines and AI nowadays.

2.3.2 AI Evolutions and Growth

Theorists and researchers like Allen Newell, Cliff Shaw, and Herbert Simon's Logic Theorist prove that Turing's theory question is possible in concepts, and nowadays developers and programmers prove its development and applications, but not all of them are possible (Anyoha, 2017). Artificial intelligence nowadays may think and act the way a normal human being would, but it is still lacking in a few aspects, like cognitive abilities and consciousness.

AI starts off as a computer or machine automation tool to solve complex mathematical problems, creating a thinking machine (Queensland University, 2019). But as time grows, innovations and technology take over, and applications of automation lead to the idea of AIs in general. Automation means that a machine does a certain task repeatedly. Why automate if the computer can act the way a human does? AI can analyse, adjust, and fix itself to the extent that it can operate even without human touch. The amount of data collected and available nowadays is outstanding, with its rate increasing rapidly, leading to the possibility of gaining new knowledge almost every day. Figure 2.11 below shows the historical timeline of artificial intelligence as depicted by Queensland University.

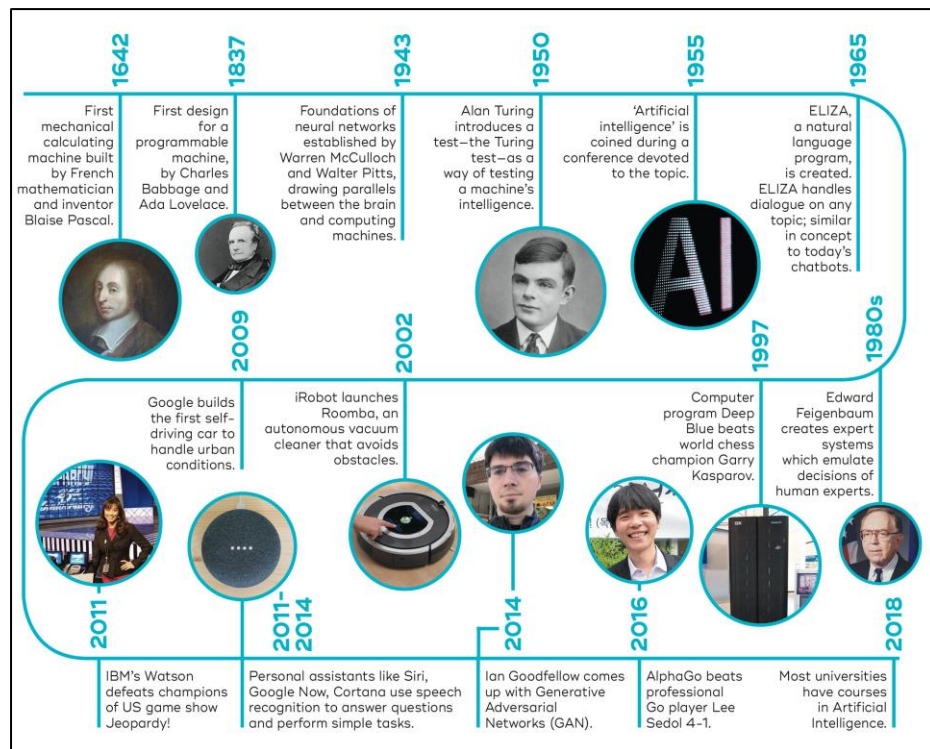


Figure 2.11 History of Artificial Intelligence

(source: Queensland University, 2019)

2.3.3 AI Affecting Job Employments

The application of AI is undoubtedly affecting all job domains and businesses. The ability to automate tasks effectively, help in decision-making, make better resource allocations, and reduce risks are the few benefits of AI applications in industries. As companies and businesses would call it “cutting corners,” utilizing AI will lead to the reduction of their physical workforces. AI is a one-time payment for lifetime implementations, why bother paying employees monthly? AI can operate all day long without resting and still maintain its performance while employees require sustenance and rest (Gardner, 2019). Aside from system services and bug repairs, AI is able to process data and carry out tasks effectively and efficiently in the long run. One fact that sums up the pace and scope of AI adoption is that ChatGPT reached 100 million users in just 60 days, whereas Instagram took about two years to reach the same milestone. This rapid rate of progress in AI development can be proven by the increase in AI patents from 2015 to 2021 (HAI, 2023).

A variety of jobs can be carried out by AI-powered technologies, such as information retrieval, logistical coordination, financial services, complex document translation, business report authoring, legal brief preparation, and even disease diagnosis. With the capability to learn and improve through the application of machine learning, AI could potentially improve the productivity and accuracy of these jobs. Acemoglu and Restrepo (2017) develop a theoretical framework to understand the impacts of AI on labour by categorizing the effects into three broad effects, which are the displacement effect, productivity effect, and reinstatement effect. This shows that AI technology implementations in businesses and companies affect physical labourers in different ways. About 47% of total US employment in service occupations is at risk of losing their jobs to AI automation over the next decade due to its high risk of computerization (Frey & Osborne, 2017). According to Autor (2015), the advent of computers has caused the distribution of jobs in the labour market to shift, with a decline in the number of medium-skilled jobs and a rise in both low-skilled and high-skilled jobs because of computerization and technological changes. Arntz et al. (2016) stated that, instead of the 9% substitution of physical labour for AI automation, 35% of jobs would undergo significant change during the next 20 years, making transformation more likely to happen.

2.3.4 Weaknesses of AI

Artificial intelligence is already having a significant impact on society nowadays, and this impact is expected to grow as technology advances. Despite all the good benefits that AI has to offer, there are still drawbacks to be addressed. Firstly, the implementation of AI is extremely costly. Developing and maintaining an AI system demands significant time, energy, and financial investment, making it a costly endeavour (Duggal, 2021).

2.4 Finding Suitable Data

Understanding the impact of AI on jobs require certain data attributes and records. These specific data attributes help in identifying the severeness of the AI impact, which job domain the job title belongs to and how the AI automation on jobs takes over a timeline.

2.4.1 Job Roles, Titles or Occupations

This column should contain the different job roles or occupations that are likely to be impacted by AI. This column will allow the categorization of the jobs and occupations that are most vulnerable to AI disruption, which is the core focus of the data visualization.

2.4.2 Job Domain, of Category

This column should contain the job domain or category of the job titles enlisted. This column will help in grouping job titles by their job category to see the overall AI impact on certain job domain or categories.

2.4.3 Degree of AI Impact

This column should quantify the level of impact AI is expected to have on each job role or occupation. This column will provide insight from vary degrees of AI impact across different job categories, allowing the highlights of jobs most at risk of automation and those less affected.

2.4.4 Timeframe

This column indicates the timeframe for these AI impacts. This will allow the contextualization of the data and show how the impact of AI on jobs is expected to evolve over time, providing a more comprehensive understanding of the issue.

2.5 Technology Acceptance Model (TAM)

The Technology Acceptance Model is a framework used to assess users' perceptions and experiences towards a specific technology and its utilization. Perceived usefulness and perceived ease of use are the two key factors that influence technology acceptance. Perceived usefulness refers to the belief that the technology tested helps the user perform jobs and accomplish goals. Simply put, users are more likely to adopt a technology if they think it is practical and simple to use. This technology acceptance model will be used after the dashboard of this project's research is developed in order to measure the dashboard's usefulness and satisfaction. Figure 2.12 shows an illustration of the technology acceptance model's flow.

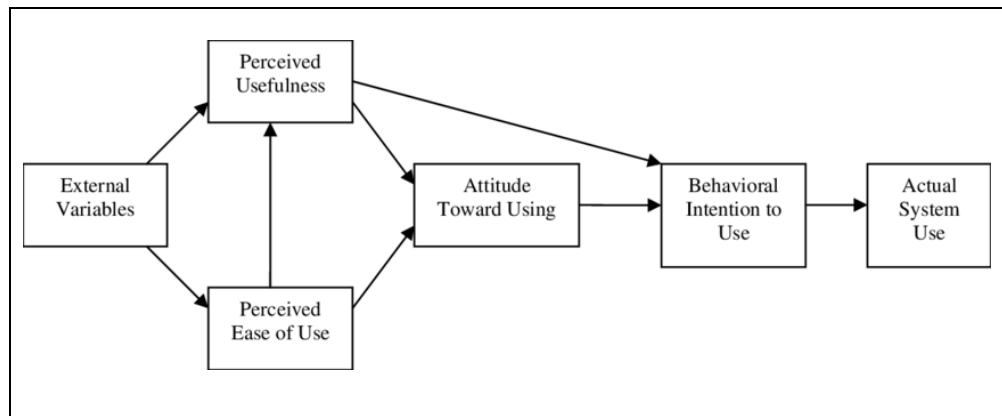


Figure 2.12 Technology Acceptance Model (TAM)

(source: Miller & Khera, 2010)

2.5.1 Perceived Usefulness (PU)

Perceived usefulness is described as the extent to which an individual thinks that utilising a specific technology would be beneficial (Burgess & Worthington, 2021). Their intentions to use the technology increase in line with their perceived usefulness. Questionnaires are oftentimes used to assess perceived usefulness.

2.5.2 Perceived Ease of Use (PEU)

Perceived ease of use refers to the extent to which an individual believes that utilising the specific technology would be easy or almost effortless (Burgess & Worthington, 2021). Perceived ease of use is also commonly assessed with questionnaires. The more the individual feels at ease when using the technology, the more likely they are to utilise it.

2.5.3 Attitude Towards Using (ATT)

The attitude towards using refers to the user's overall attitude or evaluation of using a particular technology or system. This attitude is influenced by the user's perceptions of the technology's usefulness and ease of use, which in turn determine the user's intention to use the technology (Sulistiyo et. Al, 2022). The TAM posits that a positive attitude towards using a technology leads to a higher intention to use it, and ultimately to actual usage behaviour.

2.5.4 Behavioural Intention to Use (BI)

Behavioural intention to use represents the strength of a person's intention to use a particular technology or system. It determined by the two factors of TAM which are perceived usefulness and perceived ease of use, which is the immediate determinant of usage behaviour (Marikyan, 2023). The stronger a person's BI, the more likely they are to actually use the technology.

2.5.5 Actual System Use (AU)

Actual System Use is the final aspect of the technology acceptance model. It refers to the actual usage behaviour of the system by the user. This aspect is the ultimate dependent variable in TAM, as it represents the real-world outcome of the user's intention to use the system (Marikyan, 2023).

2.6 Related Study

Related study are reviews of relevant past studies related to the project title. This includes analyzing research gaps that the current study aims to address and how they relate to the current project title. Table 2.4 below shows the articles and past research that was used in this project.

Title	Author, Year	Keywords	Issues	Method	What is Interesting
Smart Technology, Artificial Intelligence, Robotics, and Algorithms: Employees' perceptions of our future workplace	David Brougham, Jarrod Haar, 2018	Career Planning, Change, Technology, Disruptive Technology, Employees	Perceptions of the future workplace with STARA, its impact on employees, concerns about Automation	Modified, Analysis of Variance (ANOVA)	As of 2018, the awareness towards STARA among employees is relatively low, leading to perceptions of technology replacing employees is low.
Influences of Artificial Intelligence (AI) awareness on career competency and job burnout	Haiyan Kong, Yue Yuan, Yehuda Baruch, Naipeng Bu, Xinyu Jiang, Kangping Wang, 2021	Artificial Intelligence, AI Awareness, Career Competency, Job Burnout, Organisational Commitment	The potential impact of AI on hospitality industry employees	Structural Equation Modelling (SEM)	The "STARA awareness" method was also used to aid in this research study.
Artificial Intelligence, Growth and Employment: The Role of Policy	Philippe Aghion, Céline Antonin, Simon Bunel, 2021	Artificial Intelligence, Growth, Policy, Automation, Robots, Employment	The argument that the effects of Artificial Intelligence and automation on growth and employment depend on the institutions and policies.	Analysis of Survey and Literature	2018 Robot automation, which needed a human touch to fix, is as much of a threat to jobs as AI does in 2023.
Testing the Employment Impact of Automation, Robots and AI: A Survey and Some Methodological Issues	Laura Barbieri, Chiara Mussida, Mariacristina Piva, Marco Vivarelli, 2019	Technology, Innovation, Employment, Skill, Task, Routine	AI and technologies are transforming societies and economies at an accelerated pace, raising concerns about the impact on employment and wage inequality.	Analysis of Survey and Literature	Robotic automation and AI automation provide different unemployment causes as robotics cannot fix themselves, unlike AI which can fix itself. Robotics still requires manual labour.
The Impact of AI and Machine Learning on Job Displacement and Employment Opportunities	Rudra Tiwari, 2023	Artificial Intelligence, Machine Learning, Job Displacement, Employment Opportunities, Automation, Economic Impact	AI does threaten present jobs and causes unemployment, but it also opens opportunities for new future jobs.	Analysis of literature reviews using a thematic approach.	Factors like the rate of technology advancement, workers' adaptation ability to new technologies, and individual sectors and occupations also affect the impact of AI on job displacement.

Table 2.3 List of Related Studies Used in this Project Research

2.6.1 Discussion of Related Study

To summarize, almost all the past research and articles uses the subjects' emotion and awareness to measure their perception of AI's presents in the working industries. Table 2.4 below show the discussion of related study.

Research Paper	Smart Technology, Artificial Intelligence, Robotics, and Algorithms (STARA) Awareness	Modified, Analysis of Variance, ANOVA	Structural Equation Modelling (SEM)	Analysis of Survey and Literature Reviews	Data Visualisation Technique	Study of Employee Perception Towards AI
Smart Technology, Artificial Intelligence, Robotics, and Algorithms (STARA): Employees' perceptions of our future workplace	✓	✓		✓	✓	...✓
Influences of Artificial Intelligence (AI) awareness on career competency and job burnout	✓		✓	✓	✓	✓
Artificial Intelligence, Growth and Employment: The Role of Policy				✓	✓	
Testing the Employment Impact of Automation, Robots and AI: A Survey and Some Methodological Issues				✓	✓	
The Impact of AI and Machine Learning on Job Displacement and Employment Opportunities				✓		✓

Table 2.4 Discussion of Related Study Used in this Project Research

2.7 Summary

This literature review covers the overall topics associated with the title Data Visualisation of The Impact of Radical Artificial Intelligence Technology Towards Jobs. A literature review is important because it provides a foundation of knowledge on the topic, allowing further assessment of the current state of the research and identifying missing requirements. It helps in evaluating the success and failure of the assessment and research methods, providing them with a better understanding of the theory used and justified. The literature review also helps in analysing the significance of the additional research done relating to the research title. Figure 2.13 below shows the mind map of the chapter 2 contents.

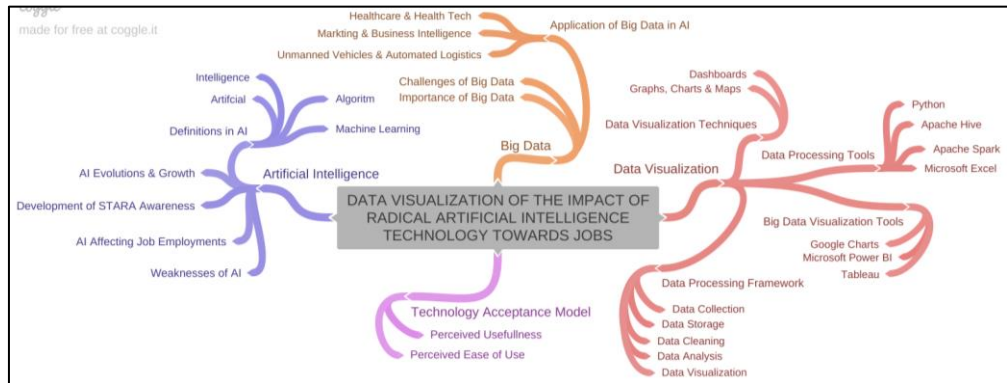


Figure 2.13 Mind Map of Chapter 2

CHAPTER 3

RESEARCH METHODOLOGY

This chapter covered the project's methodology and provided a description of each phase of the methodology. Research methodology refers to the steps or methods used to identify, select, process, and analyse information on a subject.

3.1 Research Methodology

This research project was done using the Agile methodology. Although it was designed for team-based development, solo developers can benefit from Agile methods like short cycles, continuous improvement, and iterative development since they are flexible, adaptable, and take feedback into account (Johnson, 2012). While pair programming and certain aspects of Scrum components might not be applicable to individual developers, the core principles of the Agile method can still be beneficial, such as adapting to changes and building functional products in briefs. In the case of this project, Agile methodology's cross-collaboration allows me to consistently gain feedback from my supervisor and improve the dashboard as it grows. Figure 3.1 shows the overview of the flow of an Agile methodology.

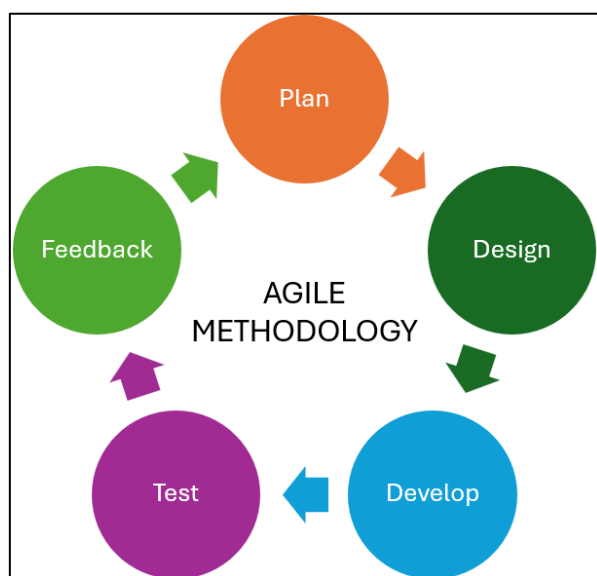


Figure 3.1 Overview of the Agile Methodology

There are five main phases in this research project's development. These five phases were planning, designing, implementation, testing, and evaluation. Figure 3.2 below shows the agile methodology phases of this project's development.

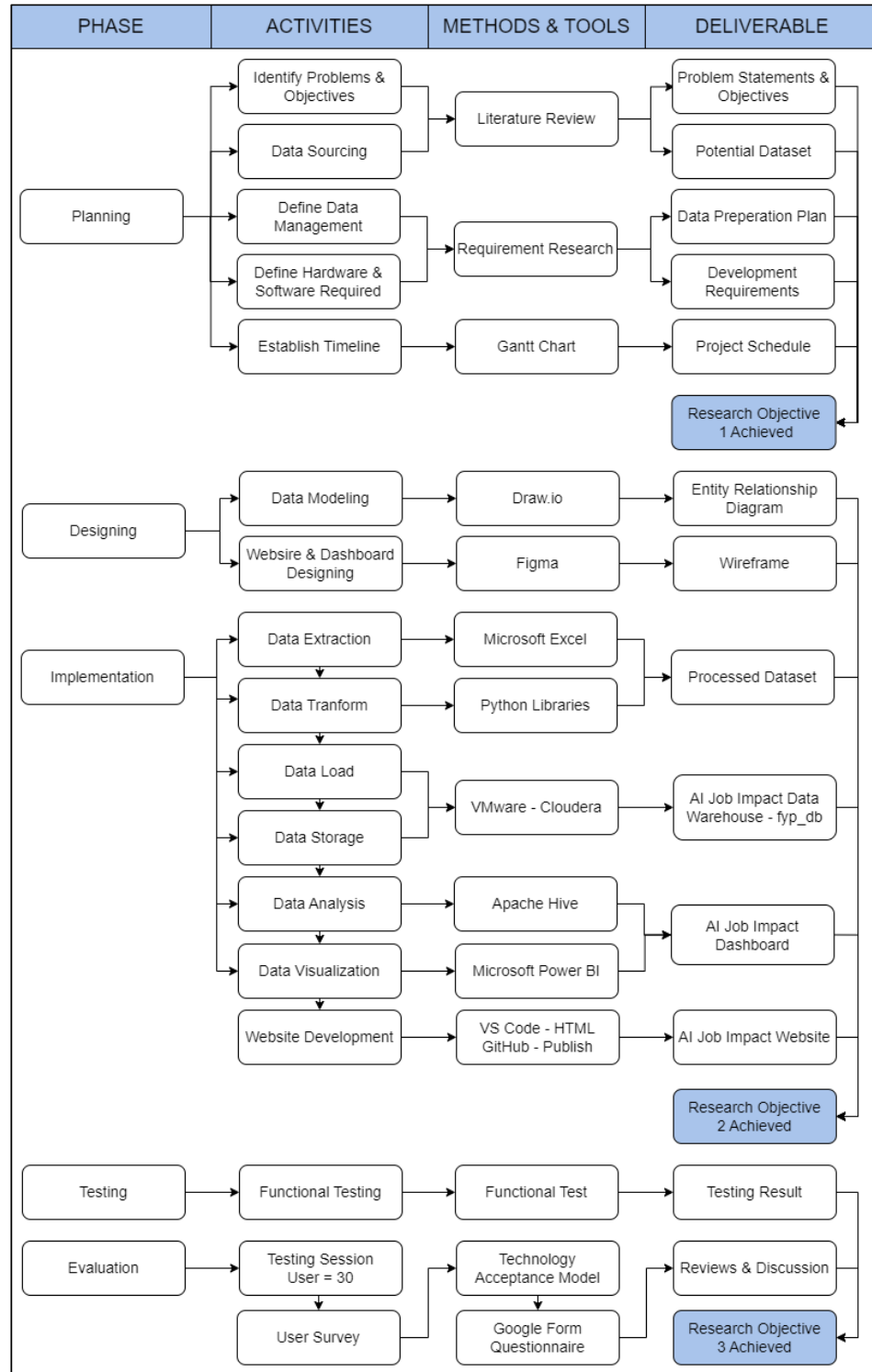


Figure 3.2 Phases in the Project Development

3.2 Planning

Planning was the first phase in project development and was important because it determined the foundation of the project and ensured that it was feasible, beneficial, and possible to develop. It involved discussions of this project's main idea and its requirements to be developed. Activities listed during this phase will be done by researching previous research, online journals, articles, and related websites. These activities include comparing related studies and data to identify commonly used methods and hypotheses, summarising existing knowledge to prevent duplications, identifying the project's requirements, including software and hardware, planning for data processing, and establishing a development schedule. This will also provide the foundation of knowledge for the topic covered in this project. All the research that is done will be recorded in the literature review in Chapter 2 of this project report. This phase will help in identifying the problem statements, scope, objectives, potential dataset, data processing plan, development requirements, and schedule.

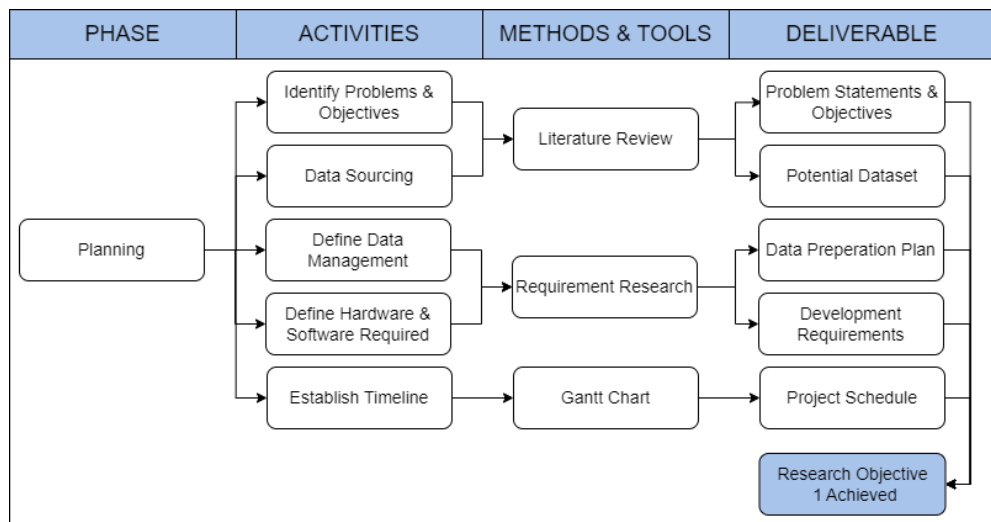


Figure 3.3 Activities in the Planning Phase

Figure 3.3 above shows the activities that will be done in the planning phase of this research project. By doing this phase, research objective 1, which is to determine the data required to analyse the impacts of radical artificial intelligence technology towards jobs will be achieved.

3.2.1 Identify Problems & Objectives

The title of the project is Data Visualisation of The Impact of Radical Artificial Intelligence Technology Towards Jobs. This project aims to develop a web-based data visualisation dashboard about job categories or domains that is automated by AI. This title was chosen because AI is still a developing technology, and the topic is still in discussion about whether it is truly beneficial or risky. However, many existing jobs have already been taken by AI for automation, leading to concerns and unemployment. Proper research to determine the exact number of jobs automated has not been done thus far, but research about the awareness of the possibility of AI automating employers' jobs was made a few times over the past few years during the peak development of AI. Knowing which jobs are being automated by AI is important for future planning and helps improve employees' skills by training them to adapt. As the goal of the project is properly defined, finding a suitable dataset to be used for visualisation is crucial to the success of the project's development. The literature review was also made as this project's base of knowledge and as a reference to be used in the future development of this project.

3.2.2 Data Sourcing

Data sourcing involves finding, selecting, and acquiring the necessary data to be used in a project. The potential dataset identified during this phase was named "The AI Job Threat Index," made by a Kaggle user with the name Tensor Boy. The original data was extracted from a website called "There's an AI for That," processed, and then published on Kaggle. There were 3 datasets in which the name indicated the year it was recorded, which were 2022, 2023, and 2024. The datasets were in the format of a CSV file, containing over 4700 data rows each. The datasets contain a list of job titles, job domains, percentages of AI impact, job tasks, and AI models. There are some noticeable errors in the dataset, such as the fact that the job listed does not match correctly to its job domains or job categories. It is important to take note

of the pre-processing of the dataset before it is used in this project. Figure 3.4 below shows a preview of the website source of the dataset that will be used in this project.



Figure 3.4 Preview of the Website
(source: There’s An AI For That, n.d.)

3.2.3 Define Data Management

Data management is the process of gathering, organising, storing, and preserving the data that is created and collected. Defining data management refers to the planning of data processing. Data management is important to ensure data is properly secured and maintained, and the quality of the quality of the data is properly preserved for future reference. This is crucial in data visualisation because it ensures that the data used for analysis and visualisation is accurate, consistent, and reliable. This also includes the planning of the data processing framework. The data processing framework for this project consists of five frameworks which are data extraction, data transformation, data loading, data analysis, and finally data visualisation.

Data extraction is the process of gathering data from multiple sources. In this research project, the datasets were extracted from the website “There Is An AI For That.” The datasets were in the form of CSV files. Upon manual inspection, the datasets contain errors that are important to take note of for the

data cleaning process. Data extraction was done using Microsoft Excel in CSV format.

Data transformation, or data cleaning, is a crucial part of the data processing framework. This process is required to remove null values, errors, and inaccuracies from the datasets, like when the job listed does not match correctly to its job domains or job categories. This is important for delivering accurate data visualisation. Data transformation and cleaning were done using the Python programming language in the software Jupyter Notebook.

For data loading, the datasets were stored in the primary data storage or data warehouse of this research project, which is the Hadoop Distributed File System (HDFS) in the Cloudera environment, and will be backed up on a secondary storage, which is Google Cloud. The datasets were saved and stored in three different forms, which were raw datasets, pre-processed datasets, and processed datasets. This is to ensure that backtracking is possible when problems occur. The data warehouse is important to ensure that the data is well-kept and ready for analysis and visualisation. Data loading will be done using Apache Hive and Apache Spark.

Data analysis is the process of extracting useful information from the processed datasets that were loaded into the data warehouse. In this project, Apache HiveQL was used in order to create queries, analyse, and extract meaningful information for data visualisation. And finally, data visualisation is the process of transforming the extracted datasets or information into visual charts or graphs. Microsoft Power BI was used in the data visualisation process. Figure 3.5 below shows the data processing framework designed for this project.

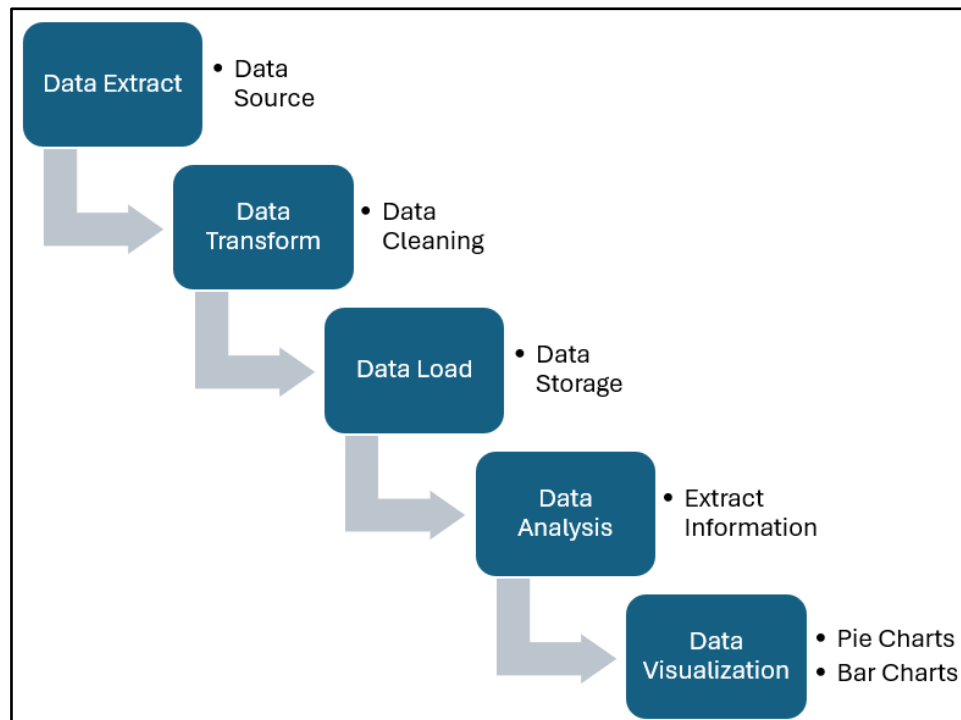


Figure 3.5 Data Processing Framework Design

3.2.4 Define Hardware & Software Requirements

Proper hardware and software will be required to ensure that this project is developed and delivered effectively. The research done in the literature review will determine these requirements. Table 3.1 and Table 3.2 below show the hardware and software requirements for this project.

Hardware	Specification
Big Data Lab Personal Computer	• Windows 10 • Intel Core i5-9500 • 24 GM Ram
Asus ROG Strix G15	• Windows 11 Pro • AMD Ryzen 7, 6000 Series Processor • NVIDIA GeForce RTX 3050 • 16 GM Ram, DDR5, 4800MHz

Table 3.1 Hardware Requirements

Software	Description
Zotero	Reference management tool.
Draw.io	Online diagrams designing tool.
Figma	Software to design prototypes and wireframes.
Python Programming	The programming language used for data cleaning and processing.
Microsoft Excel	Software to create and manage spreadsheet files.
Hadoop Distributed File System	The primary storage when using the Hadoop applications.
Apache Hadoop	Software used to store and process datasets.
Apache Hive	Software for data analysis that supports fault tolerance.
Microsoft Power BI	Software for data visualisation.
Google Form	Online creation tool for survey questionnaires and forms.

Table 3.2 Software Requirements

3.2.1 Establish Timeline

Establishing timelines or developing schedules using the Gantt chart helps track the specific tasks in project development. The vertical axis of the Gantt chart shows the list of activities, grouped by their phases, to be done in this project development, while the horizontal axis shows the date in weeks and months. The development of this project is scheduled for two UiTM semesters, which is 12 months, starting in October 2023 and ending in September 2024. For a detailed view of this project's Gantt chart, please refer to Appendix A.

3.3 Designing

Designing is the second phase of development. The designing phase is where the layout and interface will be designed for the dashboard, which will be built

before it becomes the final product. This phase also includes designing the data model, website, and dashboard of the system that will be developed.

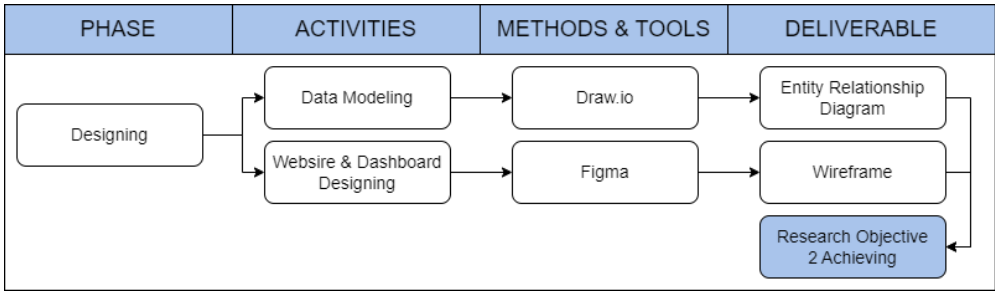


Figure 3.6 Activities in the Designing Phase

Figure 3.6 shows the activities that will be done in the design phase of this project. Research objective 2, which is to design and develop dashboards that include dynamic visualisations of the impacts of radical artificial intelligence technology on jobs using Figma and Microsoft Power BI, will be achieved after completing this phase.

3.3.1 Data Modelling

Data modelling is the process of visualising data and its relationships within an information system. Data modelling will determine the data that was used for the visualised charts. The deliverable for this activity will be the entity relationship diagram based on the relationship between the data used in this project. Sketches for data modelling can easily be done using the templates from the website tool called Draw.io.

3.3.2 Dashboard Designing

Designing the dashboard and the visualisations for this project will be done using Figma. A complete wireframe of the dashboard and the website is the deliverable for this phase. By doing the dashboard and website design in this phase, you get an idea of the elements that should be in the actual page design for the web-based system when it is developed.

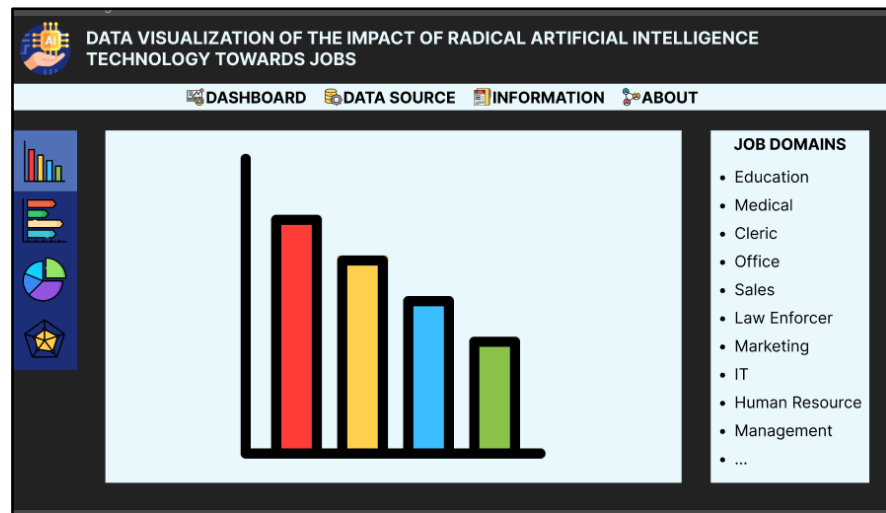


Figure 3.7 Early design of website dashboard page

Figure 3.7 above shows the early design of the website and dashboard. The website will contain four pages. The first page is the dashboard and visualisation of the impact of radical artificial intelligence technology toward jobs. The second page is the data source, which will display the website source of the datasets. The third page is the information page, containing information about artificial intelligence and future job risks. And lastly, the About page contains information about the developer.

3.4 Implementing

The implementation phase, also known as the development phase, is the third phase of this project's development process. During this phase, all the planning and designing from the previous phases will be executed in order to develop the complete project data warehouse, data visualisations, and the web-based dashboard system. Figure 3.8 shows the activities that will be done in the implementation phase of this project.

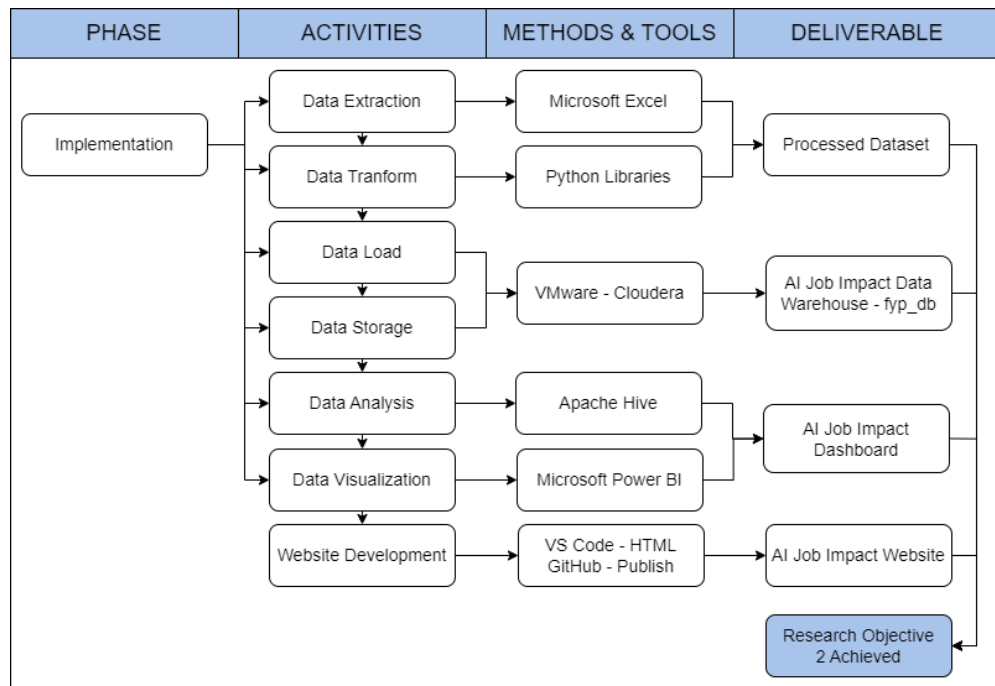


Figure 3.8 Activities in the Implementation Phase

3.4.1 Data Extract & Transform

Data extraction and transformation refers to the process of extracting data from its sources, cleaning them from any errors and inconsistencies, and transforming them into suitable formats for future usage. The datasets that were used in this research project were obtained from the website “There’s an AI for That.” The pre-processed datasets were named “The AI Job Threat Index” and were published on the website Kaggle by the user Tensor Boy. However, the dataset on the Kaggle site was insufficient. Through the website source, I was able to contact the website administrator and retrieve some raw datasets. The new datasets will be filtered, cleaned, deduplicated, validated, and authenticated with their source to ensure their legitimacy and accuracy. Figure 3.9 shows a sample of one of the datasets that will be used for this project.

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	839	Communications Manager	Communication & PR	86%	258	1823
3	4641	Web Search Evaluator	Data & IT	82%	175	1161
4	653	Chief Strategy Officer	Leadership & Strategy	81%	133	933
5	1086	Data Entry Operator	Leadership & Strategy	81%	221	1550
6	1081	Data Clerk	Data & IT	79%	230	1306
7	4228	Supply Clerk	Supply Chain & Logistics	79%	80	575
8	1082	Data Collector	Data & IT	77%	197	1596
9	1787	General Clerk	Hospitality	77%	157	1282
10	486	Business Relationship Manager	Communication & PR	76%	137	895
11	713	Clerical Assistant	Administrative & Clerical	76%	185	1228
12	863	Compliance Officer	Medical & Healthcare	76%	133	893
13	1483	Executive Assistant	Leadership & Strategy	76%	209	1536
14	4395	Therapist	Medical & Healthcare	75%	78	678
15	1185	Business Development Manager	Communication & PR	74%	118	888

Figure 3.9 Sample of the Dataset

Upon manual inspection, the dataset contains errors like miscategorized and duplicated job domains. The job domains also needed to be renamed into a single word for easier accessibility. Short forms like “CEO” in the job titles column need to be removed to reduce redundancy. For legitimacy, the data needed to be validated with the website source, as it updates consistently.

3.4.1 Data Load & Storage

After the datasets were cleaned, they were loaded into the project’s data warehouse. The data warehouse is a centralized repository that combines and stores data from various resources in a single location, making it easier to analyse and draw insights for creating reports, data analysis, and data visualisation. During this phase, Apache Hive and Apache Spark were used to create the query and the project’s data warehouse, which is in the Cloudera environment. The data warehouse for this research project was named “AI Job Impact Data Warehouse.”.

3.4.1 Data Analysis & Visualisation

Datasets from the storage will then be analysed to extract useful information for data visualisation. The data visualisation and dashboard were created using Microsoft Power BI using the design developed in the previous design phase. The visualised dashboard that was created was called the AI Job Impact

Dashboard. The content of the dashboard includes bar graphs about the percentage of AI taking over each job domain. Comparisons between different job domains will be created using an interactive pie chart so that users can choose which job domains they would like to compare. Users should also be able to filter out job domains based on their preferences, making the dashboard list only the requested job domains. Figure 3.10 below shows an example of a bar chart of the IT Jobs domain against AI Impact.

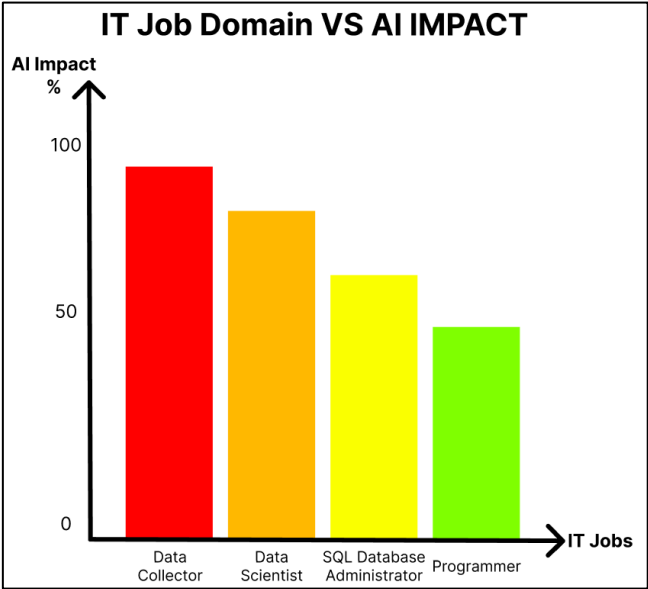


Figure 3.10 Ideate Bar Chart of IT Jobs Domain against AI Impact

3.4.2 Website Development

The website was developed using the programming languages HTML and JavaScript, based on the design that will be created using Figma. The website will contain the original link to the dataset source website, an information page about artificial intelligence, and the project data visualisation dashboard. A login feature is not required and can be tedious because the project will be developed for public usage, and no necessary data is required to be stored about the users. Various web hosting and content management systems are available and free to use, like 000WebHost, InfinityFree, and GitHub. This research project will use the GitHub hosting feature to publish the website.

3.5 Testing

The testing phase is the fourth phase of the project development process. Because this project adopts the agile methodology, the testing phase is not separate, but rather a continuous activity throughout the development process. During the testing phase, the developed dashboard and website are put to the test. These tests include the website and dashboard functionality. Figure 3.11 shows the activities that will be done in the testing phase.

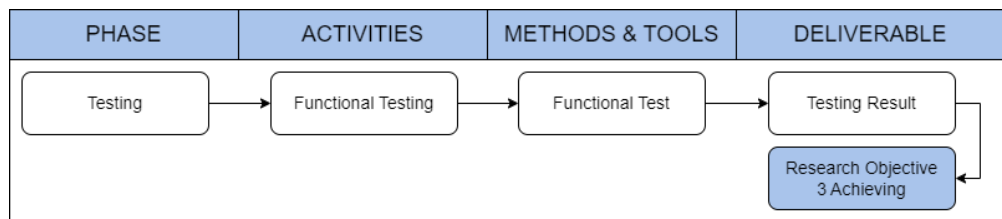


Figure 3.11 Activities in the Testing Phase

3.5.1 Functional Testing

The primary purpose of functional testing is to find and resolve any faults or errors that may impair the functioning and performance of the developing system, ensuring that the system works. The functional test will be carried out by selecting a few users to test the system, rather than having automated testing software. The testing results help in improving the dashboard and website functionality. The functional testing will be conducted for 1 month, with the target user for at least 30 participants. This functional testing and the user survey will be released simultaneously to ensure that the participants test the developed dashboard and fill in the user survey for more accurate feedback.

3.6 Evaluation

Evaluation is the fifth and last phase of the project development process. The evaluation phase is also not considered a separate phase and rather performs continuously throughout the project development process. This phase focuses on collecting feedback from public users regarding the dashboard's

accessibility and usability and determining if the research project’s objectives are fulfilled.

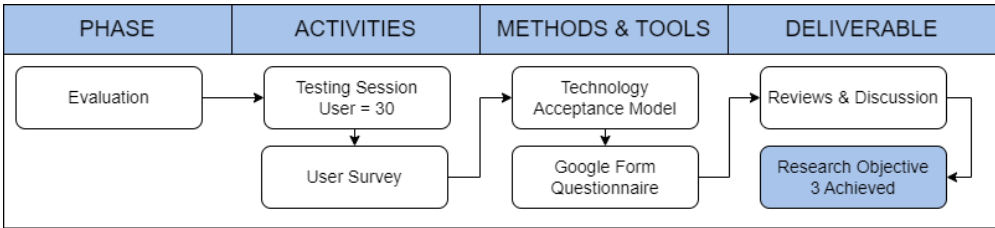


Figure 3.12 Activities in the Evaluation Phase

Figure 3.12 above shows the activities that will be done in the evaluation phase. Research objective 3 will be achieved, which is to evaluate the user acceptance of the developed web-based dashboard using the Technology Acceptance Model (TAM) after this phase is completed.

3.6.1 User Survey

The developed dashboard and website will be published to the public and will be evaluated by the public. The user survey will be carried out using the method of the of the Technology Acceptance Model, TAM. Using TAM helps us understand how users perceive the ease of use, usefulness, and overall user experience of the developed dashboard and website. Measuring the success of the deliverable that the dashboard and website provided through their functionality. The survey will be created using Google Form questionnaires and shared using emails, messaging applications, and social media platforms. For the sample questionnaires that will be used in this evaluation phase, please refer to the appendix. The deliverable of this phase, which is the users’ reviews and discussions, will be analysed and used to improve the web-based dashboard. The user survey will be released for 1 month, with the target user for at least 30 participants. This user survey and the functional testing will be conducted simultaneously to ensure that the participants can test the developed dashboard and fill in the user survey for a more accurate feedback.

3.7 Summary

This chapter shows how the process of completing the visualised dashboard of the Radical Impact of AI Towards Job Unemployment is done. This research project will use the Agile methodology and consist of five phases. The planning phase is where the requirements of this research project are defined, the designing phase gives an idea of how the website and dashboard will look like, the implementing phase is the development of the dashboard and website, the testing phase is to identify failures, and the evaluation phase is collecting user reviews to improve the dashboard. Each phase's activities, tools, and methods that will be used were defined in detail to help in achieving the deliverables and research objectives.

CHAPTER 4

CONSTRUCTION

This chapter discusses in detail the development of the Data Visualisation of the Impact of Radical Artificial Intelligence Technology Toward Jobs Dashboards using Microsoft Power BI. This phase also includes the requirement analysis, which describes the dashboard's functional and non-functional requirements as well as its software and hardware requirements. Detailed explanations of the development techniques and interface design are also provided in this chapter.

4.1 Requirement Analysis

This process is the first step in developing the AI Job Impact Dashboard. It serves as a blueprint for outlining the essential features and capabilities that the dashboard must have to fulfil its objectives effectively. From software to hardware requirements, requirement analysis forms the backbone of the construction process, ensuring efficient development and an effective platform for analysing and visualizing the data for the dashboard and website.

4.1.1 Functional Requirements

Functional requirements define what the developed system and its components should do to meet the user's needs in order to complete their desired duties. It describes the specific features and functions that must be implemented for the system and dashboard to operate correctly, enabling users to achieve their goals. Table 4.1 below shows the finalized functional requirements for the AI Job Impact Dashboard.

Functional Requirement	Chart Type	Column Used
Comparison between Impact Conditions by the count of Job Titles.	Donut Chart	Job Title, Impact Condition
To show the average of AI Impact for the year 2024.	Gauge Chart	AI Impact
To show the progression of AI Models over the years.	Line Chart	AI Models, Year
Comparison between Job Categories by the percentage of AI Impact.	Pie Chart	Job Category, AI Impact
To show the sum of Job Tasks by Job Category.	Vertical Bar Chart	Job Category, Job Tasks
To show the sum of AI Models by Job Category.	Horizontal Bar Chart	Job Category, AI Models
To show the progression of AI Impact over the years by Job Category.	Line Chart	AI Impact, Year, Job Category
Comparison between Job Titles by the percentage of AI Impact.	Pie Chart	Job Title, AI Impact
To show the progression of AI Impact over the years by Job Titles.	Line Chart	Job Title, AI Impact, Year
To show the distribution of Job Titles between Job Tasks, and AI Models.	Scatter Plot	Job Title, Job Tasks, AI Impact
To show the grouping of Job Titles by Impact Condition.	Heat Map	Job Titles, Impact Condition

Table 4.1 Finalized Functional Requirement

4.1.2 Non-functional Requirements

Non-functional requirements define the developed system and how its components should perform. It focuses on attributes like speed, security, reliability, and usability rather than what the system does. It outlines the quality attributes and constraints that guide the design and operation of the solution, ensuring that the system operates effectively and efficiently and meets the users' expectations. The non-functional requirements for the AI Job Impact Web-Based Dashboard are:

- i. **Compatibility**
The website should be able to run on browsers of multiple operating systems like Windows and Mac, without suffering failures or performance issues.
- ii. **Performance**
The website should be able to render and load the pages within 5 seconds or less.
- iii. **Availability**
The website should be available and accessible to the public with minimal downtime errors.
- iv. **Reliability**
The website should be easy to navigate with a clear and straightforward interface using a consistent layout design for all the pages.
- v. **Usability**
The dashboard should enable users to do comparisons by both job domains and job titles and visualise them on the dashboard.
- vi. **Maintainable**
The website should be able to be modified and updated by the administrators to fix any errors and further improve the system.

4.1.1 Software Requirements

Software requirements list all the software and applications needed in order to develop this project. Identifying suitable software and applications to use ensures efficiency during the process of development and delivering the system to the users. Table 4.2 below shows a list of software that was used for this project.

Process	Software	Description	Usage in Project
Documentation	Zotero	An open-source reference management software to collect, organize, annotate, cite, and share research.	Reference Guide
Designing	Figma	A digital canvas for designing websites, mobile applications, or any digital experience.	Designing wireframe
	Draw.io	An online diagram software for making diagrams like flowcharts, process diagrams, organisational charts, and network diagrams.	Data Modelling
Data Preparation	Microsoft Excel	Spreadsheet software made by Microsoft which is used to manage data entry, calculate computation analytics, and generate charts and tables.	Manage dataset
	Anaconda Distribution	Management and deployment for scientific computing using Python, R, and other supported programming languages.	Data cleaning and processing
	Python Programming	A programming language with a large community, used for web development, machine learning, data analysis, and automation.	Handling data processing.

	Cloudera	A data lake and unified analytics platform for accessible data-driven insights using Apache Hadoop and Apache Hive.	Handling queries and data warehouse.
Data Visualisation	Microsoft Power BI	Software service to transform diverse data into visual interactions.	Data analysis and visualisation
Web Development	Visual Studio Code	Versatile source-code editor with debugging, syntax highlighting and Git integration.	Developing website
	GitHub	A software development platform and cloud-hosted development environment where users can contribute their codes.	Publishing website

Table 4.2 Finalized Software Requirements

4.1.2 Hardware Requirements

The hardware requirements for this research project were divided into two phases. These two phases are hardware requirements for dashboard and website development and hardware requirements for data preparation and analysis. The hardware used for dashboard and website development was my personal laptop, which was an Asus ROG Strix G15. The hardware used for data preparation and analysis was my personal computer (PC) and the Big Data Lab computer. Both of these hardware devices have powerful processing power and a significant amount of RAM in order to perform data processing and analysis. Tables 4.3 and 4.4 below show the specifications of the hardware requirements for the developer.

Hardware	Components	Specification
ROG Strix G15	CPU	AMD Ryzen 7, 6000 Series
	GPU	Nvidia GeForce RTX 3050
	OS	Windows 11 Pro
	RAM	16GB, DDR5, 4800MHz
	ROM	1.5TB, SSD, 300GB free space

Table 4.3 Hardware Requirements for Dashboard & Website Development

Hardware	Components	Specification
OptiPlex 3070 (Big Data Lab)	CPU	Intel i5-7000
	GPU	Windows 10
	OS	Windows 11 Pro

Table 4.4 Hardware Requirements for Data Analysis and Visualisation

4.1.3 Datasets

A dataset is a collection of data, often organized in a tabular pattern, that pertains to a specific topic or analysis, consisting of separate sets of information treated as a single unit by a computer, and typically stored in various formats such as CSV, and XLSX. Identifying a suitable dataset to be used in this dataset is important in ensuring that the information conveyed is accurate and trustworthy for the users as they use it as references.

4.1.3.1 Data Sources

The purpose of a data visualisation dashboard is to deliver important and complicated information in a way that is easy to understand. Knowing the origin of the dataset used was important to ensure data integrity and legitimacy when developing the dashboard later. The dataset needs to be regularly up-to-date with the source to ensure data relevance, accuracy and the latest information was included in the visualisation.

Job title	AI Impact	Tasks	AIs
Communications Manager	98%	365	5,933
Data Collector	95%	299	4,534
Data Entry	95%	325	4,936
Mail Clerk	95%	193	2,693
Compliance Officer	92%	194	3,141
Chief Executive Officer (CEO)	91%	153	2,510
Therapist	91%	114	2,012

Figure 4.1 Website Source of the Datasets
(source: There’s An AI For That, n.d.)

The dataset used in this project originated from a website called “THERE’S AN AI FOR THAT.” This website was created to help users identify suitable types of AI to help them complete their desired tasks. It has a search feature featuring almost all types of AI that were ever enlisted or available online. There are also other AI features on the website, such as text-to-speech, an image generator, and a writing generator. Identifying the source of the dataset used was important for data legitimacy and future references.

There were 3 CSV files indicated by the years 2022, 2023, and 2024. Each dataset contains six columns and approximately 4800 rows. Contained within these files are comprehensive data examining the general relationship between job titles and the percentage of their impacts by artificial intelligence (AI). Each entry meticulously categorizes job titles, shedding light on the AI impacts, providing a proximation count for human job tasks, AI models developed, and the domain to which each job belongs.

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	839	Communications Manager	Communication & PR	86%	258	1823
3	4641	Web Search Evaluator	Data & IT	82%	175	1161
4	653	Chief Strategy Officer	Leadership & Strategy	81%	133	933
5	1086	Data Entry Operator	Leadership & Strategy	81%	221	1550
6	1081	Data Clerk	Data & IT	79%	230	1306
7	4228	Supply Clerk	Supply Chain & Logistics	79%	80	575
8	1082	Data Collector	Data & IT	77%	197	1596
9	1787	General Clerk	Hospitality	77%	157	1282
10	486	Business Relationship Manager	Communication & PR	76%	137	895
11	713	Clerical Assistant	Administrative & Clerical	76%	185	1228
12	863	Compliance Officer	Medical & Healthcare	76%	133	893
13	1483	Executive Assistant	Leadership & Strategy	76%	209	1536
14	4395	Therapist	Medical & Healthcare	75%	78	678

Figure 4.2 Quick View of One of the Extracted Datasets

Figure 4.2 above shows a quick view of one of the extracted datasets. Upon manual inspection of these datasets, they contain errors such as unsupported format in Microsoft Power BI, lengthy decimal points that would be difficult for visualisation, incomplete data in rows, and null values. All these anomalies will be handled and explained in detail during the ETL, or data processing, phase.

4.1.3.1 Data Dictionary

A data dictionary refers to a centralized repository of metadata that provides information about the data within a dataset or database, which includes definitions, relationships, and attributes. It serves as a reference guide in understanding and managing the data structure and content of the dataset. The datasets used in this project contain 6 columns which were the ID, Job Title, Domain, AI Impact, Tasks, and AI Models. Table 4.5 below shows the data dictionary of the dataset, including its description and data type.

Column Name	Description	Data Type
ID	Identification number that was assigned to be unique for each row.	String
Job Titles	Names of different job roles.	String
Domain	The category or industry to which each job title belongs.	String
AI Impact	Percentage of AI's influence on each respective job title.	Numerical
Tasks	Numerical count of human-performed tasks associated with each job title.	Numerical
AI Models	Count of AI models or systems implemented or associated with each job title.	Numerical

Table 4.5 Data Dictionary of the Datasets

4.2 Designing Process

The design process provides blueprints and guidelines for what the system should look like before it is even developed. It also ensures effective planning, execution, development, and delivery of the system.

4.2.1 Data Modelling

Data modelling using an ERD, or entity relationship diagram, helps in showing the connections between different entities within the database of the system. It involves identifying the entities, their attributes, and the relationships between all of them. ERDs are used to create a logical structure for databases, which helps in organizing and managing data effectively. Figure 4.04 below shows the ERD design connecting all the datasets.

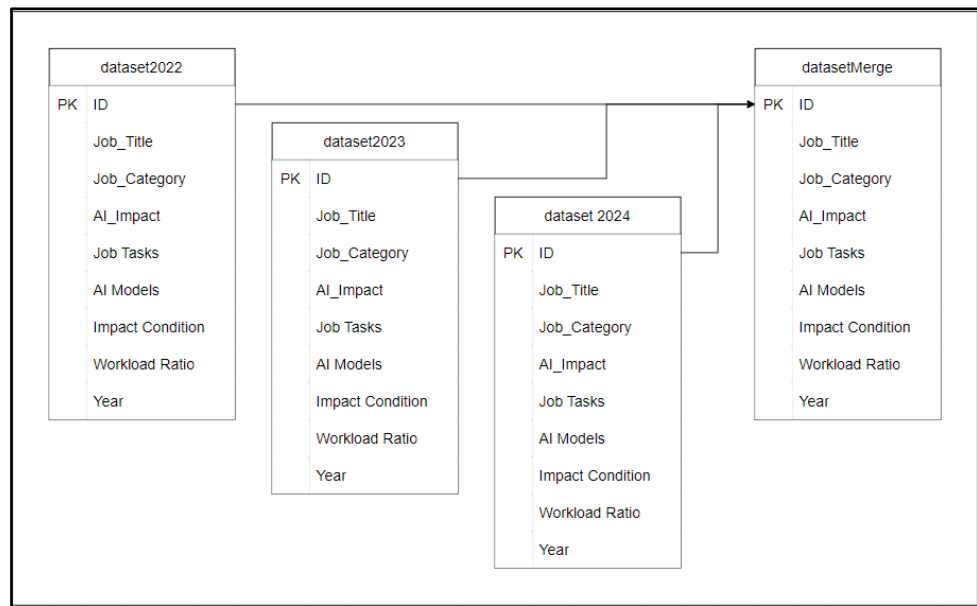


Figure 4.3 ERD Connecting All the Datasets

4.2.2 Use Case Diagram

Generally, there are two types of users that will interact with the system and dashboard, public users and administrators. On the client side, public users, including companies and agencies, can view the statistics, dashboards, and news published on the website, and they can submit any feedback that will be directed to my work email. On the server side, aside from viewing the statistics, dashboard, news, and feedback, administrators can edit the website, update the news list, create new charts for the dashboard, and refine the datasets used in the visualisations. Figure 4.4 below shows the Use Case diagram of the Data Visualisation of the Impact of Radical Artificial Technology toward Jobs.

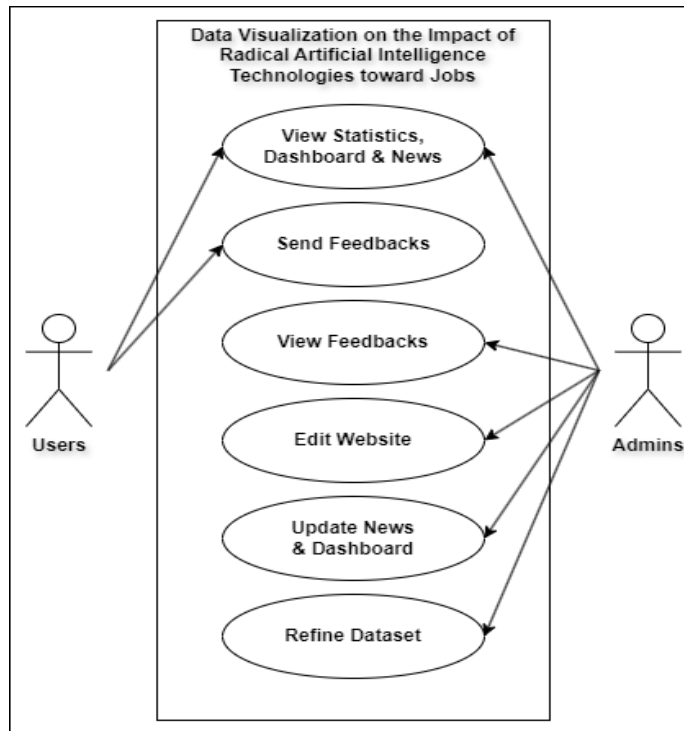


Figure 4.4 Use Case Diagram of the System

4.2.1 Standardized Design

Standardization in design refers to the process of establishing and adhering to a set of standards for the creation and production of the dashboard and website. This approach aims to ensure consistency, efficiency, and quality across various aspects of the designs. The theme of this research project is mainly about artificial intelligence and technologies.



Figure 4.5 Design Colour Scheme

Figure 4.5 above shows the colour scheme and its colour code that was used for website and dashboard design. The suitable ascent colour for artificial

intelligence would be light blue, symbolizing information and knowledge. The primary colour is white, as it helps in lighting all the contents, and dark grey is the secondary colour because it is a contrast and blends perfectly with the primary and ascent colours. The colours for miscellaneous items like hover and buttons were just slightly darker and lighter tones than the primary, secondary, and accent colours. This colour scheme was applied to both the dashboard and the website design.

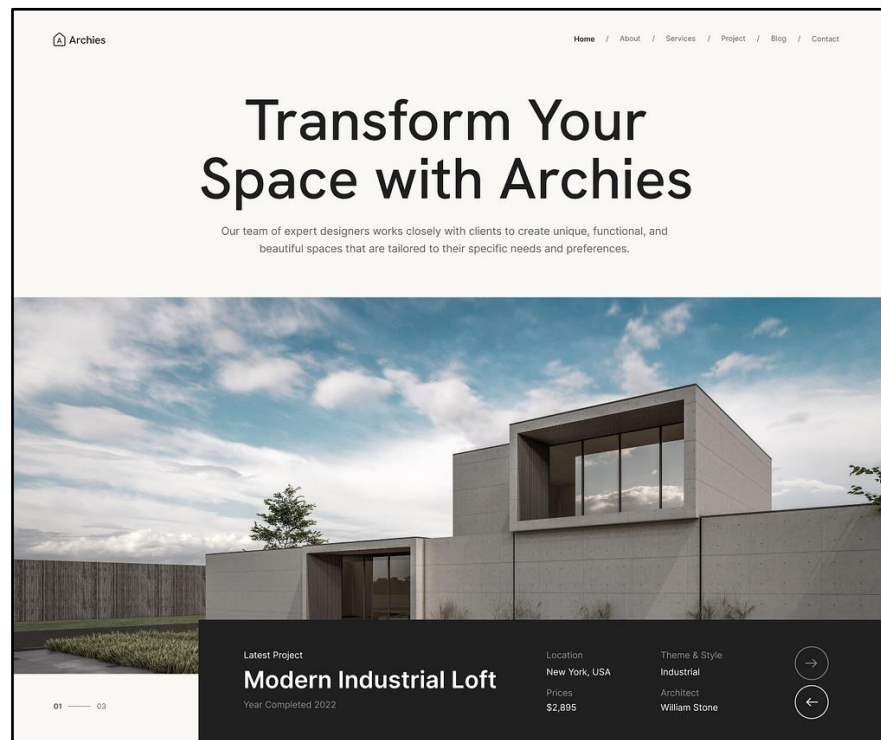


Figure 4.6 Example of Minimalist Design
(source: Ahmad S. Afandi, n.d.)

A simple and minimalist design was used for the design of the interface. The website had a floating navigation button that would follow the users as they scrolled down a specific page and navigated to all the other pages. This will always provide accessibility for the users if they decide to go to the other pages quickly. Figure 4.6 above shows an example of a minimalist design for an architecture website.

4.2.2 Dashboard Wireframe

The dashboard wireframe for this project was designed using Figma. It is a representation of the dashboard's layout, where each piece of content is strategically placed to gain the user's attention. It also serves as a blueprint for the design and development of the dashboard.

4.2.2.1 Overview Page

The overview tab is where users will get general knowledge about the dashboard. It should display all the notable data from columns such as the total number of job titles, job tasks, and AI models that will be used in the dashboard. It serves as a foundation of knowledge for users to easily understand all the upcoming charts and graphs that will be displayed on the other tab. Included on this page is also a line chart showing the progression of the data over a timeframe, a pie chart showing an overall comparison between a few impactful data, and a bar chart making a comparison between different groups of data that makes each group unique from the other. Figure 4.7 below shows the wireframe design for the overview page.

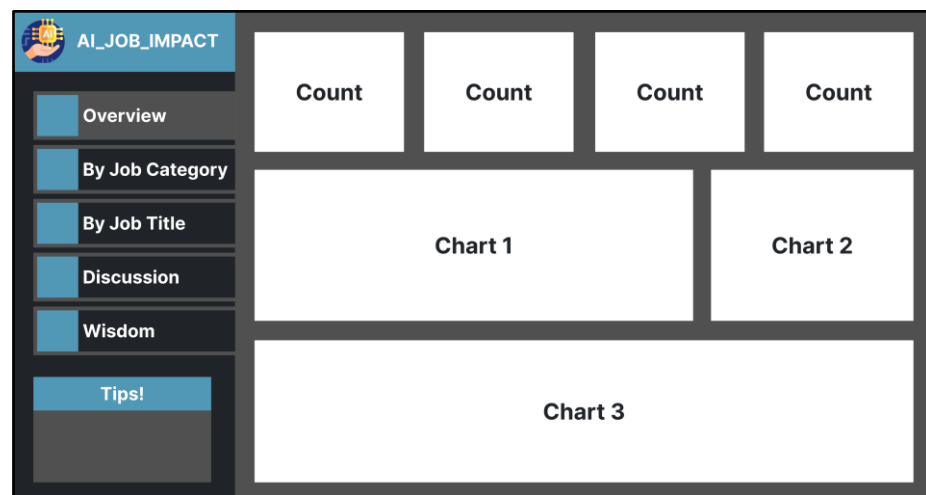


Figure 4.7 Overview Page Wireframe

4.2.2.2 By Job Category Page

This page is specifically focused on grouping the data by the column job category. All the charts and graphs that are displayed on this page have a common column that acts as a connection and relationship between all of them.

Having a filter for the job category column is important for this page as it is the main topic of discussion. Figure 4.8 below shows the wireframe design for the job category page.

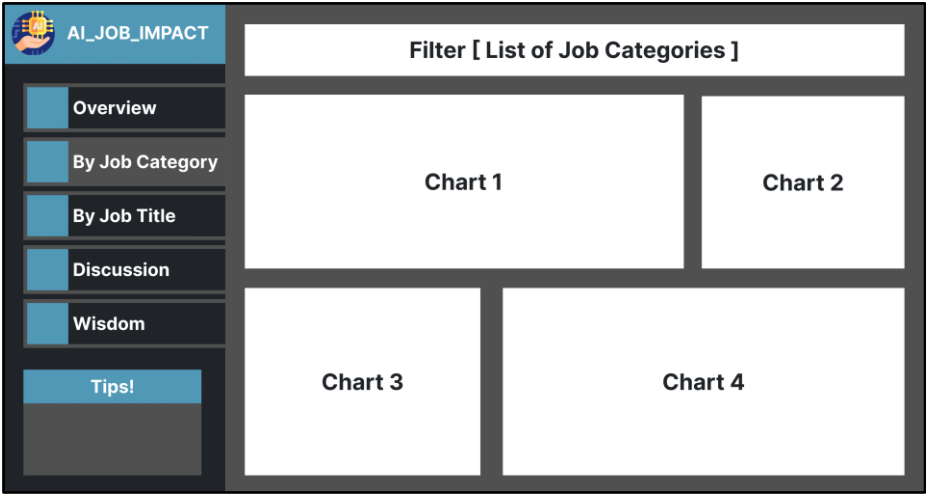


Figure 4.8 By Job Category Page Wireframe

4.2.2.3 By Job Title Page

Similar to the design of the previous page, this page, however, focuses on grouping the data by the column job title, which is much more detailed than the job category. All the charts and graphs that are displayed on this page have a common column that acts as a connection and relationship between all of them. The information displayed in the charts and graphs on this page is specifically filtered according to the job titles selected. Figure 4.9 below shows the wireframe design for the job title page.

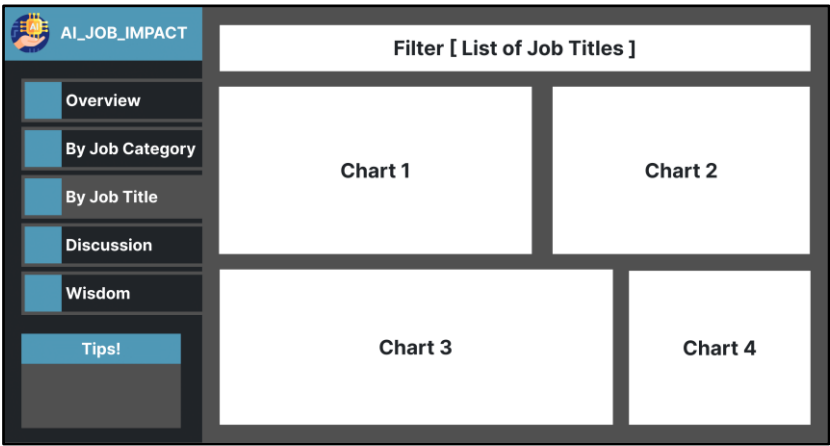


Figure 4.9 By Job Title Page Wireframe

4.2.2.4 Prediction Discussion Page

This page focuses on showing future predictions based on the dataset. The main chart that will be displayed at large on this page is a line chart that shows the progression of the dataset over a period of time. A text box will be added to provide an explanation, evidence, and references. A smaller chart will be placed to act as visual supporting evidence for the prediction made. Figure 4.10 below shows the wireframe design for the discussion page.

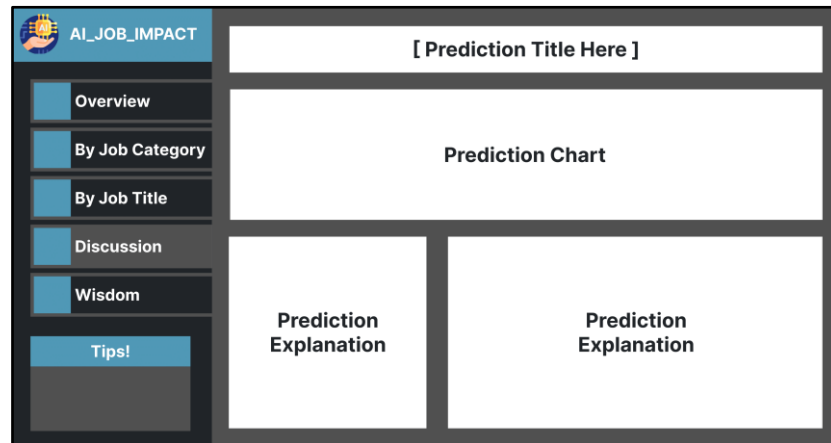


Figure 4.10 Discussion Page Wireframe

4.2.2.5 Wisdom Page

This page provides an explanation to the users about certain information that was not properly explained to the users. It has instructions about how to navigate and use the dashboard. There is also a FAQ section on the page. Figure 4.11 below shows the design of the wisdom page.

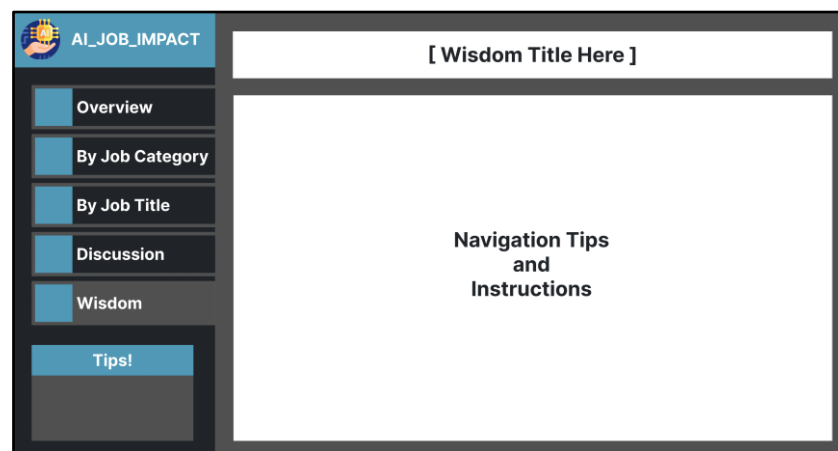


Figure 4.11 Wisdom Page Wireframe

4.2.3 Website Wireframe

The website wireframe ensures that the website that will be informative, deliverable, and user-friendly. Informative means that the website provides useful information, deliverable which includes various elements, such as content, and functionality to achieve specific goals or objectives, and user-friendly as easy navigation. This website wireframe was designed using Figma.

4.2.3.1 Home Page

The home page is where all the general ideas about the website are presented. The page is divided into 3 different sections, which are the header section, the definition section, and the slider section. The header provides information about the website title as well as a few small cards to display related information. Figure 4.11 below shows the home page design of the AI Job Impact Website.

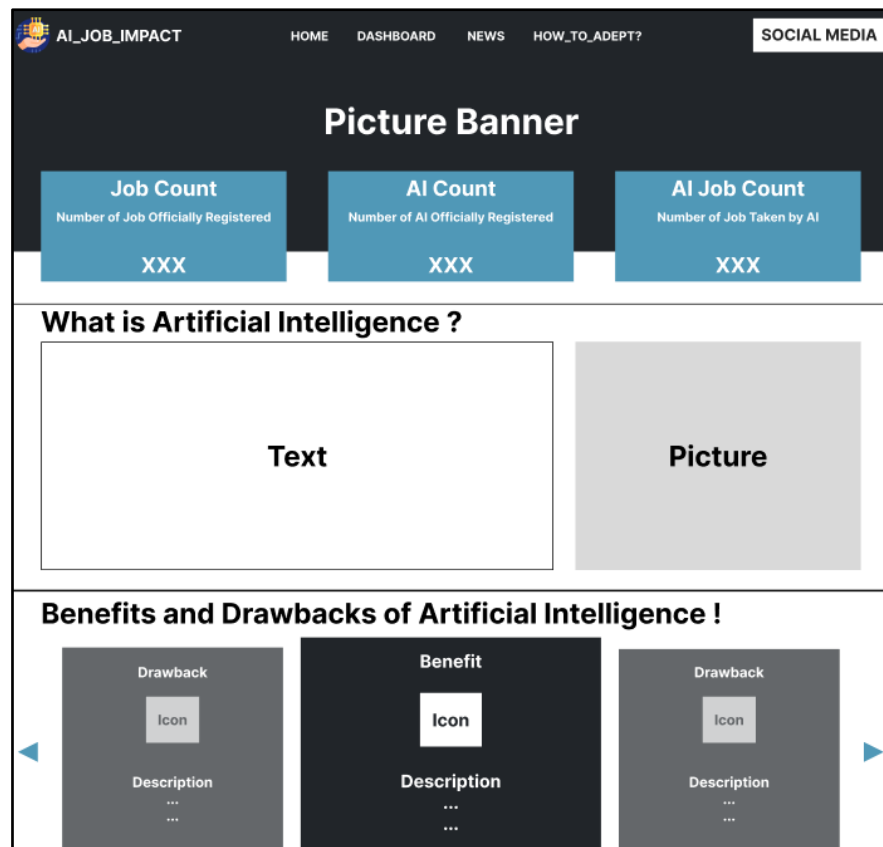


Figure 4.12 Home Page Website Design

4.2.3.2 Dashboard Page

The dashboard is the main function of website development. This page was where the core part of this research project was displayed, which is the data visualisation dashboard that was made using Microsoft Power BI. Included on this page are sections related to the dashboard, such as instructions, data sourcing, and a feedback form. Figure 4.12 below shows the dashboard page design of the AI Job Impact Website.

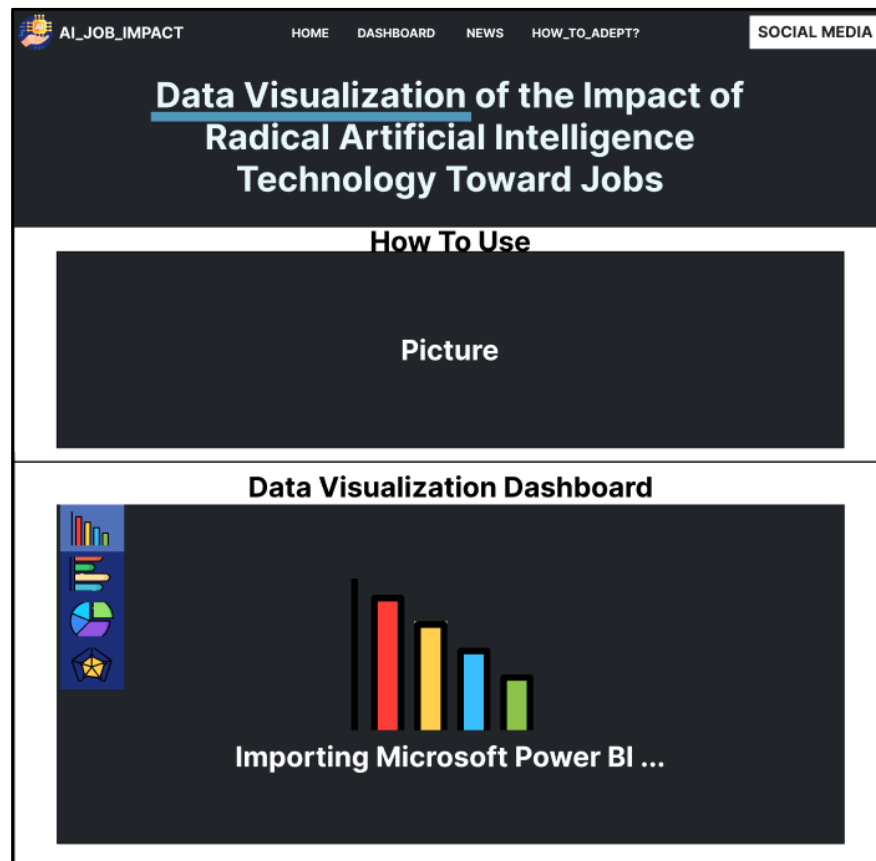


Figure 4.12 Dashboard Page Website Design

4.2.3.3 News Page

Figure 4.13 below shows the news page design of the AI Job Impact website. This news page provides users with official news related to artificial intelligence technologies that was highlighted by me. All the news is displayed on cards. Each card will display the news title, its platform, the author, and a hyperlink to redirect users to the page of the news.

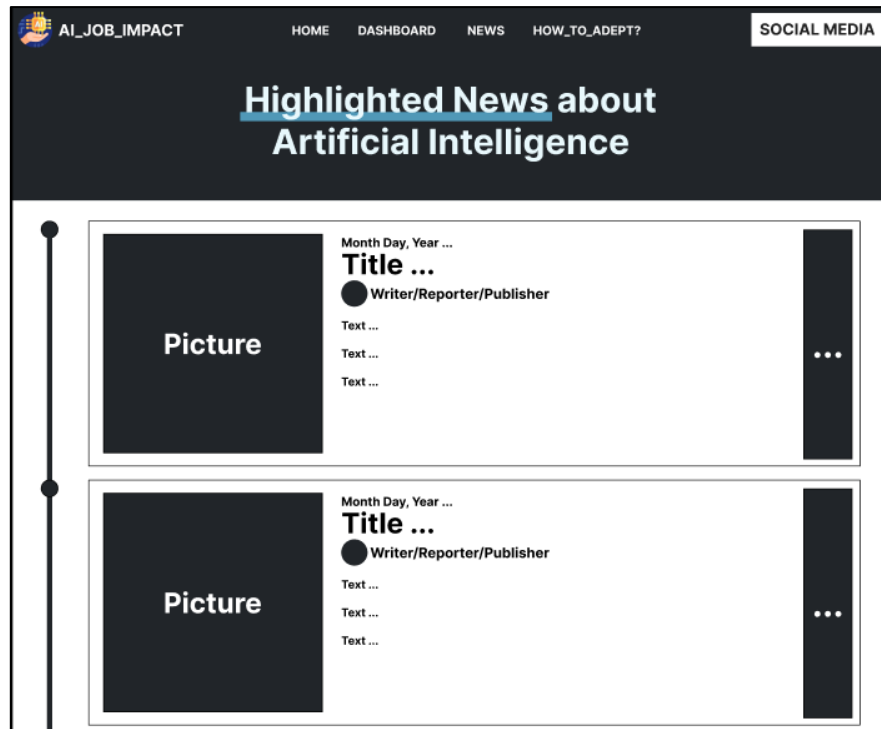


Figure 4.13 News Page Website Design

4.2.3.4 Advice Page

Figure 4.14 below shows the advice page design of the AI Job Impact website. This page will display a list of advice and suggestions for the user to overcome the impact of artificial intelligence on jobs. Included in this page is also the feedback form for the user to fill in.

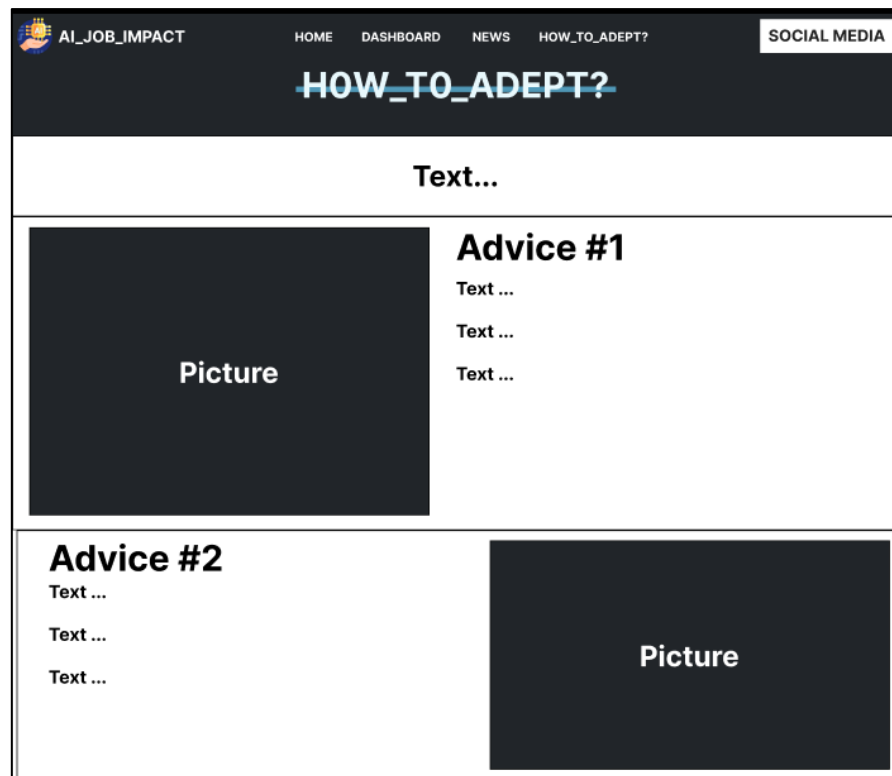


Figure 4.14 Advice Page Website Design

4.3 Data Preparation

Big data processing involves the use of advanced technologies and techniques to collect, store, manage, and analyse massive data volumes, often in real-time, to extract valuable insights, patterns, and trends. This process requires the integration of various tools, platforms, and frameworks, such as Hadoop, and Spark, to enable distributed processing, parallel computing, and scalability. The goal of big data processing is to unlock the potential of data, transform it into actionable insights that can drive business decisions, improve operational efficiency, and create new opportunities for innovation and growth.

4.3.1 Data Extract

The first step in data preparation in the ETL process is the letter E, which stands for data extraction. The datasets used in this research project were carefully reviewed by me and my supervisor to ensure that they were trustworthy and contained useful columns. We were able to identify the source of the datasets,

which linked to the website "There's an AI for That." This website was created to help users identify suitable types of AI models to help them complete their desired tasks. It keeps track of the number of AI models created, total job counts, and the percentage of AI impact on each of the enlisted job titles. Link to the website is as below:

Job Impact Index: <https://theresanaiforthat.com/job-impact-index>

4.3.1.1 Data Extraction from Source

The datasets that were used in this research project were extracted from the website "There Is An AI For That." Since the dataset published on the Kaggle website was insufficient, I contacted the website administrator through the contact form available on the website. A reply was received stating that some of the data was confidential and had been removed from the datasets. Attached to the reply was a zip file. Figure 4.15 shows the contact form available on the website.

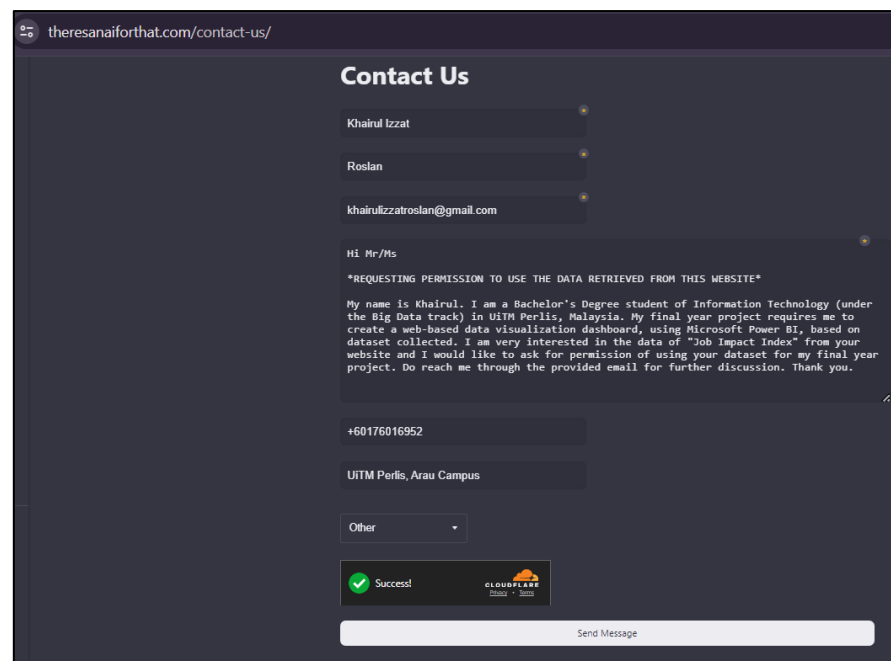


Figure 4.15 Contacting the Website Administrator

For safety precautions, the zip file was scanned using Microsoft Defender, and it was not contaminated by viruses. When unzipped, the zip file contains three separate datasets named AI_Impact_2022, AI_Impact_2023, and

AI_Impact_2024. The name itself indicates that the datasets were collected in 2022, 2023, and 2024. Figures 4.16, 4.17, and 4.18 show an overview of the extracted datasets.

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	839	Communications Manager	Communication & PR	0.86	258	1823
3	4641	Web Search Evaluator	Data & IT	0.82	175	1161
4	653	Chief Strategy Officer	Leadership & Strategy	0.81	133	933
5	1086	Data Entry Operator	Leadership & Strategy	0.81	221	1550
6	1081	Data Clerk	Data & IT	0.79	230	1306
7	4228	Supply Clerk	Supply Chain & Logistics	0.79	80	575
8	1082	Data Collector	Data & IT	0.77	197	1596
9	1787	General Clerk	Hospitality	0.77	157	1282
10	486	Business Relationship Manager	Communication & PR	0.76	137	895
11	713	Clerical Assistant	Administrative & Clerical	0.76	185	1228
12	863	Compliance Officer	Medical & Healthcare	0.76	133	893
13	1483	Executive Assistant	Leadership & Strategy	0.76	209	1536
14	4395	Therapist	Medical & Healthcare	0.75	78	678
15	1165	Development Manager	Leadership & Strategy	0.74	149	883
16	2482	Mail Clerk	Administrative & Clerical	0.74	154	935
17	3267	Pressman	Communication & PR	0.73	55	738
18	652	Chief Security Officer	Law Enforcement	0.72	97	624
19	1115	Delivery Driver	Supply Chain & Logistics	0.72	14	201
20	61	Admin	Administrative & Clerical	0.71	164	1426
21	321	Banquet Manager	Hospitality	0.71	72	453
22	414	Bricklayer	Construction	0.71	1	18
23	639	Chief Executive Officer (CEO)	Leadership & Strategy	0.71	105	611
24	1741	Front Desk Receptionist	Construction	0.71	157	1058
25	3482	QA Automation Engineer	Sales & Marketing	0.7	110	985
26	3527	Quantitative Analyst	Law Enforcement	0.7	5	28
27	645	Chief Learning Officer	Data & IT	0.69	112	751
28	1087	Data Entry Operator	Data & IT	0.69	220	1211

Figure 4.16 Overview of the dataset AI_Impact_2022

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	839	Communications Manager	Communication & PR	92%	307	2138
3	653	Chief Strategy Officer	Leadership & Strategy	89%	156	1107
4	61	Admin	Administrative & Clerical	88%	204	1767
5	2482	Mail Clerk	Administrative & Clerical	88%	172	1105
6	1082	Data Collector	Data & IT	87%	239	1779
7	1165	Development Manager	Leadership & Strategy	87%	175	1230
8	1787	General Clerk	Hospitality	86%	197	1599
9	4395	Therapist	Medical & Healthcare	86%	93	823
10	4641	Web Search Evaluator	Data & IT	85%	199	1419
11	4228	Supply Clerk	Supply Chain & Logistics	83%	98	789
12	1086	Data Entry Operator	Leadership & Strategy	82%	264	2037
13	2483	Mail Handler	Communication & PR	82%	157	1174
14	4618	Warehouse Worker	Supply Chain & Logistics	82%	61	540
15	639	Chief Executive Officer (CEO)	Leadership & Strategy	81%	124	817
16	189	Assistant	Communication & PR	80%	210	1275
17	321	Banquet Manager	Hospitality	80%	86	590
18	486	Business Relationship Manager	Communication & PR	80%	167	1063
19	633	Chief Communications Officer	Communication & PR	80%	247	1846
20	1081	Data Clerk	Data & IT	80%	279	1634
21	1483	Executive Assistant	Leadership & Strategy	80%	238	1805
22	2484	Mail Processing	Supply Chain & Logistics	79%	153	1113
23	2689	Medical Transcriptionist	Medical & Healthcare	79%	155	1025
24	3267	Pressman	Communication & PR	79%	67	889
25	4545	Vba Developer	Data & IT	79%	168	1485
26	715	Clerk Typist	Administrative & Clerical	78%	212	1651
27	863	Compliance Officer	Medical & Healthcare	78%	156	1175
28	1087	Data Entry Operator	Data & IT	78%	220	1211

Figure 4.17 Overview of the dataset AI_Impact_2023

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	839	Communications Manager	Communication & PR	98%	365	2546
3	1082	Data Collector	Data & IT	95%	299	2148
4	1084	Data Entry	Data & IT	95%	325	2278
5	2482	Mail Clerk	Administrative & Clerical	95%	193	1366
6	863	Compliance Officer	Medical & Healthcare	92%	194	1369
7	639	Chief Executive Officer (CEO)	Leadership & Strategy	91%	153	1135
8	61	Admin	Administrative & Clerical	90%	253	2014
9	653	Chief Strategy Officer	Leadership & Strategy	90%	184	1284
10	1081	Data Clerk	Data & IT	90%	317	2258
11	1085	Data Entry Clerk	Administrative & Clerical	90%	328	2305
12	1086	Data Entry Operator	Leadership & Strategy	90%	329	2311
13	1087	Data Entry Specialist	Data & IT	90%	334	2358
14	1787	General Clerk	Hospitality	90%	242	1903
15	2483	Mail Handler	Communication & PR	90%	188	1350
16	2689	Medical Transcriptionist	Medical & Healthcare	90%	189	1265
17	4186	Street Sweeper	Law Enforcement	90%	3	82
18	4228	Supply Clerk	Supply Chain & Logistics	90%	115	901
19	4395	Therapist	Medical & Healthcare	90%	114	983
20	4618	Warehouse Worker	Supply Chain & Logistics	90%	68	607
21	4641	Web Search Evaluator	Data & IT	90%	243	1599
22	1165	Development Manager	Leadership & Strategy	89%	217	1558
23	652	Chief Security Officer	Law Enforcement	88%	142	1116
24	1115	Delivery Driver	Supply Chain & Logistics	88%	20	309
25	1567	File Clerk	Administrative & Clerical	88%	228	1797
26	2484	Mail Processing	Supply Chain & Logistics	88%	189	1336
27	1231	Director Of Operations	Administrative & Clerical	87%	198	1397
28	1676	Visual Production Designer	Data & IT	87%	248	1724

Figure 4.18 Overview of the dataset AI_Impact_2024

The file format of the datasets was CSV files, which are a common data format for any spreadsheet programme or software. The CSV file format, or comma-separated value file, refers to a file where the data is input as data separated by commas. To describe the datasets, the columns on all three datasets were the same, which were ID, job titles, domain, AI impact, tasks, and AI models.

4.3.1.2 Data Extraction using Python Libraries

The datasets will go through the data cleaning process using Pandas libraries within the Jupyter Notebook environment. Figure 4.19 below shows relevant Python libraries and the datasets that were imported into the Jupyter Notebook environment.

```

Importing Libraries

In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import random

Importing Datasets

In [2]: df1 = pd.read_csv('AI_Impact_2022.csv')
df2 = pd.read_csv('AI_Impact_2023.csv')
df3 = pd.read_csv('AI_Impact_2024.csv')

```

Figure 4.19 Importing Relevant Libraries and Datasets

In [9]:	df1.head()																																															
Out[9]:	<table> <tr> <th></th><th>ID</th><th>Job titles</th><th>Domain</th><th>AI impact</th><th>Tasks</th><th>AI models</th></tr> <tr> <td>0</td><td>1</td><td>3D Animator</td><td>Data & IT</td><td>0.37</td><td>90</td><td>559</td></tr> <tr> <td>1</td><td>2</td><td>3D Artist</td><td>Data & IT</td><td>0.39</td><td>111</td><td>681</td></tr> <tr> <td>2</td><td>3</td><td>3D Designer</td><td>Data & IT</td><td>0.22</td><td>209</td><td>1427</td></tr> <tr> <td>3</td><td>4</td><td>3D Modeler</td><td>Data & IT</td><td>0.46</td><td>132</td><td>1017</td></tr> <tr> <td>4</td><td>5</td><td>911 Dispatcher</td><td>Law Enforcement</td><td>0.55</td><td>96</td><td>616</td></tr> </table>							ID	Job titles	Domain	AI impact	Tasks	AI models	0	1	3D Animator	Data & IT	0.37	90	559	1	2	3D Artist	Data & IT	0.39	111	681	2	3	3D Designer	Data & IT	0.22	209	1427	3	4	3D Modeler	Data & IT	0.46	132	1017	4	5	911 Dispatcher	Law Enforcement	0.55	96	616
	ID	Job titles	Domain	AI impact	Tasks	AI models																																										
0	1	3D Animator	Data & IT	0.37	90	559																																										
1	2	3D Artist	Data & IT	0.39	111	681																																										
2	3	3D Designer	Data & IT	0.22	209	1427																																										
3	4	3D Modeler	Data & IT	0.46	132	1017																																										
4	5	911 Dispatcher	Law Enforcement	0.55	96	616																																										

Figure 4.20 Checking the Datasets

Figure 4.20 shows one of the datasets being checked using the head() function of the Python libraries. This function shows five entries from the top of the dataset. This process was to ensure that the datasets imported into the Jupyter Notebook environment were correct. This process was repeated for all three datasets by replacing df1 with df2 and df3. All three datasets were correctly imported into the Jupyter Notebook Environment.

4.3.2 Data Transform

The second data preparation step of the ETL process is T, which means data transformation. Data transformation is the process of correcting the datasets, which includes changing the data format, dealing with missing values by deleting or recalculating them, and detecting inaccurate, redundant, or incomplete data rows.

4.3.2.1 Missing Values

The first step in data transformation was to identify missing values. This Python code effectively analyses a dataset by calculating and displaying the percentage of missing values in each column. This process was repeated for all 3 of the datasets, by replacing the df1 with df2 and df3.

In [12]:	<pre>for col in df1.columns: pct_missing = np.mean(df1[col].isnull()) print('{} - {}'.format(col, round(pct_missing*100)))</pre>					
	<pre>ID - 0% Job titles - 0% Domain - 0% AI impact - 0% Tasks - 0% AI models - 0%</pre>					

Figure 4.21 Percentage of Missing Values

Figure 4.21 above shows the percentage of missing values from each column. The initial inspection of all three datasets showed no missing values. This is a positive finding and allows us to proceed with further data transformation without needing to handle missing data.

4.3.2.1 Rename Existing Columns

As all the corrections have been made to all the inconsistent data rows, the next step is to rename the columns. While the data visualisation itself relies on the data, not the column names, using clear and descriptive column names significantly improves our ability to understand the dataset and interpret the visualisation accurately.

```
In [15]: print(df1.columns)
Index(['ID', 'Job titles', 'Domain', 'AI impact', 'Tasks', 'AI models'], dtype='object')

In [16]: df1.rename(columns={'Job titles': 'Job Titles'}, inplace=True)
df1.rename(columns={'Domain': 'Job Category'}, inplace=True)
df1.rename(columns={'AI impact': 'AI Impact'}, inplace=True)
df1.rename(columns={'Tasks': 'Job Tasks'}, inplace=True)
df1.rename(columns={'AI models': 'AI Models'}, inplace=True)
print(df1.columns)
Index(['ID', 'Job Titles', 'Job Category', 'AI Impact', 'Job Tasks',
      'AI Models'],
      dtype='object')
```

Figure 4.22 Renaming the columns

Figure 4.22 above shows the process of changing the columns' names. All the column names require changing except for the ID column. This process was repeated for all three datasets by replacing df1 with df2 and df3.

4.3.2.2 Create New Columns

While the datasets and their contents are now cleaned and ready for visualisation, the limited number of columns available restricts the variety of charts that can be created. In this process, three new columns were created in which the data values were driven from the other columns.

```
In [22]: df1['Impact Condition'] = ''
df1.loc[(df1['AI Impact'] >= 0) & (df1['AI Impact'] < 0.2), 'Impact Condition'] = 'Minimal'
df1.loc[(df1['AI Impact'] >= 0.2) & (df1['AI Impact'] < 0.5), 'Impact Condition'] = 'Moderate'
df1.loc[(df1['AI Impact'] >= 0.5) & (df1['AI Impact'] < 0.8), 'Impact Condition'] = 'Significant'
df1.loc[(df1['AI Impact'] >= 0.8) & (df1['AI Impact'] < 0.9), 'Impact Condition'] = 'Severe'
df1.loc[(df1['AI Impact'] >= 0.9), 'Impact Condition'] = 'Critical'
```

Figure 4.23 New column named Impact Condition

Figure 4.23 above shows the programming aspect of the new column that was made using Python. The first new column was a categorical data column driven by interval comparison for the AI impact column. The new column name was Impact Condition, and the data values were minimal, moderate, significant, severe, and critical. This process was repeated for the other two datasets, df2 and df3.

Condition 1	Condition 2	Assigned Data
-	AI Impact < 0.2	Minimal
AI Impact >= 0.2	AI Impact < 0.5	Moderate
AI Impact >= 0.5	AI Impact < 0.8	Significant
AI Impact >= 0.8	AI Impact < 0.9	Severe
AI Impact >= 0.9	-	Critical

Table 4.6 Condition for New Data Value

Table 4.6 above shows the conditions used when assigning the data value for the column impact condition. This column will help in classifying the job title by how critical the condition of the AI impact was that affected the job title.

```
In [25]: df1['Workload Ratio'] = df1['Job Tasks'] / df1['AI Models']
df1['Workload Ratio'] = df1['Workload Ratio'].round(3)
```

Figure 4.24 New Column named Workload Ratio

The second new column was the workload ratio. This column's data value was calculated 4.24 above shows the new column workload ratio being made using Python. by calculating the workload ratio by using the column job tasks and AI models. Figure 4.24 above shows the new column workload ratio being made using Python. For the calculation aspects, the workload ratio was calculated by dividing the data value from the column job tasks and AI models. This process was repeated for df2 and df3.

4.3.2.3 Concatenating the Datasets

Concatenating, also referred to as combining datasets, is the process of joining multiple datasets into a single, larger dataset. This is done by stacking the datasets onto each other, creating a chain. This allows for richer data analysis and the creation of more informative visualisations.

```
In [21]: df1['Year'] = 2022  
         df2['Year'] = 2023  
         df3['Year'] = 2024
```

Figure 4.25 New Column named Year

The first step for concatenating the AI impact datasets was to create a new column in each dataset. The new column was named Year, where the data value was indicated by the different years of the dataset recorded. Figure 4.25 above shows the creation of a new column, Year, and the value that was assigned to it. For example, data frame df1 originates from the AI_Impact_2022.csv dataset, thus the Year column exclusively contains the value 2022.

```
In [29]: df4 = pd.concat([df1, df2, df3], ignore_index=True)
```

Figure 4.26 Concatenate the Datasets

The Python command that will be used to concatenate the data frame is "pd.concat." This command was imported from the Pandas libraries. Figure 4.26 above shows how the "pd.concat" command was used. All the datasets were stacked on top of each other, with df1 on top, df2 in the middle, and df3 on the bottom. This new data frame was assigned as df4.

```
In [30]: df4 = df4.sort_values(by='Job Titles')
```

Figure 4.27 Sorting the Dataset

Figure 4.27 above shows the process of sorting the new dataset, df4. Rearranging or sorting the new dataset by the column Job Title will help in detecting any data errors that may occur.

In [31]: df4.head()

Out[31]:

	ID	Job Titles	Job Category	AI Impact	Job Tasks	AI Models	Year	Impact Condition	Workload Ratio
0	1	3D Animator	Data & IT	0.37	90	559	2022	Moderate	0.161
4707	1	3D Animator	Data & IT	0.45	113	762	2023	Moderate	0.148
9414	1	3D Animator	Data & IT	0.50	132	1041	2024	Significant	0.127
1	2	3D Artist	Data & IT	0.39	111	681	2022	Moderate	0.163
4708	2	3D Artist	Data & IT	0.45	136	952	2023	Moderate	0.143

Figure 4.28 Reviewing the Dataset Arrangement

Figure 4.28 above shows the arrangements of the dataset df4 after sorting. Similar job titles will stack onto each other, where all stacking groups should be in 3 pairs since the dataset df4 was the combination of 3 other datasets.

4.3.2.1 Before After Comparison

Before

	A	B	C	D	E	F
1	ID	Job titles	Domain	AI impact	Tasks	AI models
2	1	3D Animator	Data & IT	0.37	90	559
3	2	3D Artist	Data & IT	0.39	111	681
4	3	3D Designer	Data & IT	0.22	209	1427
5	4	3D Modeler	Data & IT	0.46	132	1017
6	5	911 Dispatcher	Law Enforcement	0.55	96	616
7	6	911 Operator	Law Enforcement	0.49	96	570
8	7	Aba Therapist	Hospitality	0.08	281	1145
9	8	Abap Developer	Data & IT	0.51	173	1109
10	9	Academic Administrator	Administrative & Clerical	0.08	638	2153

After

	A	B	C	D	E	F	G	H	I	J
1	ID	Job Titles	Job Category	AI Impact	Job Tasks	AI Models	Year	Impact Condition	Workload Ratio	
2	0	1 3D Animator	Data & IT	0.37	90	559	2022	Moderate		0.161
3	1	2 3D Artist	Data & IT	0.39	111	681	2022	Moderate		0.163
4	2	3 3D Designer	Data & IT	0.22	209	1427	2022	Moderate		0.146
5	3	4 3D Modeler	Data & IT	0.46	132	1017	2022	Moderate		0.13
6	4	5 911 Dispatcher	Law Enforcement	0.55	96	616	2022	Significant		0.156
7	5	6 911 Operator	Law Enforcement	0.49	96	570	2022	Moderate		0.168
8	6	7 Aba Therapist	Hospitality	0.08	281	1145	2022	Minimal		0.245
9	7	8 Abap Developer	Data & IT	0.51	173	1109	2022	Significant		0.156
10	8	9 Academic Administrator	Administrative & Clerical	0.08	638	2153	2022	Minimal		0.296
11	9	10 Academic Advisor	Administrative & Clerical	0.14	602	2323	2022	Minimal		0.259
12	10	11 Academic Counselor	Administrative & Clerical	0.13	616	2325	2022	Minimal		0.265

Figure 4.29 Finalized and Cleaned of dataset2022.csv

Before

ID	Job titles	Domain	AI impact	Tasks	AI models
1	1 3D Animator	Data & IT	0.45	113	762
2	2 3D Artist	Data & IT	0.45	136	952
3	3 3D Designer	Data & IT	0.22	234	1673
4	4 3D Modeler	Data & IT	0.49	165	1228
5	5 911 Dispatcher	Law Enforcement	0.59	111	719
6	6 911 Operator	Law Enforcement	0.54	119	730
7	7 Aba Therapist	Hospitality	0.1	328	1435
8	8 Abap Developer	Data & IT	0.54	214	1284
9	9 Academic Administrator	Administrative & Clerical	0.09	717	2773
10	10 Academic Advisor	Administrative & Clerical	0.15	727	3054

After

ID	Job Titles	Job Category	AI Impact	Job Tasks	AI Models	Year	Impact Condition	Workload Ratio
0	1 3D Animator	Data & IT	0.45	113	762	2023	Moderate	0.148
1	2 3D Artist	Data & IT	0.45	136	952	2023	Moderate	0.143
2	3 3D Designer	Data & IT	0.22	234	1673	2023	Moderate	0.14
3	4 3D Modeler	Data & IT	0.49	165	1228	2023	Moderate	0.134
4	5 911 Dispatcher	Law Enforcement	0.59	111	719	2023	Significant	0.154
5	6 911 Operator	Law Enforcement	0.54	119	730	2023	Significant	0.163
6	7 Aba Therapist	Hospitality	0.1	328	1435	2023	Minimal	0.229
7	8 Abap Developer	Data & IT	0.54	214	1284	2023	Significant	0.167
8	9 Academic Administrator	Administrative & Clerical	0.09	717	2773	2023	Minimal	0.259
9	10 Academic Advisor	Administrative & Clerical	0.15	727	3054	2023	Minimal	0.238
10	11 Academic Counselor	Administrative & Clerical	0.14	742	2795	2023	Minimal	0.265

Figure 4.30 Finalized and Cleaned of dataset2023.csv

Before

ID	Job titles	Domain	AI impact	Tasks	AI models
1	1 3D Animator	Data & IT	0.5	132	1041
2	2 3D Artist	Data & IT	0.5	156	1190
3	3 3D Designer	Data & IT	0.25	272	1956
4	4 3D Modeler	Data & IT	0.6	204	1735
5	5 911 Dispatcher	Law Enforcement	0.6	133	1008
6	6 911 Operator	Law Enforcement	0.6	133	1007
7	7 Aba Therapist	Hospitality	0.1	379	1898
8	8 Abap Developer	Data & IT	0.65	242	1696
9	9 Academic Administrator	Administrative & Clerical	0.1	858	3596
10	10 Academic Advisor	Administrative & Clerical	0.15	905	3602
11	10 Academic Advisor	Administrative & Clerical	0.15	905	3602

After

ID	Job Titles	Job Category	AI Impact	Job Tasks	AI Models	Year	Impact Condition	Workload Ratio
0	1 3D Animator	Data & IT	0.5	132	1041	2024	Significant	0.127
1	2 3D Artist	Data & IT	0.5	156	1190	2024	Significant	0.131
2	3 3D Designer	Data & IT	0.25	272	1956	2024	Moderate	0.139
3	4 3D Modeler	Data & IT	0.6	204	1735	2024	Significant	0.118
4	5 911 Dispatcher	Law Enforcement	0.6	133	1008	2024	Significant	0.132
5	6 911 Operator	Law Enforcement	0.6	133	1007	2024	Significant	0.132
6	7 Aba Therapist	Hospitality	0.1	379	1898	2024	Minimal	0.2
7	8 Abap Developer	Data & IT	0.65	242	1696	2024	Significant	0.143
8	9 Academic Administrator	Administrative & Clerical	0.1	858	3596	2024	Minimal	0.239
9	10 Academic Advisor	Administrative & Clerical	0.15	905	3602	2024	Minimal	0.251
10	11 Academic Counselor	Administrative & Clerical	0.15	885	3544	2024	Minimal	0.25

Figure 4.31 Finalized and Cleaned of dataset2024.csv

ID	Job Titles	Job Category	AI Impact	Job Tasks	AI Models	Year	Impact Condition	AI Workload Ratio
0	1 3D Animator	Data & IT	0.37	90	559	2022	Moderate	0.161
1	1 3D Animator	Data & IT	0.45	113	762	2023	Moderate	0.148
2	1 3D Animator	Data & IT	0.5	132	1041	2024	Significant	0.127
3	2 3D Artist	Data & IT	0.39	111	681	2022	Moderate	0.163
4	2 3D Artist	Data & IT	0.45	136	952	2023	Moderate	0.143
5	2 3D Artist	Data & IT	0.5	156	1190	2024	Significant	0.131
6	3 3D Designer	Data & IT	0.22	209	1427	2022	Moderate	0.146
7	3 3D Designer	Data & IT	0.22	234	1673	2023	Moderate	0.14
8	3 3D Designer	Data & IT	0.25	272	1956	2024	Moderate	0.139
9	4 3D Modeler	Data & IT	0.46	132	1017	2022	Moderate	0.13
10	4 3D Modeler	Data & IT	0.49	165	1228	2023	Moderate	0.134
11	4 3D Modeler	Data & IT	0.6	204	1735	2024	Significant	0.118
12	5 911 Dispatcher	Law Enforcement	0.55	96	616	2022	Significant	0.156
13	5 911 Dispatcher	Law Enforcement	0.59	111	719	2023	Significant	0.154
14	5 911 Dispatcher	Law Enforcement	0.6	133	1008	2024	Significant	0.132
15	6 911 Operator	Law Enforcement	0.54	119	730	2023	Significant	0.163
16	6 911 Operator	Law Enforcement	0.6	133	1007	2024	Significant	0.132
17	6 911 Operator	Law Enforcement	0.49	96	570	2022	Moderate	0.168

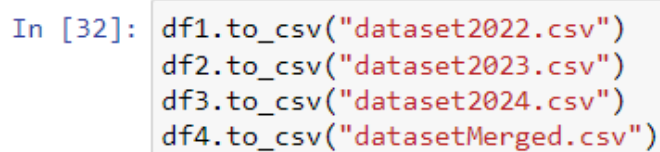
Figure 4.32 Finalized and Cleaned of datasetMerge.csv

4.3.3 Data Load

The final data preparation of the ETL process is the letter L, which is the data load. Data load refers to loading the data into the data storage or warehouse. The data warehouse for this research project is Apache Hive in the Cloudera environment. After the data transformation is done, the datasets must be exported out of the Jupyter Notebook environment and transferred to the Cloudera local storage.

4.3.3.1 Exporting Datasets

Before loading the datasets into Apache Hive, the processed datasets needed to be exported from the Jupyter Notebook environment. Figure 4.33 below shows the process of exporting the datasets that had been transformed out of the Jupyter environment and into local storage



```
In [32]: df1.to_csv("dataset2022.csv")
         df2.to_csv("dataset2023.csv")
         df3.to_csv("dataset2024.csv")
         df4.to_csv("datasetMerged.csv")
```

Figure 4.33 Exporting Datasets from Jupyter Environment

4.3.3.2 Load datasets into Apache Hive

After the datasets had successfully been exported out of the Jupyter Notebook environment, the next step was to load the dataset into the data warehouse, which was Apache Hive in the Cloudera environment. All the datasets needed to be moved from PC local storage into Cloudera local storage. This process was done by having Cloudera run on VMware, copying the dataset files, choosing a designated location, in this case, the FYP folder inside the folder Documents, and pasting the dataset files into the FYP folder. Figure 4.34 below shows the location of the datasets inside the Cloudera Local Storage.

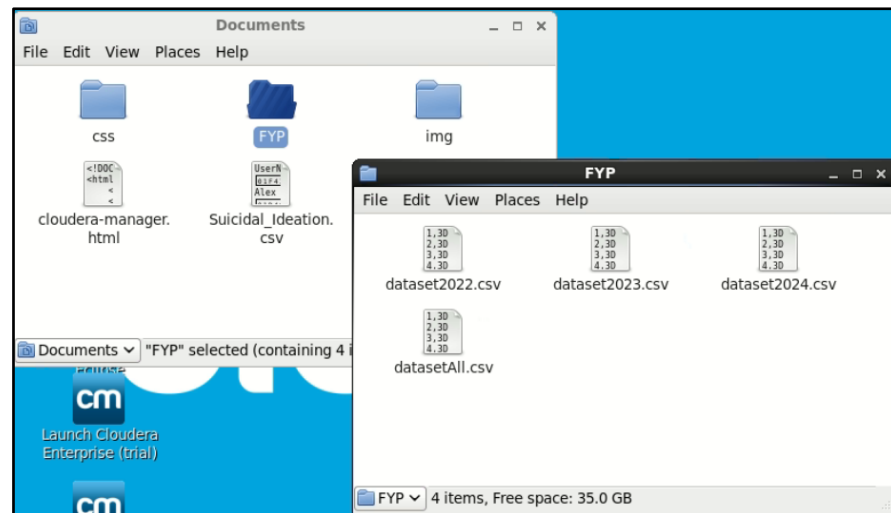


Figure 4.34 Datasets in Cloudera Local Storage

After activating the Cloudera services, a new database for this research project needed to be created to store all the data. The new database was named fyp_db. Figure 4.35 below shows that the database fyp_db was successfully created.

+	-----	+	+
	database_name		
+	-----	+	+
	default		
	fyp_db		
	retail_db		
	retail_db2		
	suicide		
	training		
	usdeath		
+	-----	+	+
7 rows selected (0.026 seconds)			

Figure 4.35 Create Database fyp_db

The next step was to create the tables for each of the datasets in the fyp_db database. This was to ensure that all the queries would take place within the tables and only use columns from inside the database. Each of the tables will consist of nine columns, which are ID, job category, job titles, AI impact, job tasks, ai models, impact condition, workload ratio, and year. The arrangement of the columns must be the same as the column inside the dataset. Figure 4.36 below shows the SQL query used to create all the tables. Figure 4.37 shows the evidence that all four tables were created.

```

create table dataset2022 ( ID string, job_title string, job_category string,
ai_impact float, job_tasks int, ai_models int, year int, impact_condition string, workload_ratio float)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';

create table dataset2023 (ID string, job_title string, job_category string,
ai_impact float, job_tasks int, ai_models int, year int, impact_condition string, workload_ratio float)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';

create table dataset2024 ( ID string, job_title string, job_category string,
ai_impact float, job_tasks int, ai_models int, year int, impact_condition string, workload_ratio float)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';

create table datasetAll ( ID string, job_title string, job_category string,
ai_impact float, job_tasks int, ai_models int, year int, impact_condition string, workload_ratio float)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';

```

Figure 4.36 SQL Command Used to Create the Tables

```

+-----+-----+
|  tab_name  |
+-----+-----+
| dataset2022 |
| dataset2023 |
| dataset2024 |
| datasetall  |
+-----+-----+
4 rows selected (0.039 seconds)

```

Figure 4.37 Tables for Datasets Created Inside fyp_db

1b755434c8): describe dataset2022

INFO : Starting task [Stage-0:DDL] in serial mod

INFO : Completed executing command(queryId=hive_

e9-ad65-1b1b755434c8); Time taken: 0.017 seconds

INFO : OK

col_name	data_type	comment
id	string	
job_title	string	
job_category	string	
ai_impact	float	
job_tasks	int	
ai_models	int	
year	int	
impact_condition	string	
workload_ratio	float	

f43adcaa74): describe dataset2023

INFO : Starting task [Stage-0:DDL] in serial mod

INFO : Completed executing command(queryId=hive_

71-b346-51f43adcaa74); Time taken: 0.015 seconds

INFO : OK

col_name	data_type	comment
id	string	
job_title	string	
job_category	string	
ai_impact	float	
job_tasks	int	
ai_models	int	
year	int	
impact_condition	string	
workload_ratio	float	

8b544cbcb4): describe dataset2024

INFO : Starting task [Stage-0:DDL] in serial mod

INFO : Completed executing command(queryId=hive_

0d-9633-948b544cbcb4); Time taken: 0.014 seconds

INFO : OK

col_name	data_type	comment
id	string	
job_title	string	
job_category	string	
ai_impact	float	
job_tasks	int	
ai_models	int	
year	int	
impact_condition	string	
workload_ratio	float	

3b8d23eba6): describe datasetall

INFO : Starting task [Stage-0:DDL] in serial mod

INFO : Completed executing command(queryId=hive_

69-8a41-073b8d23eba6); Time taken: 0.015 seconds

INFO : OK

col_name	data_type	comment
id	string	
job_title	string	
job_category	string	
ai_impact	float	
job_tasks	int	
ai_models	int	
year	int	
impact_condition	string	
workload_ratio	float	

Figure 4.38 Describe All the Tables

The database and the table are now ready to accept the datasets. Figure 4.38 below shows the SQL command used to load the data into the tables of fyp_db. The status of the process shows that the data was successfully loaded into the table.

```

0: jdbc:hive2://localhost:10000> LOAD DATA LOCAL INPATH '/home/cloudera/Documents/FYP/dataset2022.csv' INTO TABLE dataset2022
;
INFO : Compiling command(queryId=hive_20240609230202_5b8db6f1-0153-4b37-ac10-83b049c4dbd8): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2022.csv' INTO TABLE dataset2022
INFO : Semantic Analysis Completed
INFO : Returning Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20240609230202_5b8db6f1-0153-4b37-ac10-83b049c4dbd8); Time taken: 0.024 seco
nds
INFO : Executing command(queryId=hive_20240609230202_5b8db6f1-0153-4b37-ac10-83b049c4dbd8): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2022.csv' INTO TABLE dataset2022
INFO : Starting task (Stage-0:W0M0) in serial mode
INFO : Loading data to table fyp_db.dataset2022 from file:/home/cloudera/Documents/FYP/dataset2022.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Table fyp_db.dataset2022 stats: [numFiles=2, numRows=0, totalSize=709132, rawDataSize=0]
INFO : Completed loading command(queryId=hive_20240609230202_5b8db6f1-0153-4b37-ac10-83b049c4dbd8); Time taken: 0.319 seco
nds
INFO : OK
No rows affected (0.366 seconds)

0: jdbc:hive2://localhost:10000> LOAD DATA LOCAL INPATH '/home/cloudera/Documents/FYP/dataset2023.csv' INTO TABLE dataset2023
;
INFO : Compiling command(queryId=hive_20240609230202_e657a2ed-69ac-4d80-932c-6dbd35d5faa51): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2023.csv' INTO TABLE dataset2023
INFO : Semantic Analysis Completed
INFO : Returning Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20240609230202_e657a2ed-69ac-4d80-932c-6dbd35d5faa51); Time taken: 0.098 seco
nds
INFO : Executing command(queryId=hive_20240609230202_e657a2ed-69ac-4d80-932c-6dbd35d5faa51): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2023.csv' INTO TABLE dataset2023
INFO : Starting task (Stage-0:W0M0) in serial mode
INFO : Loading data to table fyp_db.dataset2023 from file:/home/cloudera/Documents/FYP/dataset2023.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Table fyp_db.dataset2023 stats: [numFiles=2, totalSize=712682]
INFO : Completed loading command(queryId=hive_20240609230202_e657a2ed-69ac-4d80-932c-6dbd35d5faa51); Time taken: 0.278 seco
nds
INFO : OK
No rows affected (0.397 seconds)

0: jdbc:hive2://localhost:10000> LOAD DATA LOCAL INPATH '/home/cloudera/Documents/FYP/dataset2024.csv' INTO TABLE dataset2024
;
INFO : Compiling command(queryId=hive_20240609230202_01d0ee32-97b3-4425-8654-65cfe326561a): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2024.csv' INTO TABLE dataset2024
INFO : Semantic Analysis Completed
INFO : Returning Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20240609230202_01d0ee32-97b3-4425-8654-65cfe326561a); Time taken: 0.098 seco
nds
INFO : Executing command(queryId=hive_20240609230202_01d0ee32-97b3-4425-8654-65cfe326561a): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/dataset2024.csv' INTO TABLE dataset2024
INFO : Starting task (Stage-0:W0M0) in serial mode
INFO : Loading data to table fyp_db.dataset2024 from file:/home/cloudera/Documents/FYP/dataset2024.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Table fyp_db.dataset2024 stats: [numFiles=2, totalSize=712678]
INFO : Completed loading command(queryId=hive_20240609230202_01d0ee32-97b3-4425-8654-65cfe326561a); Time taken: 0.254 seco
nds
INFO : OK
No rows affected (0.374 seconds)

0: jdbc:hive2://localhost:10000> LOAD DATA LOCAL INPATH '/home/cloudera/Documents/FYP/datasetAll.csv' INTO TABLE datasetall;
;
INFO : Compiling command(queryId=hive_20240609231212_ffeida0b-160e-4893-8b60-b3aa3567bbb0): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/datasetAll.csv' INTO TABLE datasetall
INFO : Semantic Analysis Completed
INFO : Returning Hive schema: Schema(fieldSchemas:null, properties:null)
INFO : Completed compiling command(queryId=hive_20240609231212_ffeida0b-160e-4893-8b60-b3aa3567bbb0); Time taken: 0.02 secon
ds
INFO : Executing command(queryId=hive_20240609231212_ffeida0b-160e-4893-8b60-b3aa3567bbb0): LOAD DATA LOCAL INPATH '/home/cl
oudera/Documents/FYP/datasetAll.csv' INTO TABLE datasetall
INFO : Starting task (Stage-0:W0M0) in serial mode
INFO : Loading data to table fyp_db.datasetall from file:/home/cloudera/Documents/FYP/datasetAll.csv
INFO : Starting task [Stage-1:STATS] in serial mode
INFO : Table fyp_db.datasetall stats: [numFiles=3, totalSize=3201738]
INFO : Completed loading command(queryId=hive_20240609231212_ffeida0b-160e-4893-8b60-b3aa3567bbb0); Time taken: 0.691 seco
nds
INFO : OK
No rows affected (0.738 seconds)

```

Figure 4.39 Loading Data into the Tables

4.4 Data Analysis

Data analysis involves the process of collecting, organising, exploring, and interpreting data from the datasets to uncover underlying patterns, trends, and relationships. The primary goal of data analysis is to extract useful and meaningful information from the datasets, which are then used to create visualisation charts that effectively communicate the information to the target audience through the dashboard. Data analysis and querying were done using Apache Hive in the Cloudera environment.

4.4.1 Total Job Titles by Impact Condition

Table 4.7 below shows the HiveQL query that was used to display the count of job titles by impact condition. This query counts the total number of job titles and group them by impact condition from the table dataset2024.

```
SELECT impact_condition AS Impact_Condition,  
COUNT (id) AS Job_Titles  
FROM dataset2024  
GROUP BY Impact_Condition;
```

Table 4.7 Data Query 1

4.4.2 Average AI Impact by Year

Table 4.8 below shows the HiveQL query that was used to display the average AI impact by years. This query calculates the average AI impact by years from the table datasetall.

```
SELECT year, AVG (ai_impact) AS AVG_AI_Impact  
FROM datasetall  
GROUP BY year  
ORDER BY year;
```

Table 4.8 Data Query 2

4.4.3 Average AI Impact by Job Category

Table 4.9 below shows the HiveQL query that was used to display the average AI impact by job category and year. This query calculates the average AI impact of each job category for every year from the table datasetall.

```
SELECT job_category, AVG (ai_impact) AS AVG_AI_Impact, year
FROM datasetall
GROUP BY job_category
ORDER BY year DESC;
```

Table 4.9 Data Query 3

4.4.4 Total Job Tasks by Job Category

Table 4.10 below shows the HiveQL query that was used to display sum of job tasks by job category. This query counts the total number of job task for each job category from the table dataset2024.

```
SELECT job_category, SUM (job_tasks) AS Total_Job_Task
FROM dataset2024
GROUP BY job_category
ORDER BY Total_Job_Task DESC;
```

Table 4.10 Data Query 4

4.4.5 Total AI Models by Job Category

Table 4.11 below shows the HiveQL query that was used to display sum AI models by job category. This query counts the total number of AI models for each job category from the table dataset2024.

```
SELECT job_category, SUM (ai_models) AS Total_AI_Model
FROM dataset2024
GROUP BY job_category
ORDER BY Total_AI_Model DESC;
```

Table 4.11 Data Query 5

4.4.6 AI Impact by Job Title

Table 4.12 below shows the HiveQL query that was used to display AI models by job title. This query counts the total number of AI models for each job title from the table dataset2024.

```
SELECT job_title, ai_impact AS Impact_Percentage, year
FROM datasetall
GROUP BY job_title
ORDER BY year DESC;
```

Table 4.12 Data Query 6

4.4.7 Job Tasks by Job Title

Table 4.13 below shows the HiveQL query that was used to display job tasks by job title. This query counts the total number of job tasks for each job title from the table dataset2024.

```
SELECT job_title, job_tasks
FROM dataset2024
ORDER BY job_tasks DESC;
```

Table 4.13 Data Query 7

4.4.8 AI Models by Job Title

Table 4.14 below shows the HiveQL query that was used to display AI models by job title. This query counts the total number of AI models for each job title from the table dataset2024.

```
SELECT job_title, ai_models
FROM dataset2024
ORDER BY ai_models DESC;
```

Table 4.14 Data Query 8

4.4.9 Job Tasks and AI Models by Job Title

Table 4.14 below shows the HiveQL query that was used to display the job tasks and AI models by job title. This query counts the total number of job task and AI models for each job title from the table dataset2024.

```
SELECT job_title, ai_models, job_tasks  
FROM dataset2024  
ORDER BY job_title DESC;
```

Table 4.15 Data Query 9

4.5 Visualisation & Dashboard

This section focuses on the development of the data analysis, visualisation charts, and dashboard. The title of the dashboard is AI Job Impact Dashboard.

4.5.1 Data Visualisation using Microsoft Power BI

The data visualisation software used for this research project was Microsoft Power BI. Microsoft Power BI offers a comprehensive set of features that make it well-suited for handling large datasets and integrating with other Microsoft tools. The visualised dashboard can be directly published from this platform. Users can also use the dashboard file extension link to import the dashboard to personal websites. Users can also edit their dataset directly from the platform by using the table view and model view.

4.5.1 Data Visualisation Dashboard

The AI Job Impact dashboard contains five pages which are the overview page, job category page, job title page, prediction discussion page, and wisdom page. The link to the dashboard is as follows:

Dashboard Link:

<https://app.powerbi.com/view?r=eyJrIjoiYmYxN2JkOTktNzYzNy00NmU4LWI4MGItMWVmNzFhZTc5MWM1IiwidCI6ImNkY2JiMGUyLTlmZWEtNGY1NC04NjcwLTk3MjcwNzc5N2FkYSIsImMiOiJEWfQ%3D%3D>

Short Link:

<https://tinyurl.com/687nkw93>

4.5.1.1 Overview Tab

The overview page of the AI Job Impact dashboard contains 4 cards and 3 charts. It serves as a foundation of knowledge for users to easily understand all the upcoming charts and graphs that will be displayed on the other tab. The four cards were displayed below the page title, containing the count of the total number of job categories, job titles, job tasks, and AI models. The doughnut charts show the comparison between the impact conditions, while the gauge chart shows the average AI impact for the year 2024. The line chart shows the progression of the sum of the AI models registered over the course of 3 years. Figure 4.40 below shows the overview page of the AI Job Impact dashboard.



Figure 4.40 Dashboard Overview Page

4.5.1.2 Job Category Page

The second page of the AI Job Impact dashboard was the Job Category page. This page focuses on displaying data columns that were filtered by job category. All the charts and graphs that were displayed on this page had a

common column, which is the job category, that acts as a connection and relationship between all of the charts on the page. There were 4 charts used on this page, which were, a pie chart, 2 bar charts, and a line chart. The pie chart displays the comparison of AI impact between different job categories. The vertical bar chart shows the total job tasks, while the horizontal bar chart shows the total AI models for each of the selected job categories. The line chart shows the average AI impact of the job categories for the years 2022 to 2024. Figure 4.41 below shows the job category page of the AI Job Impact dashboard.

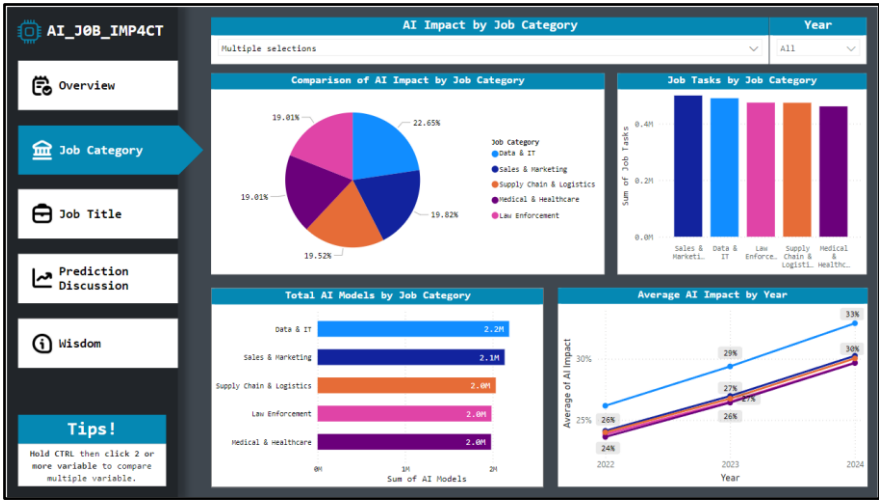


Figure 4.41 Dashboard Job Category Page

4.5.1.3 Job Title Page

The third page of the dashboard was the job title page. This page focuses on grouping the data by the column job title, which is much more detailed than the job category. All the charts and graphs that were displayed on this page also had a common column, which was the job title, that acted as a connection and relationship between all of the charts on the page. The pie chart shows the comparison of AI impact between different selected job titles. The line chart shows the progression of AI impact for each job title from 2022 to 2024. The scatter plot shows the connections between the total count of job tasks and AI models for each job title. And finally, the heatmap displays the distribution of the impact conditions of the job titles. Figure 4.42 below shows the job title page of the AI Job Impact dashboard.

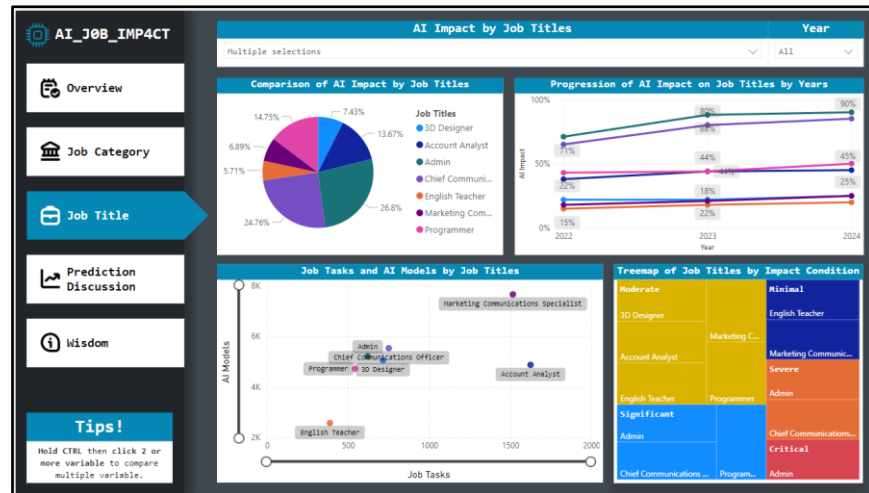


Figure 4.42 Dashboard Job Title Page

4.5.1.4 Prediction Discussion Page

The fourth page of the AI Job Impact dashboard was the prediction discussion page. This page focuses on showing predictions of the AI impact for the year 2024. The main chart that was displayed at large on this page was a line chart that shows the progression of the AI impact from 2022 to 2024. The text provides an explanation of the prediction, providing evidence and references. Below the text boxes were two smaller charts that acted as visual supporting evidence for the prediction. Figure 4.43 below shows the prediction discussion page of the AI Job Impact dashboard.

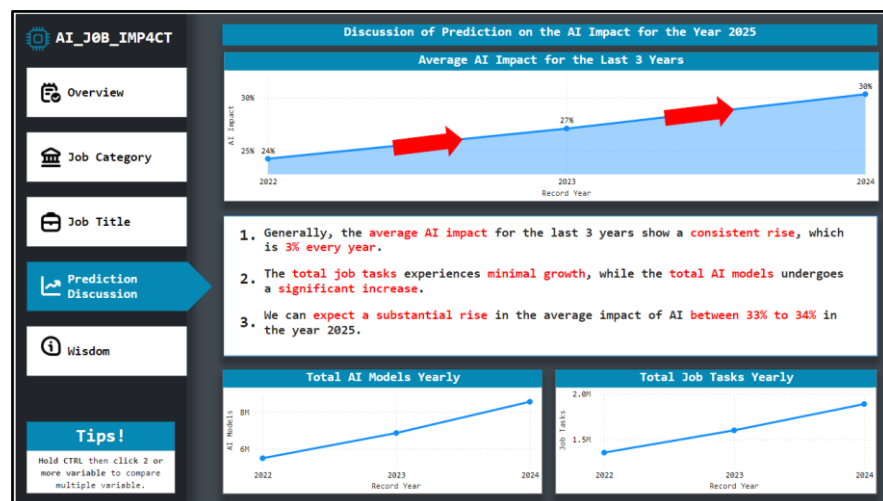


Figure 4.43 Dashboard Prediction Discussion Page

4.5.1.1 Wisdom Page

The fifth page of the AI Job Impact Dashboard is the wisdom page. This page is used to teach users about how to use the AI Job Impact Dashboard. A preview of the dashboard is in the middle of the page, with arrows and instructions surrounding it. The number on the red bookmarks indicates where the user should temper first. In ascending order, users choose a website page from the navigation bar on the left side of the dashboard, choose a page filter on the top left side, choose a year filter on the top right side, and lastly, users can hover over the charts to display related information and interact with the chart by clicking it to further filter the page. There is no data or chart displayed on this page. Figure 4.44 below shows the developed wisdom page of the dashboard.

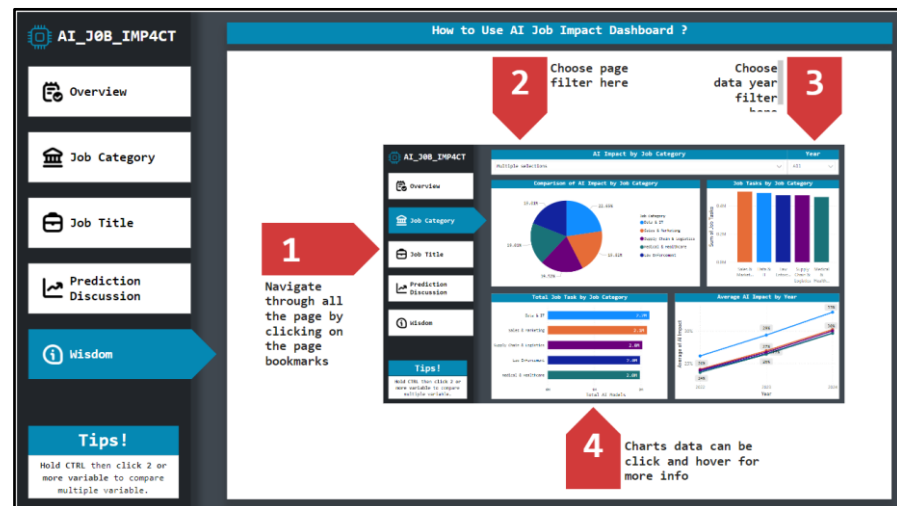


Figure 4.44 Dashboard Wisdom Page

4.6 Website Development

This section focuses on the development of the AI Job Impact website. The website contains four pages, which are the home page, dashboard page, news page, and advice page.

4.6.1 Developing Website

The website was developed using Visual Studio Code. The programming language includes HTML, JavaScript, and CSS. The general idea or wireframe of the website had already been designed during the design phase. The website was tested using a local host before it was published to a web hosting service.

4.6.1.1 Home Page

This is the AI Job Impact website's home page. It contains four sections, which are the header section, the definition section, the comparison section, and the slider section. This page serves to provide users with general ideas about the topic and theme of the website and dashboard. Figure 4.45 below shows the home page of the AI Job Impact website.

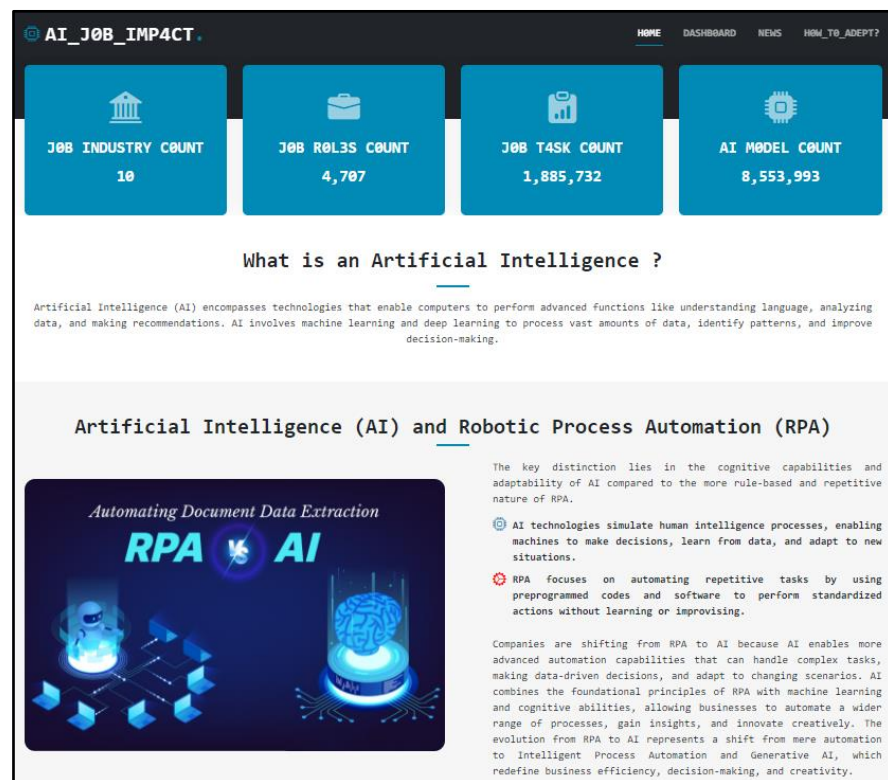


Figure 4.45 Developed Home Page

The header sections contain a banner and a button to redirect users to the dashboard page. The definition section contains the definition of artificial intelligence, while the comparison section contains a discussion about the comparison between artificial intelligence and robotic process automation. The

slider sections were used to display the benefits and drawbacks of artificial intelligence by using the card slider. At the bottom of the page, there was a banner and button that led to the dashboard page.

4.6.1.2 Dashboard Page

The dashboard page is the main attraction of the AI Job Impact website. It contains the data visualisation dashboard that was created using Microsoft Power BI. There were 3 sections on this page. Figure 4.46 below shows the dashboard page of the AI Job Impact website.



Figure 4.46 Developed Dashboard Page

The first section was the dashboard section, containing the AI Job Impact dashboard. Users can freely navigate through the dashboard by clicking the menu option on the left of the dashboard. The second section was the data sourcing section. It provides information about the data used for the dashboard and a button to redirect users to the website “There’s an AI for That,” which

was the source of the dataset. The third section was the FAQ section, where all the common questions had already been provided with answers.

4.6.1.3 News Page

The third page is the news page. This page contains news related to the title and theme of Artificial Intelligence, which were displayed on cards. Users can click on the title or the news article to redirect to the source of the news. Figure 4.47 below shows the news page of the AI Job Impact website.

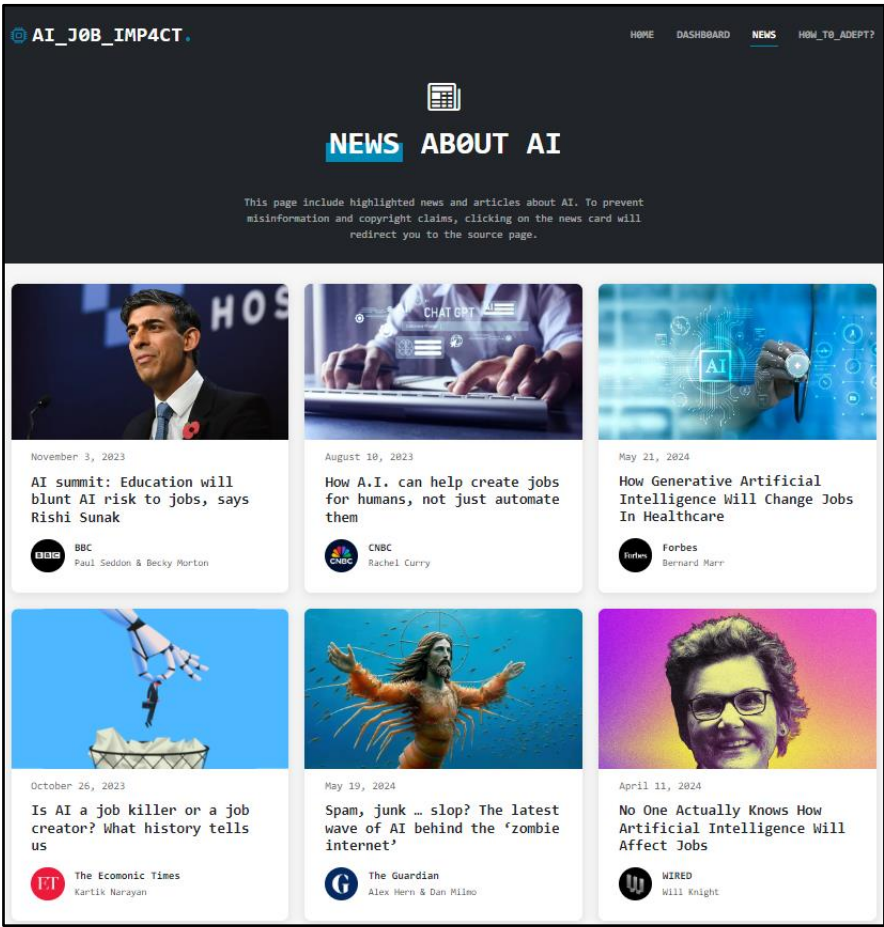


Figure 4.47 Developed News Page

4.6.1.1 Advice Page

The last page is the advice page. On this page, a list of advice and suggestions is displayed on cards. This page is designed to help users overcome their anxieties about artificial intelligence taking over their jobs. There are a total of five advice cards that were listed. Figure 4.48 below shows the advice page of the AI Job Impact website.

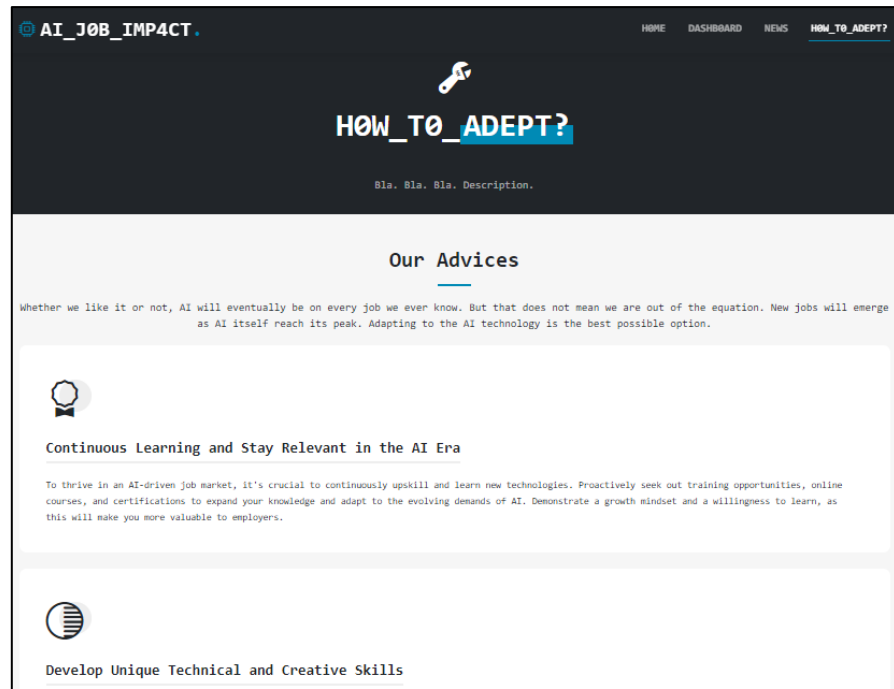


Figure 4.48 Developed Advice Page

4.6.1 Hosting Website

The final step in the development phase was to publish the website. For this research project, the website and dashboard were published using the GitHub website hosting feature. To do this, the website files needed to be uploaded onto the GitHub cloud server. The AI Job Impact website was designed using Visual Studio Code, and all the website files were synchronized to my personal GitHub account using an internal extension. The AI Job Impact dashboard is inherently integrated into the website's structure through its code.

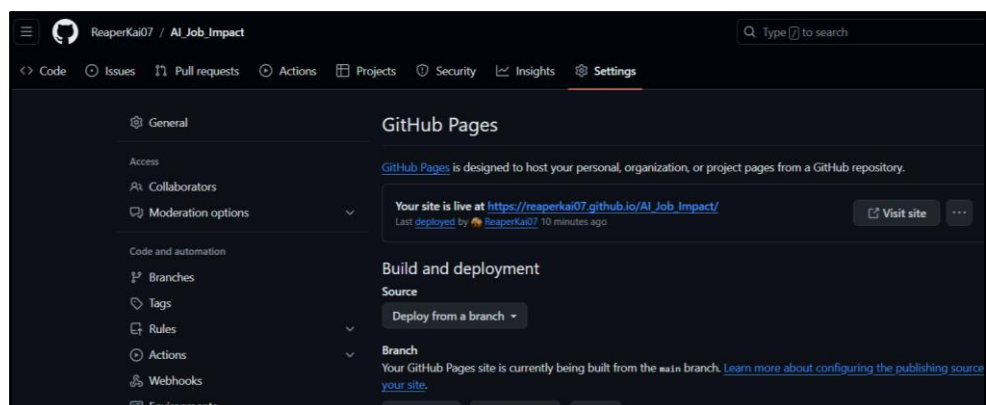


Figure 4.49 GitHub Web Hosting Feature

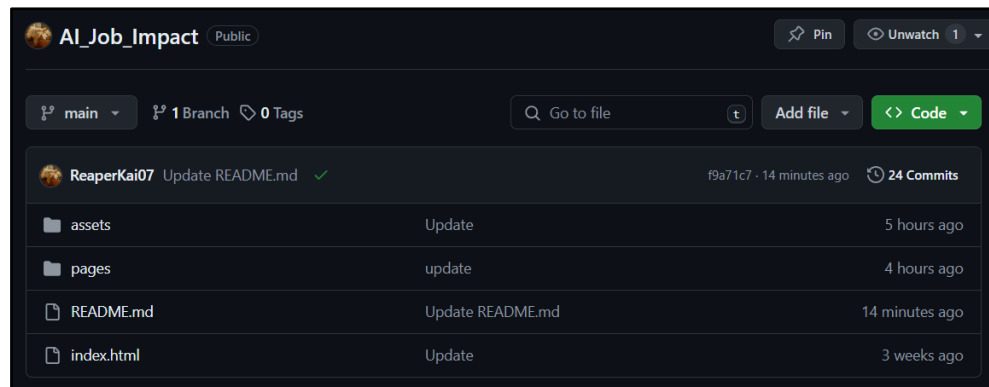


Figure 4.50 AI Job Impact Website Files on GitHub

Figures 4.49 and 4.50 above show that the AI Job Impact website files were uploaded to the GitHub cloud storage and the website hosting feature was activated. Below is the link to the website:

Website Link: https://reaperkai07.github.io/AI_Job_Impact/

4.7 Summary

This chapter covers the whole process of developing this research project, including the requirement analysis, designing process, dashboard, and website development. This chapter also offered a thorough explanation of the various stages involved in the ETL process or data preparation. A variety of software and tools were used during this phase, such as Figma, Microsoft Power BI, and Visual Studio Code. The output of this phase was the deployable AI Job Impact website and dashboard.

CHAPTER 5

RESULTS AND FINDINGS

This chapter delves into the results, outcome, and analysis of this research project's testing phase. In this chapter, the method used during the testing was underlined, the design for the users was clarified, and the test result was analysed for further improvements to the dashboard.

5.1 Technology Acceptance Model (TAM)

The user survey and dashboard evaluation for this research project were carried out using the Technology Acceptance Model (TAM). TAM takes measures of five aspects, which are perceived ease of use (PEU), perceived usefulness (PU), behavioural intentions (BI), attitude (ATT), and actual use (AU). These 5 aspects help in analyzing the significance of the interaction between humans and technology, which is the AI Job Impact dashboard.

5.2 Questionnaire

The user survey was created using Google Forms. The survey was separated into 3 sections which were User Information, Dashboard Survey and Comments & Recommendation. Participants were given brief introductions to the survey with links to the dashboard for the user to test. The survey will take approximately 5 minutes to complete. For a detailed view of the user survey, please refer to Appendix B. Below is the link to the user survey:

User Survey: <https://forms.gle/4cpsSyyMfB3uKtdY6>

The first section is the user information, which is the demographic part of the user survey. This section contains 5 questions. This section focuses on gaining information about the user, including their general information about AI and their experience using data visualisation dashboards. Users are required to

answer all the questions in this section. The second section is the main content of the user survey. This section contains the questionnaire to measure the effectiveness of the dashboard. Using the TAM method, the questionnaire was grouped into 3 subsections, which are Part 1: Perceived Ease of Use, Part 2: Dashboard Survey, and Part 3: Comment & Recommendation. Each part consists of 3 to 4 questions. It is mandatory to answer all the questions in this section. The third section is the open-ended question, which users were not required to answer but were encouraged to. In this section, there are 2 questions which are, asking for the user's comment and their suggestions to improve the system. Figure 5.1 below shows the measurement scale that was used in the user survey.

Scale	Description
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Table 5.1 User Survey Measurement Scale

5.3 Research Environment

The questionnaire that was clarified in the previous subsection was then distributed to the public, especially to a few targeted groups of the audience. The first targeted community group was the job-seeking community. This group consisted of individuals who are currently seeking jobs, with the chance that some of them are unfamiliar with AI technology and unaware that AI is currently automating some of the jobs for which they are seeking employment. Next is the UiTM student community, especially for courses that were not from the College of Computing, Informatics, and Mathematics (KPPIM), as most of them do not know how AI technology works or how AI is impacting their future careers. The survey was also distributed to the public in order to gain

unbiased feedback and measure the dashboard's usability and effectiveness with individuals who were not within the target audience.

5.4 Results Analysis

This survey aims to assess the usability and effectiveness of the AI Job Impact dashboard for users with diverse backgrounds. The results and feedback gained from this survey will be used to improve the developed dashboard and assess whether the research objectives of this project were achieved. Based on the figure 5.2 below, the mean scale of the overall user survey is 4. This shows that participants are generally satisfied with the AI Job Impact dashboard.

Questionnaire		Scale					Means
		1	2	3	4	5	
Part 1	Perceived Ease of Use (PEU)						
1	It is easy for me to navigate and use the AI Job Impact dashboard.	0	0	0	3	33	4.92
2	The information displayed in the dashboard is clear and easy to understand.	0	0	6	13	17	4.31
3	The dashboard does not require me to learn any complex functionalities.	0	0	1	9	26	4.44
Part 2	Perceived Usefulness (PU)						
4	The dashboard enhances my knowledge about how AI can influence a job title.	0	0	6	22	8	4.06
5	I am able to gain information on how much AI is affecting current job titles.	0	2	3	17	14	4.11
6	The dashboard shows the connection between AI impact, job task, and AI models clearly.	0	0	1	11	24	4.64
Part 3	Behavioural Intension (BI) & Attitude (ATT)						
7	I like using AI Job Impact dashboard as my sources of information about AI.	0	1	13	20	2	3.64
8	I believe that using the dashboard helps in understanding the impact of AI on jobs.	0	0	3	25	8	4.14
9	I am satisfied with the AI Job Impact dashboard.	0	0	4	18	14	4.28
10	I would recommend this dashboard to others.	0	0	2	7	27	4.69

Table 5.2 Overview of Dashboard Survey Section Result

5.4.1 User Information

By completing this section, participants agreed to contribute their data for the survey analysis. A total of 36 respondents of various backgrounds, including their gender, age group, occupation, and industry, participated in this survey. This section of the survey serves as a foundation of information about the background of the participants and their general knowledge regarding the dashboard topic. Figure 5.1 indicates that out of 36 participants, 24 (66.7%) were male, and the remaining 23 (33.3%) were female.

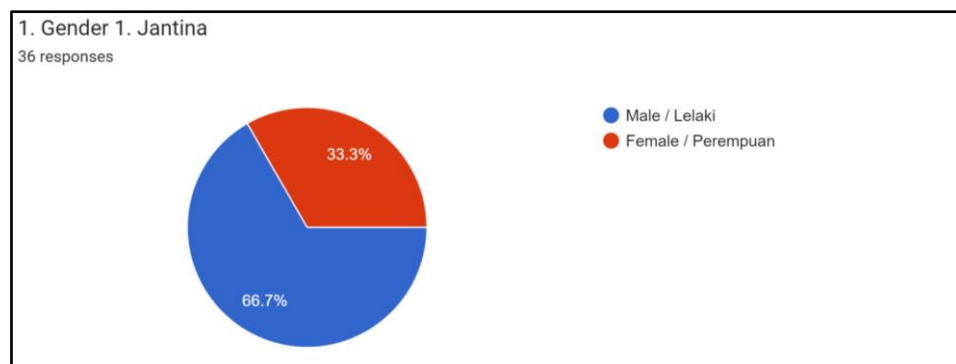


Figure 5.1 User Information Question 1

Figure 5.2 shows a pie chart of age groups. Based on the pie chart, 16 (44.4%) participants were between the ages of 18 and 25, and 10 (27.8%) participants were between the ages of 26 and 35, while the remaining 10 (27.8%) participants were evenly distributed for the group ages of 36 to 45 and 46 and above. We can conclude that the largest age group of the participants falls between 18 and 25 years old, representing nearly half of the response.

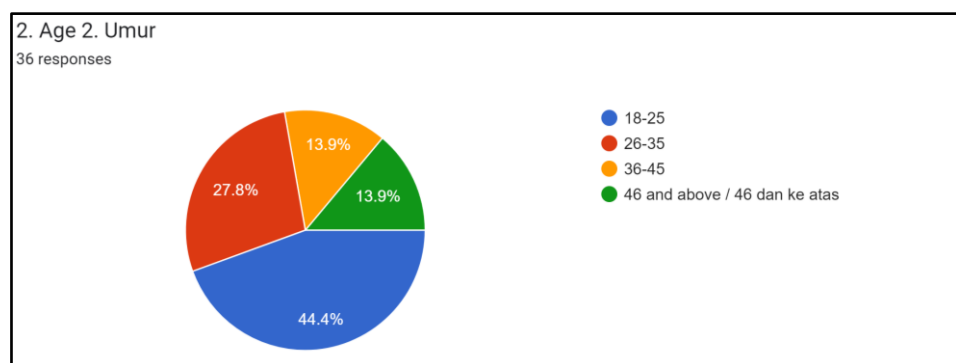


Figure 5.2 User information Question 2

Figure 5.3 shows the distribution of participants by occupation status. There are four occupation statuses which are, student, employed, unemployed, and pension. 20 (55.6%) participants were students, 8 (22.2%) participants were employed, 5 (13.9%) participants were pensioners, and the remaining 3 (8.3%) participants were unemployed. To conclude, the majority of participants were students, which took half of the pie chart distribution.

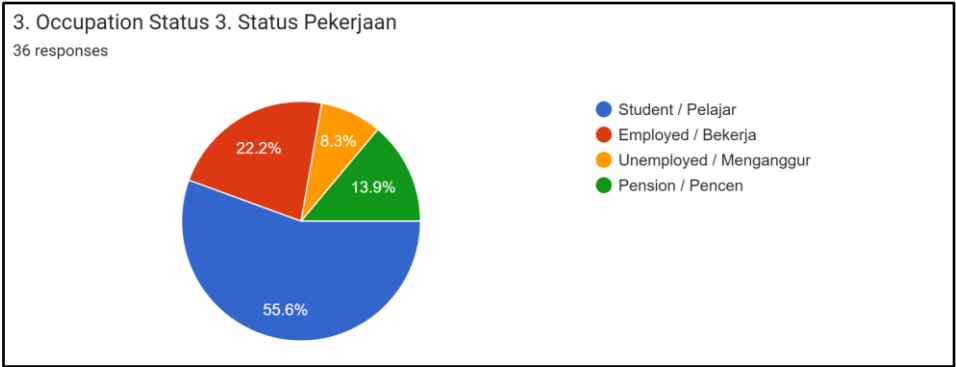


Figure 5.3 User information Question 3

There are a total of 10 industries listed in the user survey and dashboard. Figure 5.4 indicates that 12 (33.3%) of participants were in the data and IT industry, 6 (16.7%) were from the medical and healthcare industry, sales and marketing, and leadership and strategy industries both have an even number of participants, which were 4 (11.1%), law enforcement, and supply chain and logistics both have a total of 3 (8.3%) participants, 2 (5.6%) participants are from the construction industry, while the remaining 2 (5.6%) participant are evenly from administrative and clerical (2.8%), and communication and PR (2.8%).

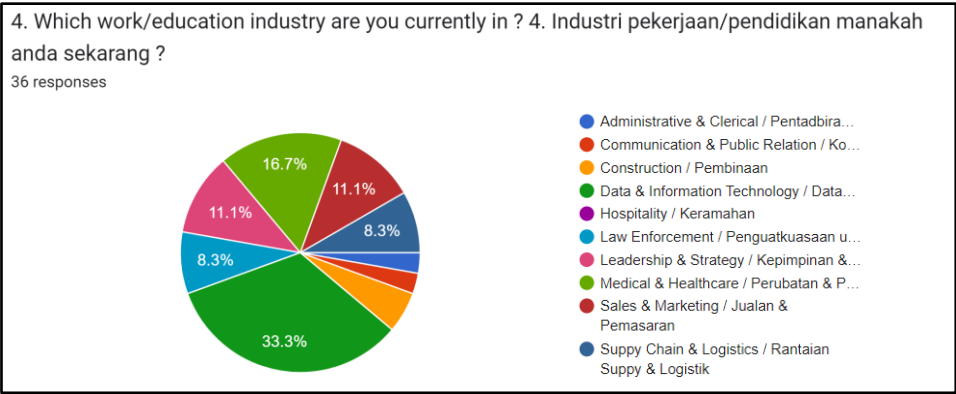


Figure 5.4 User information Question 4

Figure 5.5 shows the distribution of participants by the news platform. 33 (91.7%) participants found news regarding AI on social media, 2 (5.6%) chose a news bulletin, while the remaining 1 (2.8%) participant chose word of mouth as their news source. No participant chooses search engines as their source of information about AI. We can conclude that the majority of participants chose social media as their source platform.

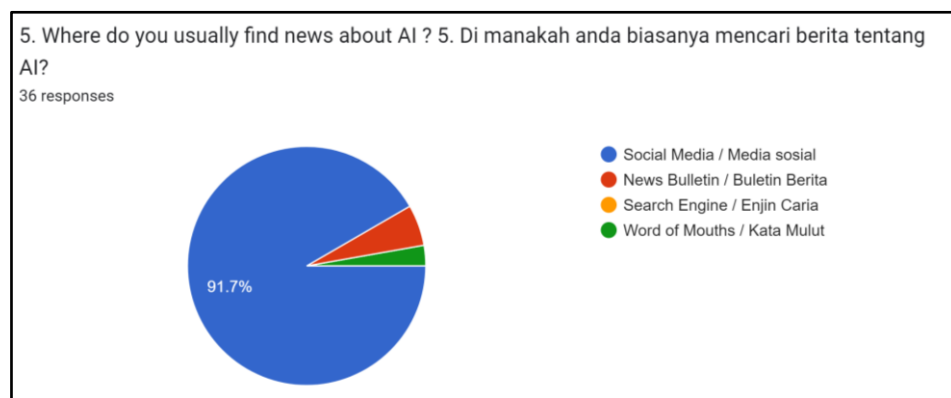


Figure 5.5 User information Question 5

The next question is to identify whether the participant is familiar with the title of the research project and dashboard. Figure 5.6 illustrates that 21 (58.3%) participants acknowledged that AI is taking over jobs, while the remaining 15 (41.7%) were not aware of this issue. In general, we can conclude that most participants are familiar with the news that AI is taking over jobs nowadays.

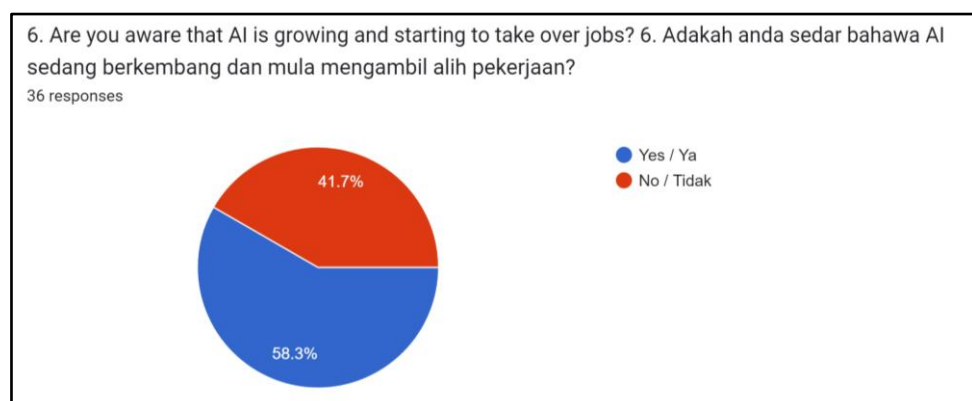


Figure 5.6 User Information Question 6

The last question in this section is to clarify participants' familiarity with a data visualisation dashboard. Based on Figure 5.7 below, 20 (55.6%) participants do not have experience using data visualisation dashboards, while the remaining 16 (44.4%) participants are familiar with them. In conclusion, most participants are first-time participants in using a data visualisation dashboard.

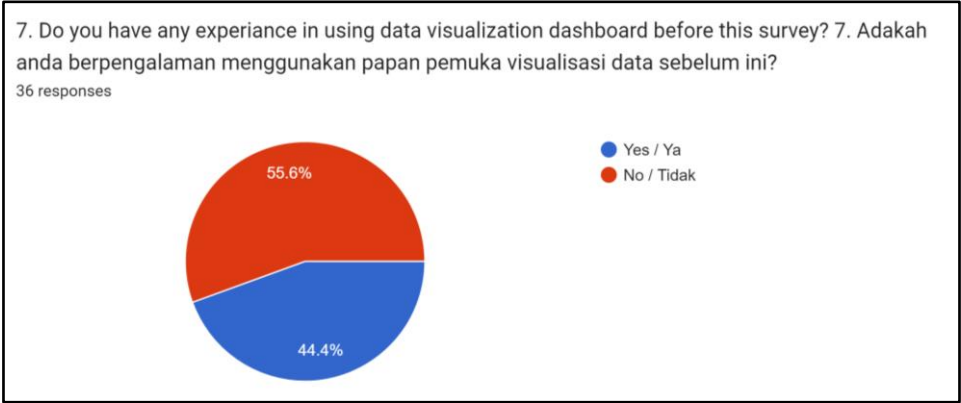


Figure 5.7 User Information Question 7

5.4.2 Perceived Ease of Use

In terms of perceived ease of use, the questionnaire was focused on evaluating whether the participants found that the dashboard met their expectations, such as the dashboard difficulty and the clear navigation. Figure 5.8 shows that 33 (91.7%) strongly agreed that the dashboard was easy to navigate, while the remaining 3 (8.3%) chose to agree. There were no neutrals or disagreements.

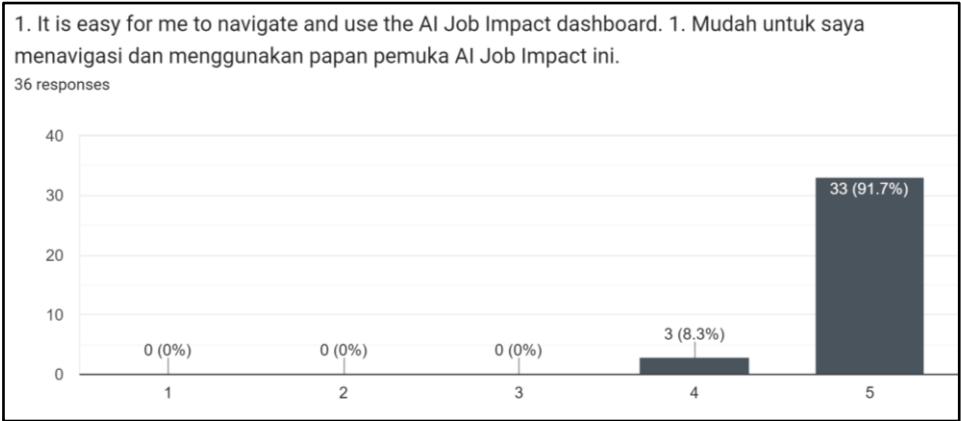


Figure 5.8 Dashboard Survey Questions 1

Figure 5.9 indicates that 17 (47.2%) participants strongly agreed that the information displayed in the dashboard is clear and easy to understand, 13 (36.1%) participants chose to agree, while the remaining 6 (16.7%) chose to be neutral to the question. Most participants chose to agree that the dashboard’s information was clear. There were no disagreements.

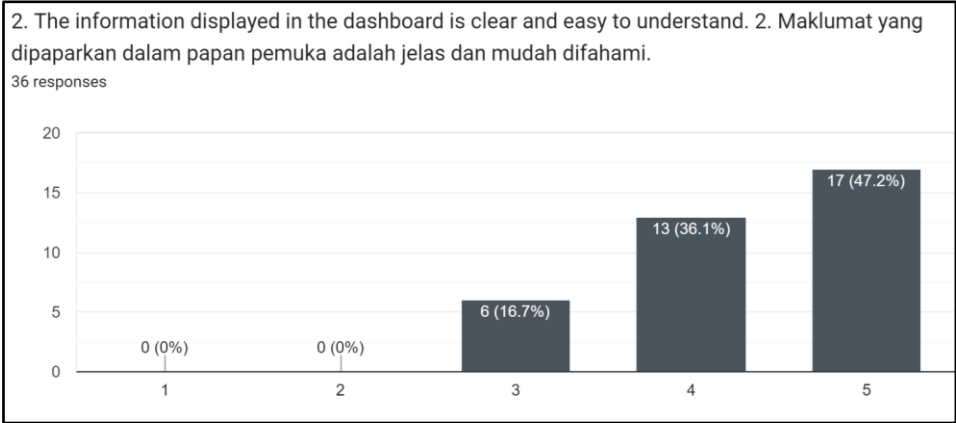


Figure 5.9 Dashboard Survey Questions 2

The third question asked about the need for complex functionality within the dashboard. Based on Figure 5.10, 26 (72.2%) choose to strongly agree that the dashboard does not require them to learn complex functionalities, 9 (25%) choose to agree, and the remaining 1 (2.8%) choose to be neutral. While most participants find the functionality of the dashboard straightforward, there are still a few who find that the functionality is complicated and hard for them to understand. There was no disagreement.

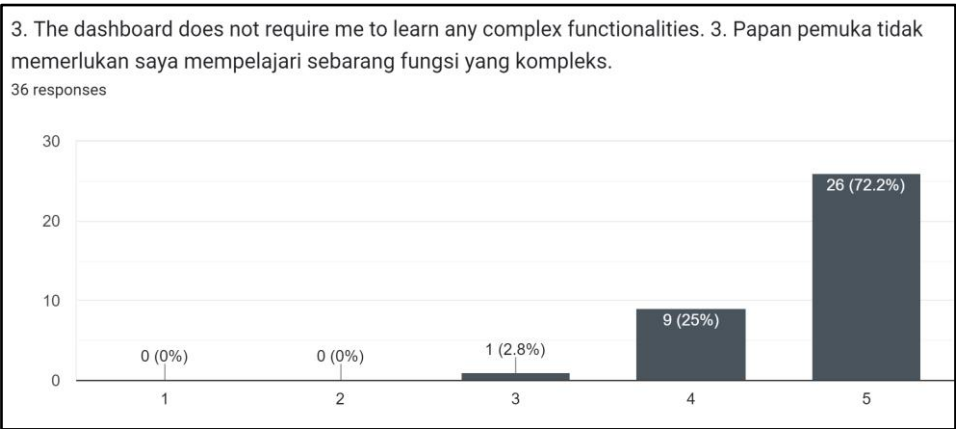


Figure 5.10 Dashboard Survey Questions 3

5.4.3 Perceived Usefulness

In terms of perceived usefulness, the questionnaire focuses on measuring how valuable and beneficial the dashboard is in order to achieve the user's specific needs and tasks. Based on Figure 5.11, 8 (22.2%) participants strongly agreed that the dashboard enhances their knowledge about how AI influences job titles, 22 (61.1%), which is the highest count, only agreed, while the remaining 6 (16.7%) chose to be neutral on this question. There is no disagreement.

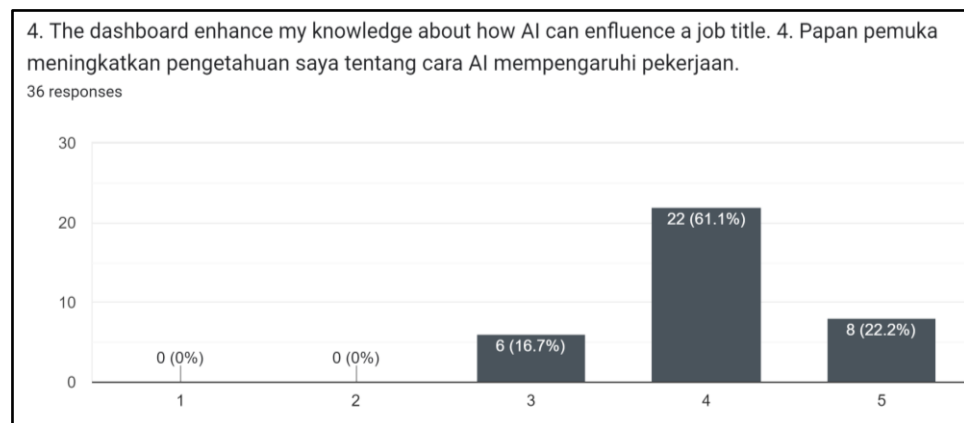


Figure 5.11 Dashboard Survey Questions 4

The next question was to clarify whether participants gained information about how much AI is affecting current job titles. 14 (38.9%) strongly agreed, while 17 (47.2%) participants only agreed. 3 (8.3%) participants were on the neutral side of the statement, while 2 (5.6%) participants disagreed with this statement.

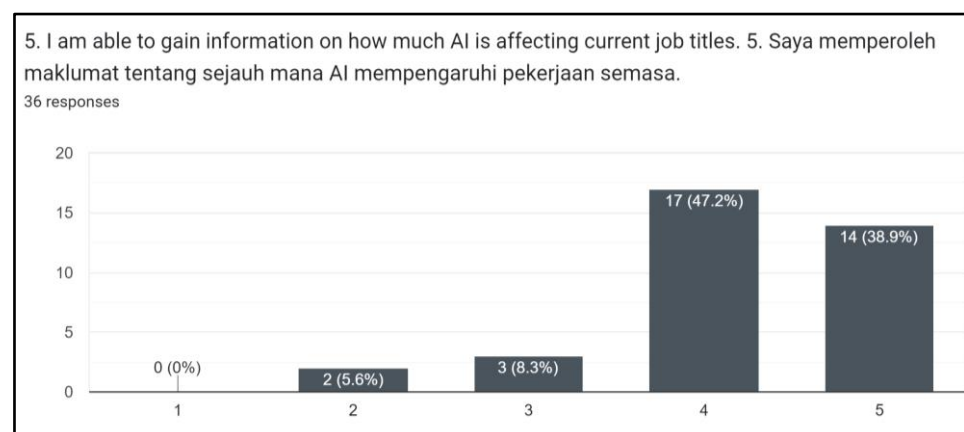


Figure 5.12 Dashboard Survey Questions 5

Figure 5.13 illustrates that 24 (66.7%) participants strongly agreed that the dashboard shows the connection between AI impact, job tasks and AI models, while 11 (30.6%) participants partially agreed, and the remaining 1 (2.8%) chose to be neutral. We can see that almost all participants agreed.

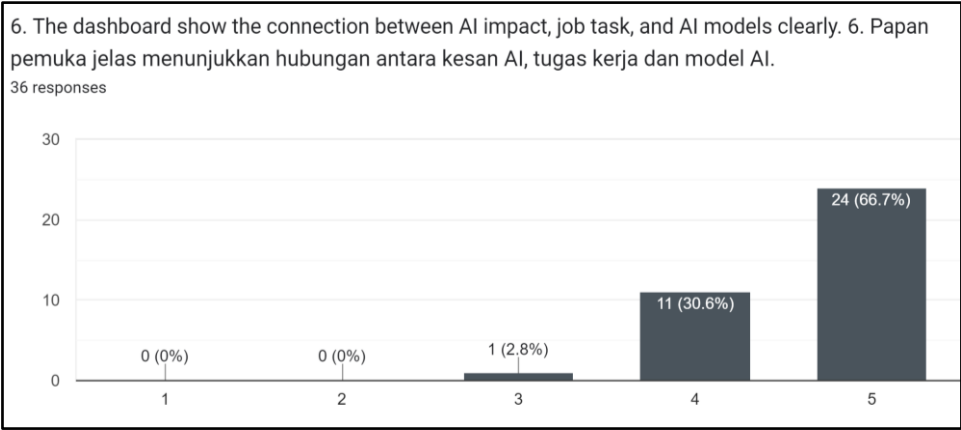


Figure 5.13 Dashboard Survey Questions 6

5.4.4 Behavioural Intentions & Attitude

This part combines the last three aspects of TAM, which are behavioural intentions (BI), attitude (ATT), and actual use (AU). It focuses on assessing participants' likelihood of adopting and using the dashboard on a regular basis. Figure 5.14 shows that 2 (5.6%) participants strongly agreed, 20 (55.6%) chose agree, 13 (36.1%) chose neutral, and 1 (2.8%) participant disagreed that they would use the dashboard as a source of information about AI.

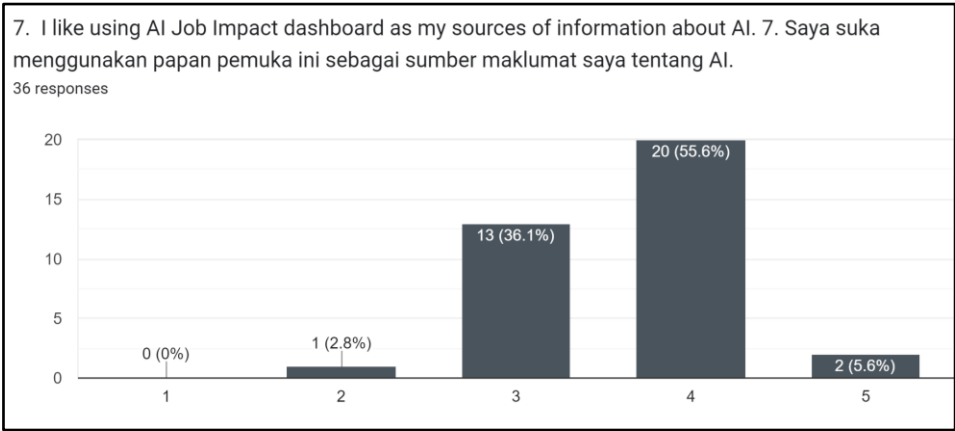


Figure 5.14 Dashboard Survey Questions 7

Question 8 clarifies whether the AI Job Impact dashboard helps users understand the impact of AI on jobs. Based on Figure 5.15, there were 8 (22.2%) participants who chose to strongly agree, 25 (69.4%) chose to agree, and the remaining 3 (8.3%) chose neutral. Most participants choose to only agree. Question 8 was used to measure the actual use (AU) of the dashboard.

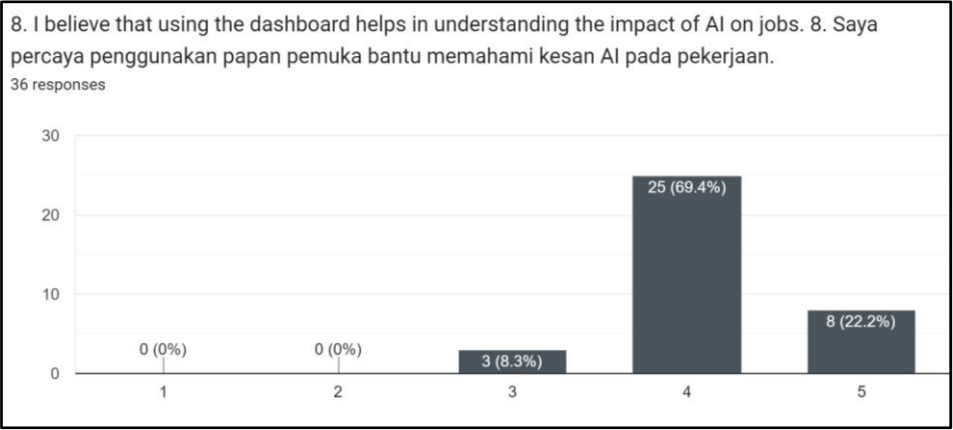


Figure 5.15 Dashboard Survey Questions 8

The overall satisfaction of the participant was measured through question 9. This question falls under the attitude (ATT) aspect of TAM. Figure 5.16 indicates that 14 (38.9%) participants strongly agreed, 18 (50%) only agreed, and the remaining 4 (11.1%) chose neutral on this statement. This suggests that a majority of the participants (88.9%) had a positive attitude towards the system, indicating a good level of satisfaction.

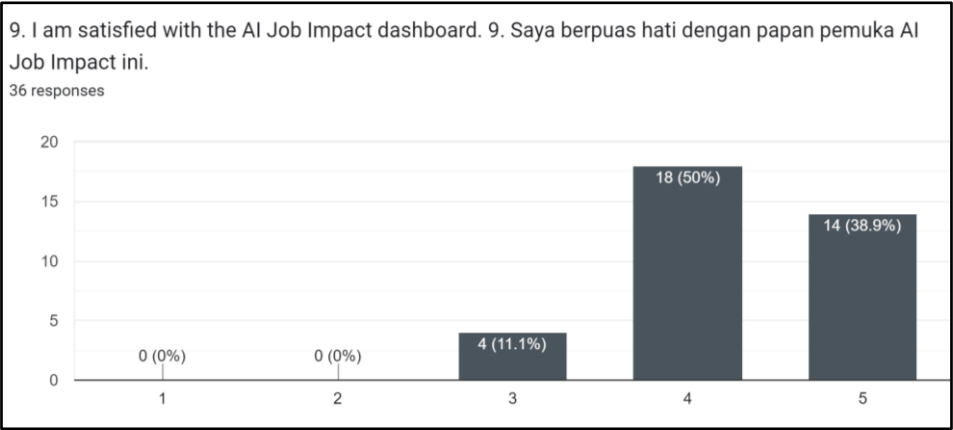


Figure 5.16 Dashboard Survey Questions 9

The last question for this part was to ask the participants if they would recommend the dashboard to others. As illustrated in Figure 5.17, 27 (75%) of participants strongly agreed with the statement, 7 (19.4%) only agreed, while the remaining 2 (5.6%) chose neutral. This suggests a very high proportion of participants (94.4%) would recommend the dashboard to others, indicating a strong potential for user adoption.

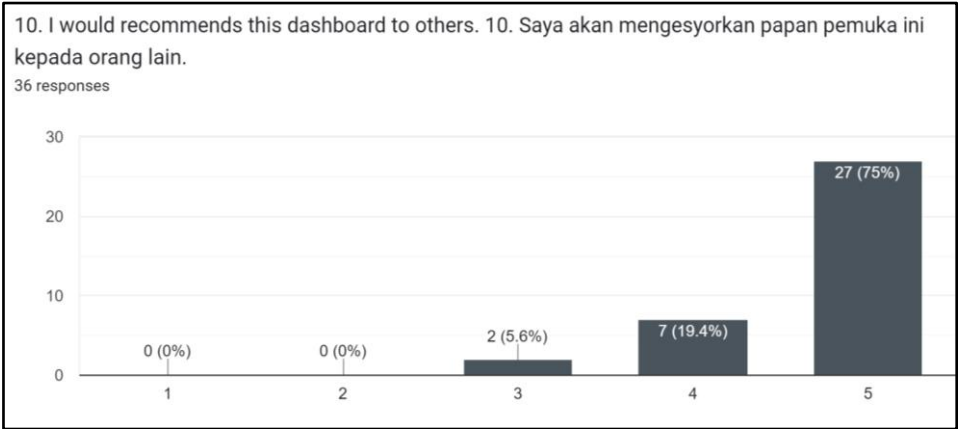


Figure 5.17 Dashboard Survey Questions 10

To conclude this user survey part, the results indicate a positive outlook on the adoption and use of the AI Job Impact dashboard. The majority of participants (61.2%) agreed or strongly agreed that they would use the dashboard as a source of information about AI, and most (91.6%) agreed or strongly agreed that the dashboard helped them understand the impact of AI on jobs. Additionally, a large majority (88.9%) had a positive attitude towards the system, and an even higher proportion (94.4%) would recommend the dashboard to others, suggesting a strong potential for user adoption.

5.4.5 Comments & Suggestions

The last section of the user survey is the comments and suggestions section. Participant are asked to leave a comment about the overall projects and give a suggestion to improve the established dashboard. Specifically, 18 out of 36 participants in the user survey contributed to this section. Table 5.3 below shows the comments and suggestions collected from the user survey.

Comment	Suggestion
<i>Dashboard design is a bit dull</i>	<i>Decorate the dashboard</i>
<i>Dashboard menarik</i>	<i>Jenis data yg ditunjukkn trlalu sedikit</i>
<i>Nice content</i>	<i>Page prediction and discussion has lots of words, it made me a bit nauseous</i>
<i>Informasi dalam dashboard sedikit</i>	<i>Cari lebih banyak data berkaitan kerana saya hanya nampak data AI Impact, Job Task dengan AI Models sahaja</i>
<i>Discussion page is confusing</i>	<i>Try to simplify the discussion</i>
<i>Maklumat agak sedikit</i>	<i>Cari lebih banyak maklumat berkaitan AI</i>
<i>Goodluck</i>	<i>Takde</i>
<i>Pengesahan pasal data tak diberi tahu</i>	<i>Sila maklumkn sumber data dlm papan pemuka</i>
<i>Page discussion agk memeningkan</i>	<i>silu buat infographic</i>
<i>gudluck fyp jat</i>	<i>-</i>
<i>Need more information</i>	<i>Clarify how the impact is calculated</i>
<i>Semoga berjaya KI</i>	<i>Tiada</i>
<i>Tak ckup detail</i>	<i>Olahkn lgi pascal chart2</i>
<i>Info trlalu general</i>	<i>Prlukn byk data</i>
<i>Job title saya x jumpe</i>	<i>+ byk lgi job title</i>
<i>I am not familiar with dashboards</i>	<i>Website need improvements</i>
<i>Need more information</i>	<i>Elaborate more</i>
<i>Mklumt x lengkap</i>	<i>Tambah charts</i>
<i>Sya tak fhm AI Model tu cmne</i>	<i>Terangkn pascal AI model</i>

Table 5.3 Comments and Suggestions Collected

5.5 Action to be Taken

Based on the comments and suggestions given by participants, further action improvements for the dashboard should be made in order to improve the existing dashboard and further boost its usability and user satisfaction. Following a comprehensive review of all the participants' feedback, the following actions have been prioritized based on their relevance, deliverables, and feasibility within the current project scope. These actions are:

- Improve the visual appeal of the dashboard by adding more colour schemes, icons, and graphics.
- Simplify the discussion page by reducing clutter images, using clear headings, and considering using infographics for better data comprehension.
- Enrich the data visualisation dashboard by including more relevant data sources and expanding the scope of AI metrics displayed.
- Clarify data sources to clearly indicate the origin of each data point used in the dashboard to enhance transparency.
- Add an explanation of how the column percentage of AI impact was calculated.
- Expand data coverage by including additional data sources on economic, social, and ethical aspects of the AI impact on jobs.
- Provide clear and concise explanations of the different types of AI models used and their role in each job title.

5.6 Summary

In conclusion, chapter 5 shows a comprehensive analysis of the results collected during the testing phase of the dashboard using the Technology Acceptance Model (TAM). This chapter covers the designed questions, target audience, test results, and analysis of the user testing survey that was carried out. Through the analysis, overall user satisfaction with the system can be identified. The comments and suggestions obtained during the testing phase are important to further improve the usability and reliability of the AI Job Impact dashboard and contribute to the achievement of this research project's objectives.

CHAPTER 6

CONCLUSION AND RECOMMENDATION

This last chapter discusses the results and outcome of this research project's completion. It evaluates the research's success in achieving the underlined research objectives, highlighting the limitations of the research projects, and making recommendations for future studies to build upon this work.

6.1 Project Conclusion

This research project, "Data Visualisation of the Impact of Radical Artificial Intelligence Technology Toward Jobs" has been developed based on three main research objectives. Chapter 2, Literature Review, helps in establishing the foundation of knowledge about the topic of this research project, artificial intelligence, and big data technologies. Requirement analysis in Chapters 3 and 4 helps in understanding all the requirements necessary for the successful development of this research project, ensuring that the final solution aligns with the project's objectives and addresses the identified user needs while considering technical and operational limitations.

The first research objective was to determine the data required to analyse the impact of radical artificial intelligence technology toward jobs. Identifying and collecting relevant data from various sources helps ensure that the analysis is comprehensive and accurate. The data collection process involves careful consideration of data quality, reliability, and relevance to the research topic. The dataset used for this research project was extracted from the website "There Is An AI For That." The website enlists new AI models that were developed and registered online and simultaneously keeps a record of the AI job impact index by calculating the total job tasks, AI models, and the percentage of AI impact on each job title.

The second research objective was to design and develop dashboards that include dynamic visualisations of the impacts of the radical artificial

intelligence technology towards jobs using Figma and Microsoft Power BI. Figma is a flexible tool used in designing the user interface and user experience for any platform, and Microsoft Power BI offers interactive and dynamic displays, allowing users to explore data in real-time and instantly adjust parameters to gain deeper insights. Based on the Gantt Chart (Appendix A), the designing phase of this project took approximately 5 weeks, while the implementation or development phase took approximately 4 months to complete.

The third research objective was to evaluate the user acceptance of the developed web-based dashboard using the Technology Acceptance Model (TAM). The TAM framework focuses on measuring the perceived ease of use, perceived usefulness, behavioural intentions, attitude, and actual use of the dashboard. Conducting functional and usability testing is crucial to ensuring the effectiveness, efficiency, and user-friendliness of the dashboard. It also helps in identifying and addressing any usability issues that users may encounter, enhancing the overall user experience, and increasing the likelihood of user adoption of the dashboard on a regular basis. The functional and usability testing of the dashboard took approximately 2 months. There were 36 participants who contributed to the user survey, and a large portion of them (32 participants) agreed that they were satisfied with the dashboard.

6.2 Project Contribution

Data Visualisation of the Impact of Radical Artificial Intelligence Technology Toward Jobs has contributed to several communities with the development of the AI Job Impact Dashboard and the research study documentation. The AI Job Impact dashboard has contributed most to the graduates' and undergraduates' communities, as they can use these insights to help them reevaluate their skills to find suitable industries that fit them and prepare for their future. Scholars and practitioners can use this research study documentation to further improve their research and boost their decision-making. In general, the development of this research project and the AI Job

Impact Dashboard helps the public communities become more aware of artificial intelligence, its technological advancement, and how AI can influence the current job scope and future expectations.

6.3 Limitation

During the development and testing phases of this project, a few problems and limitations were identified. These problems and limitations were due to a lack of expertise and a limited time frame. It is possible to overcome these problems and limitations, but it requires a long-term scope, mastery of specific techniques, and in-depth knowledge about how to apply them. These problems and limitations are:

- The data collection scope was restricted because the source only provided datasets for the last 3 years, and some of the attributes were removed due to data privacy protection from the owner.
- There was no advanced data processing method such as machine learning to reassign the attribute job titles to their correct job category because the attribute contains errors, manual cleaning will take some time since the data was large, and mastering machine learning will take much longer to implement in this project.
- The dashboard page of the website takes time to render or load up since the dashboard was embedded in the page code, and the page will only display if the dashboard has been rendered, which also took quite some time because there were many data and charts to display.
- Artificial intelligence is a rapidly changing and advancing technology, meaning that the references, related studies and this research project's documentation might be outdated quickly.

6.4 Suggestions for Future Studies

Through the analysis of this research project's problems and limitations, several avenues for future studies and improvements can be explored. Practitioners, scholars, and researchers can also use these suggestions to further improve their studies. These suggestions are:

- Leveraging rich datasets containing numerous attributes, extensive instances, and a comprehensive timeframe can provide a more holistic understanding of the topic.
- Conduct extensive research on advanced data preparation methods to ensure the highest quality data is used for analysis, leading to more reliable and insightful results.
- Improve website loading time by considering lazy loading for the dashboard, allowing the webpage to render first and then progressively loading the data and charts of the dashboard afterwards.
- Maintain rigorous fact-checking procedures and ensure continuous awareness of the most recent developments in artificial intelligence.

6.5 Summary

In conclusion, this chapter discusses the results and outcome of this research project's completion. It evaluates the research's success in achieving the underlined research objectives, highlighting the limitations of the research projects, and making recommendations for future studies to build upon this work. Generally, this project's research objective has been achieved, and overall, users are satisfied with the developed dashboard. A few problems and limitations have been identified, and corresponding action and suggestions can be taken to improve future studies.

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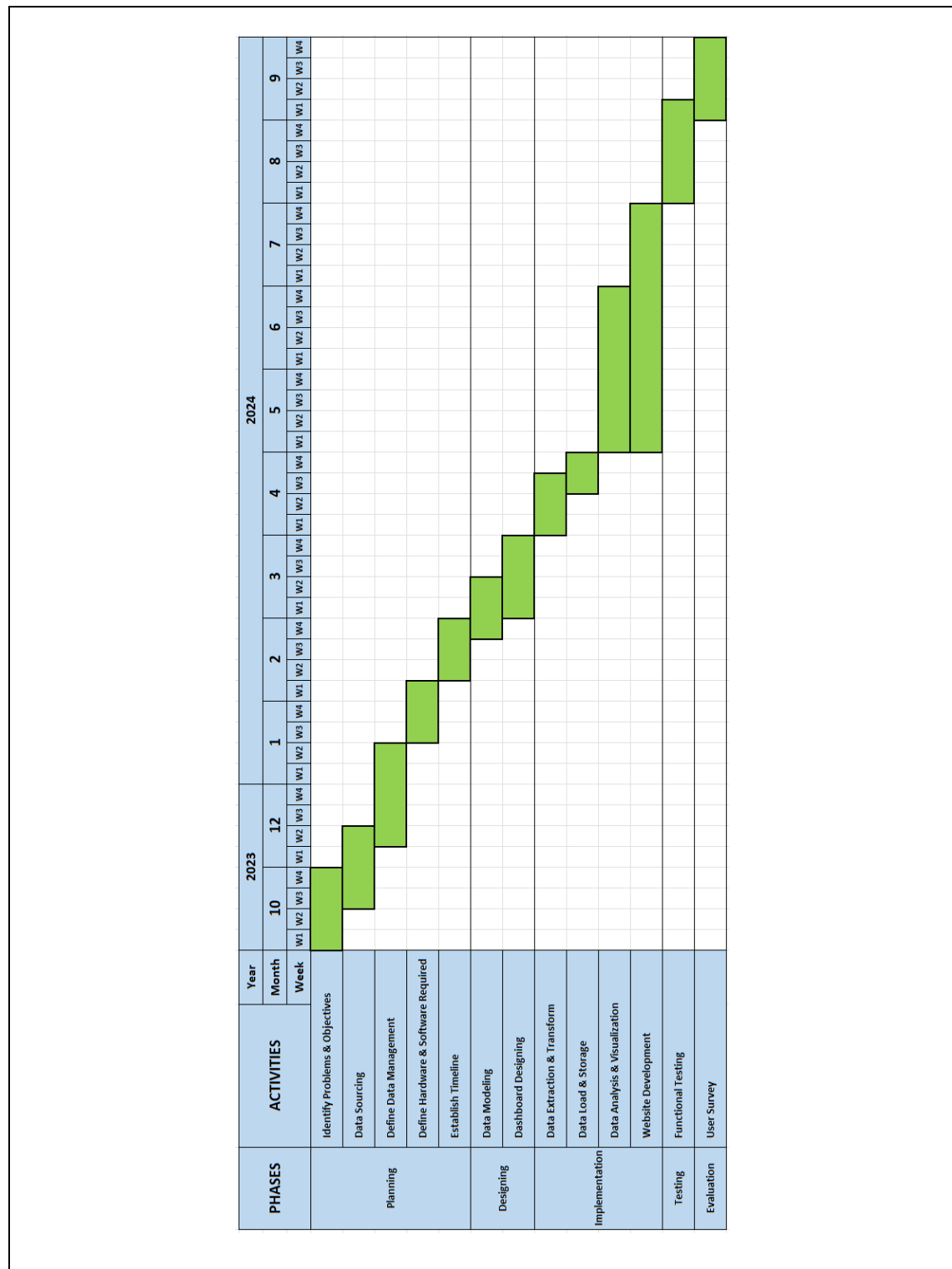
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APPENDICES

Appendix A: Project Development Gantt Chart



Appendix B.1: User Survey - Part 1

User Survey on AI Job Impact Dashboard

Assalamua'laikum, I would like to take a bit of your time.

My name is Khairul, and I am conducting a survey to receive feedbacks about my Final Year project(FYP), which is a data visualization dashboard embedded on a website. The information are as follows:

My Information

Name : Khairul Izzat Bin Roslan

Email : 2022464134@student.uitm.edu.my

University : Technology Mara University (UiTM.) Perlis, Arau Campus

Course : CS240, Bachelor's Degree in Information Technology (Hons.)

AI Job Impact Website and Dashboard

Title : Data Visualization of the Impact of Radical Artificial Intelligence Technology Toward Jobs

Description : The purpose of the AI Job Impact dashboard is to spread awareness about the overall impact of Artificial Intelligence on job categories and job titles. This impacts are aided by the counts of developed AI models and registered jobs for each job titles. The AI Job Impact website focus on providing informations about the Artificial Intelligences, highlighted news, and provide advices to overcome.

Dictionary

AI : Artificial Intelligence

Web : Website

* Indicates required question

English

Instructions

1. The following is the link to the developed dashboard website. You may navigate to the other pages.

Web-Based Dashboard Link : [AI Job Impact](#)

2. It is recommended to switch to Desktop Mode if you are browsing using mobile phone.

3. Explore the developed website and dashboard. Note down any flaws and possible improvements.

4. The survey consist of 3 separate sections which are:

A. User Information

B. Dashboard Survey

C. Comment & Recommendations

5. It is recommended that you fill-in the survey after you explore the website and dashboard.

Participation

You are required to answer all the question that is marked by *. However you can withdraw from answering the survey at any time. It is estimated that this survey will take about 5 minutes to complete. Your participation is greatly appreciated.

Data Security

The security of the data you provided will be kept private from the public and only be use for survey analysis of the effectiveness of this dashboard and website. Any feedbacks will be used to improve the website and dashboard.

Thank you for your response.

Appendix B.2: User Survey - Part 2

Bahasa Melayu

Arahan

1. Berikut adalah pautan ke laman web dan papan pemuka yang dibangunkan. Anda boleh menavigasi ke halaman lain di laman web tersebut.

Pautan Web : [AI Job Impact](#)

2. Adalah disyorkan untuk bertukar kepada Mod Desktop jika anda membuka pautan itu menggunakan telefon mudah alih.

3. Terokai laman web dan papan pemuka yang dibangunkan. Catatkan sebarang kelemahan dan kemungkinan penambahbaikan.

4. Tinjauan ini terdiri daripada 3 bahagian berasingan iaitu:

A. Maklumat Pengguna

B. Tinjauan Papan Pemuka

C. Komen & Cadangan

5. Anda disyorkan untuk mengisi tinjauan ini selepas anda meneroka laman web dan papan pemuka.

Penyertaan

Anda dikehendaki menjawab semua soalan yang bertanda *. Walau bagaimanapun anda boleh menarik diri daripada menjawab tinjauan ini pada bila-bila masa. Dianggarkan tinjauan ini akan mengambil masa kira-kira 5 minit untuk diselesaikan. Penyertaan anda amat saya hargai.

Keselamatan Data

Keselamatan data yang anda berikan akan dirahsiakan daripada orang ramai dan hanya digunakan untuk analisis tinjauan keberkesanan papan pemuka dan tapak web ini. Sebarang maklum balas akan digunakan untuk menambah baik laman web dan papan pemuka.

Terima kasih atas maklum balas anda.

A. User Information / Maklumat Pengguna

1. 1. Gender *

1. Jantina

Mark only one oval.

☐ Male / Lelaki

☐ Female / Perempuan

Appendix B.3: User Survey - Part 3

2. 2. Age *

2. *Umur*

Mark only one oval.

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46 and above / 46 dan ke atas

3. 3. Occupation Status *

3. *Status Pekerjaan*

Mark only one oval.

- ☐ Student / Pelajar
- ☐ Employed / Bekerja
- ☐ Unemployed / Menganggur
- ☐ Pension / Pencen

4. 4. Which work/education industry are you currently in ? *

4. *Industri pekerjaan/pendidikan manakah anda sekarang ?*

Mark only one oval.

- ☐ Administrative & Clerical / Pentadbiran & Perkeranian
- ☐ Communication & Public Relation / Komunikasi & Perhubungan Awam
- ☐ Construction / Pembinaan
- ☐ Data & Information Technology / Data & Teknologi Maklumat
- ☐ Hospitality / Keramahan
- ☐ Law Enforcement / Penguatkuasaan undang-undang
- ☐ Leadership & Strategy / Kepimpinan & Strategi
- ☐ Medical & Healthcare / Perubatan & Penjagaan Kesihatan
- ☐ Sales & Marketing / Jualan & Pemasaran
- ☐ Supply Chain & Logistics / Rantaian Supply & Logistik

Appendix B.4: User Survey - Part 4

5. 5. Where do you usually find news about AI ? *

5. Di manakah anda biasanya mencari berita tentang AI?

Mark only one oval.

- ☐ Social Media / Media sosial
☐ News Bulletin / Buletin Berita
☐ Search Engine / Enjin Caria
☐ Word of Mouths / Kata Mulut

6. 6. Are you aware that AI is growing and starting to take over jobs? *

6. Adakah anda sedar bahawa AI sedang berkembang dan mula mengambil alih pekerjaan?

Mark only one oval.

- ☐ Yes / Ya
☐ No / Tidak

7. 7. Do you have any experience in using data visualization dashboard before this survey? *

7. Adakah anda berpengalaman menggunakan papan pemuka visualisasi data sebelum ini?

Mark only one oval.

- ☐ Yes / Ya
☐ No / Tidak

B. Dashboard Survey / Tinjauan Papan Pemuka

This test method is called TAM or Technology Acceptance Model. It is a theory used to understand why people adopt or use new technologies. This testing method is used in evaluating the effectiveness of the developed dashboard.

Kaedah ujian ini dipanggil TAM atau Technology Acceptance Model. Ini adalah teori yang digunakan untuk memahami mengapa orang menerima pakai atau menggunakan teknologi baharu. Kaedah ujian ini digunakan dalam menilai keberkesanan papan pemuka yang dibangunkan.

Appendix B.5: User Survey - Part 5

Answering Guide / Panduan Menjawab

1. Thoroughly explore the AI Job Impact Dashboard. Link : [AI Job Impact](#)

1. Terokai Papan Pemuka Impak Kerja AI secara menyeluruh. Pautan : [AI Job Impact](#)

2. Answer all 10 questionnaires by choosing between 1 to 5. Below is the rating description.

2. Jawab kesemua 10 soal selidik dengan memilih antara 1 hingga 5. Di bawah ialah penerangan penilaian.

1 : Strongly Disagree / Sangat Tidak Bersetuju

2 : Disagree / Tidak Bersetuju

3 : Neutral

4 : Agree / Setuju

5 : Strongly Agree / Sangat Bersetuju

3. Be honest with your response as it is important to improve the dashboard.

3. Jujurlah dengan respons anda kerana ia adalah penting untuk menambah baik papan pemuka.

Part 1 : Perceived Ease of Use

Bahagian 1 : Persepsikan Kemudahan Penggunaan

8. 1. It is easy for me to navigate and use the AI Job Impact dashboard. *

1. Mudah untuk saya menavigasi dan menggunakan papan pemuka AI Job Impact ini.

Mark only one oval.

1 2 3 4 5

Strongly Disagree / Sangat Tidak Bersetuju

9. 2. The information displayed in the dashboard is clear and easy to understand. *

2. Maklumat yang dipaparkan dalam papan pemuka adalah jelas dan mudah difahami.

Mark only one oval.

1 2 3 4 5

Strongly Disagree / Sangat Tidak Bersetuju

Appendix B.6: User Survey - Part 6

10. 3. The dashboard does not require me to learn any complex functionalities. *
3. Papan pemuka tidak memerlukan saya mempelajari sebarang fungsi yang kompleks.

Mark only one oval.

1 2 3 4 5

Strongly Disagree / Sangat Bersetuju

Part 2 : Perceived Usefulness

Bahagian 2 : Persepsikan Kebergunaan

11. 4. The dashboard enhance my knowledge about how AI can enfluence a job title. *
4. Papan pemuka meningkatkan pengetahuan saya tentang cara AI mempengaruhi pekerjaan.

Mark only one oval.

1 2 3 4 5

Strongly Disagree / Sangat Bersetuju

12. 5. I am able to gain information on how much AI is affecting current job titles. *
5. Saya memperoleh maklumat tentang sejauh mana AI mempengaruhi pekerjaan semasa.

Mark only one oval.

1 2 3 4 5

Strongly Disagree / Sangat Bersetuju

Appendix B.7: User Survey - Part 7

13. 6. The dashboard show the connection between AI impact, job task, and AI models clearly. *
6. Papan pemuka jelas menunjukkan hubungan antara kesan AI, tugas kerja dan model AI.

Mark only one oval.

1 2 3 4 5

Strongly Dislike / Sangat Bersetuju

Part 3 : Behavioural & Attitude

Bahagian 3 : Tingkah Laku & Sikap

14. 7. I like using AI Job Impact dashboard as my sources of information about AI. *
7. Saya suka menggunakan papan pemuka ini sebagai sumber maklumat saya tentang AI.

Mark only one oval.

1 2 3 4 5

Strongly Dislike / Sangat Bersetuju

15. 8. I believe that using the dashboard helps in understanding the impact of AI on jobs. *
8. Saya percaya penggunaan papan pemuka bantu memahami kesan AI pada pekerjaan.

Mark only one oval.

1 2 3 4 5

Strongly Dislike / Sangat Bersetuju

Appendix B.8: User Survey - Part 8

16. 9. I am satisfied with the AI Job Impact dashboard. *
9. Saya berpuas hati dengan papan pemuka AI Job Impact ini.

Mark only one oval.

1 2 3 4 5

Strongly Dislike / Sangat Bersetuju

17. 10. I would recommends this dashboard to others. *
10. Saya akan mengesyorkan papan pemuka ini kepada orang lain.

Mark only one oval.

1 2 3 4 5

Strongly Dislike / Sangat Bersetuju

C. Comment & Recommendation / Komen & Cadangan

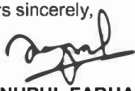
18. 1. Leave a comment.
1. Tinggalkan komen

19. 2. Provide a suggestion.
2. Berikan cadangan.

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Google Forms

Appendix C: Supervisor Approval Letter

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Bahagian Penyelidikan, Jaringan Industri Masyarakat, Alumni & Keusahawanan Universiti Teknologi MARA Cawangan Perlis Kampus Arau, 02600 Arau, Perlis Tel: (+604) 988 2017 / 2028 / 2029 / 2080 / 2302 / 2308 Email: pijmaperlis@uitm.edu.my	25 TAHUN 1974-2024 UITM PERLIS UNIVERSITI TEKNOLOGI MARA Cawangan Perlis / Kampus Arau						
Our ref : 600-UiTMPs (PJIM&A/REC-ERC 106/2024) Date : 1445H شوال 10 19 April 2024							
Khairul Izzat bin Roslan (Dr. Aznoora binti Osman) College of Computing, Informatics and Mathematics Universiti Teknologi MARA 02600 Arau PERLIS							
Dear Madam,							
APPROVAL LETTER - UiTM PERLIS BRANCH ETHICS REVIEW COMMITTEE							
Thank you for submitting your research proposal to Ethics Review Committee (ERC) of UiTM Perlis Branch under Research Ethics Committee (REC). After considering your application, the Committee approved your proposal entitled "Data Visualization of The Impact of Radical Artificial Intelligence Technology Toward Jobs."							
Details of the approval are as follows:							
<table border="1"><tr><td>Ref. number:</td><td>REC-ERC 106/2024</td></tr><tr><td>Approval Period:</td><td>02/04/2024 – 01/04/2025</td></tr><tr><td>Authorised personnel:</td><td>1. Khairul Izzat bin Roslan 2. Dr. Aznoora binti Osman</td></tr></table>	Ref. number:	REC-ERC 106/2024	Approval Period:	02/04/2024 – 01/04/2025	Authorised personnel:	1. Khairul Izzat bin Roslan 2. Dr. Aznoora binti Osman	
Ref. number:	REC-ERC 106/2024						
Approval Period:	02/04/2024 – 01/04/2025						
Authorised personnel:	1. Khairul Izzat bin Roslan 2. Dr. Aznoora binti Osman						
The UiTM Research Ethics Committee operates in accordance to the IGH Good Clinical Practice Guidelines, Malaysia Good Clinical Practice Guidelines and the Declaration of Helsinki. The approval of this project is conditional upon your continuing compliance with these guidelines and declaration.							
We draw to your attention the requirement that a report on this research, must be submitted every 12 months from the date of the approval or on the completion of the project, whichever occurs first. Failure to submit reports will result in withdrawal of consent for the project to proceed. Amendments, if any, to the study documents are to be submitted to ERC for approval.							
If you require further information, please contact ERC Secretariat at 04-988 2509 or email at jasshani@uitm.edu.my							
اوسها، تقوى، مليا MALAYSIA MADANI" "BERKHIDMAT UNTUK NEGARA"							
Yours sincerely, 							
DR. NURUL FARHA ZAINUDDIN Chairperson Ethics Review Committee (ERC) UiTM Perlis Branch							
Universiti Teknologi MARA Cawangan Perlis Kampus Arau 02600 Arau, Perlis, MALAYSIA Tel: (+604) 988 2000 Faks: (+604) 988 2019 e-mel: korporatperlis@uitm.edu.my							
50 TAHUN 1974-2024 SAMBUTAN 50TAHUN UITM CAWANGAN PERLIS UiTM							

Appendix D: Certificate of Proofreading

28/06/2024

To whom it may concern

This is to certify that the undergraduate proposal entitled

DATA VISUALIZATION OF THE IMPACT OF RADICAL ARTIFICIAL INTELLIGENCE
TECHNOLOGY TOWARD JOBS

Prepared and submitted by

KHAIRUL IZZAT BIN ROSLAN

2022464134

is a part and partial fulfillment for his/her degree in

Bachelor of Information Technology (Hons.) (CS240)

Has been edited and proof-read based on:

- spelling
- grammar
- punctuation
- sentence structure
- flow of paragraph

with the standard rules of British English language by me

DR. KHAIRUL ANWAR SEDEK

SENIOR LECTURER

KOLEJ PENGAJIAN PERKOMPUTERAN, INFORMATIK, DAN MATEMATIK

UITM CAWANGAN PERLIS,

KAMPUS ARAU, PERLIS

Sincerely yours;



DR. KHAIRUL ANWAR SEDEK
PENKYSARAH KAHANAN
PENGAJIAN SAINS PERKOMPUTERAN
KOLEJ PENGAJIAN PERKOMPUTERAN, INFORMATIK DAN MATEMATIK
UITM CAWANGAN PERLIS KAMPUS ARAU, 32600 ARAU, PERLIS

Appendix E: Similarity Report

Similarity Report	
PAPER NAME	AUTHOR
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PAGE COUNT	FILE SIZE
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Summary	